

PRACTICE SHEET

- Consider the following statements:
I. If $\theta = 1200^\circ$, then $(\sec\theta + \tan\theta)^{-1}$ is positive
II. If $\theta = 1200^\circ$, then $(\operatorname{cosec}\theta - \cot\theta)$ is negative.
Which of the statements given above is/are correct?
(a) Only I (b) Only II
(c) Both I nor II (d) Neither I nor II
- Match List I with List II and select the correct answer using the code given below the lists.

	List I		List II
A.	$\tan 15^\circ$	(i)	$-2 - \sqrt{3}$
B.	$\tan 75^\circ$	(ii)	$2 + \sqrt{3}$
C.	$\tan 105^\circ$	(ii)	$-2 + \sqrt{3}$
		(iv)	$2 - \sqrt{3}$

(a) A-(iv), B-(i), C-(ii) (b) A-(iv), B-(ii), C-(i)
(c) A-(iii), B-(ii), C-(i) (d) A-(ii), B-(i), C-(iv)
- If $A+B+C = \frac{\pi}{2}$, then what is the value of $\tan A \tan B + \tan B \tan C + \tan C \tan A$?
(a) 0
(b) 1
(c) -1
(d) $\tan A \tan B \tan C$
- What is the value of $\cos\left(\frac{\pi}{9}\right) + \cos\left(\frac{\pi}{3}\right) + \cos\left(\frac{5\pi}{9}\right) + \cos\left(\frac{7\pi}{9}\right)$?
(a) 1 (b) -1
(c) -1/2 (d) 1/2
- If $\cot\theta = 5/12$ and θ lies in the third quadrant, then what is $(2\sin\theta + 3\cos\theta)$ equal to?
(a) -4
(b) $-p^2$ for some odd prime p
(c) $\left(-\frac{q}{p}\right)$ where p is an odd prime and q is a positive integer with $\left(\frac{q}{p}\right)$ not an integer
(d) $-p$ for some odd prime p .
- If $\cot\theta = 2\cos\theta$, where $\left(\frac{\pi}{2}\right) < \theta < \pi$, then what is the value of θ ?
(a) $\frac{5\pi}{6}$ (b) $\frac{2\pi}{3}$
(c) $\frac{3\pi}{4}$ (d) $\frac{11\pi}{12}$
- If $p = \sin(989^\circ) \cos(991^\circ)$, then which one of the following is correct?
(a) p is finite and positive
(b) p is finite and negative
(c) $p = 0$
(d) p is undefined
- If $\sin x + \operatorname{cosec} x = 2$, then what is the value of $\sin^4 x + \operatorname{cosec}^4 x$?
(a) 2 (b) 4
(c) 8 (d) 16
- What is the value of $\tan(-1575^\circ)$?
(a) 1 (b) $\frac{1}{2}$
(c) 0 (d) -1
- For which acute angle θ , $\operatorname{cosec}^2\theta = 3\sqrt{3} \cot\theta - 5$?
(a) $\frac{5\pi}{12}$ (b) $\frac{\pi}{3}$
(c) $\frac{\pi}{6}$ (d) $\frac{\pi}{4}$
- If $\tan^2\theta = 2 \tan^2\phi + 1$, then which one of the following is correct?
(a) $\cos(2\theta) = \cos(2\phi) - 1$
(b) $\cos(2\theta) = \cos(2\phi) + 1$
(c) $\cos(2\theta) = [\cos(2\phi) - 1]/2$
(d) $\cos(2\theta) = [\cos(2\phi) + 1]/2$
- If $\cos\theta < \sin\theta$ and θ lies in the first quadrant, then which one of the following is correct?
(a) $0 < \theta < \frac{\pi}{4}$ (b) $\frac{\pi}{4} < \theta < \frac{\pi}{2}$
(c) $0 < \theta < \frac{\pi}{3}$ (d) $\frac{\pi}{3} < \theta < \frac{\pi}{2}$
- If $\sin^2 x + \sin^2 y = 1$, then what is the value of $\cot(x+y)$?
(a) 1 (b) $\sqrt{3}$
(c) 0 (d) $\frac{1}{\sqrt{3}}$
- If $\sin x + \sin y = a$, $\cos x + \cos y = b$, then what is the value of $\cos(x-y)$?
(a) $a^2 - 1$ (b) $b^2 - 1$
(c) $\frac{1}{2}(a^2 + b^2 - 2)$ (d) $\frac{1}{2}(a^2 + b^2)$
- Which one of the following is correct?
 $\left(1 + \cos 67\frac{1}{2}^\circ\right)\left(1 + \cos 112\frac{1}{2}^\circ\right)$ is?
(a) An irrational number and is greater than 1
(b) A rational number but not an integer
(c) An integer
(d) An irrational number and is less than 1
- What is the measure of the angle $114^\circ 35' 30''$ in radian?
(a) 1 rad (b) 2 rad
(c) 3 rad (d) 4 rad
- If $\sin A = \frac{1}{\sqrt{5}}$, $\cos B = \frac{3}{\sqrt{10}}$; where A, B being positive acute angles, then what is $(A+B)$ equal to:

- (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$
 (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$
18. What is $\sqrt{2+\sqrt{2+\sqrt{2+2\cos 4A}}}$ equal to?
 (a) $\cos A$ (b) $\cos(2A)$
 (c) $2\cos\left(\frac{A}{2}\right)$ (d) $\sqrt{2\cos A}$
19. What is the value of $\sin(1110^\circ)$?
 (a) 1 (b) $\frac{1}{2}$
 (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{\sqrt{3}}{2}$
20. If $\sin 2A = \frac{4}{5}$, then what is the value of $\tan A$ (where $0 \leq A \leq \frac{\pi}{4}$)?
 (a) 1 (b) -1
 (c) $\frac{1}{2}$ (d) 2
21. What is the value of $\frac{\cos 10^\circ - \sin 10^\circ}{\cos 10^\circ + \sin 10^\circ}$?
 (a) $\tan 35^\circ$ (b) $\tan 10^\circ$
 (c) $\frac{1}{\sqrt{2}}$ (d) 1
22. Which one of the following pairs is not correctly matched?
 (a) $\sin 2\pi$ $\sin(-2\pi)$
 (b) $\tan 45^\circ$ $\tan(-315^\circ)$
 (c) $\cot(\tan^{-1} 0.5)$ $\tan(\cot^{-1} 0.5)$
 (d) $\tan 420^\circ$ $\tan(-60^\circ)$
23. What is the value of $\sin\left(\frac{5\pi}{12}\right)$?
 (a) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ (b) $\frac{\sqrt{6}+\sqrt{2}}{4}$
 (c) $\frac{\sqrt{3}+\sqrt{2}}{4}$ (d) $\frac{\sqrt{6}+1}{2}$
24. What is the value of $\left(\sin 22\frac{1^\circ}{2} + \cos 22\frac{1^\circ}{2}\right)^4$?
 (a) $\frac{3+2\sqrt{2}}{2}$ (b) $\frac{1+2\sqrt{2}}{2}$
 (c) $\frac{3\sqrt{3}+2}{2}$ (d) 1
25. What is the value of $\sin 292\frac{1^\circ}{2}$?
 (a) $\frac{1}{3}\sqrt{2+\sqrt{3}}$ (b) $-\frac{1}{3}\sqrt{2-\sqrt{3}}$
 (c) $\frac{1}{2}\sqrt{2+\sqrt{2}}$ (d) $-\frac{1}{2}\sqrt{2+\sqrt{2}}$
26. What is the value of $\frac{\sin \theta + \cos \theta - \tan \theta}{\sec \theta + \operatorname{cosec} \theta - \cot \theta}$, when $\theta = \frac{3\pi}{4}$?
 (a) 0 (b) 1
 (c) -1 (d) None of these
27. What is the value of $\sin 15^\circ \cdot \sin 75^\circ$?
 (a) $\frac{1}{4}$ (b) $\frac{1}{8}$
 (c) $\frac{1}{16}$ (d) 1
28. What is $\frac{\sin \theta + 1}{\cos \theta}$ equal to?
 (a) $\frac{\sin \theta + \cos \theta - 1}{\sin \theta + \cos \theta + 1}$ (b) $\frac{\sin \theta + \cos \theta + 1}{\sin \theta + \cos \theta - 1}$
 (c) $\frac{\sin \theta - \cos \theta - 1}{\sin \theta + \cos \theta + 1}$ (d) $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1}$
29. If $y = \sec^2 \theta + \cos^2 \theta$, where $0 < \theta < \frac{\pi}{2}$, then when one of the following is correct?
 (a) $y = 0$ (b) $0 \leq y \leq 2$
 (c) $y \geq 2$ (d) None of these
30. What is the value of $(\sin 50^\circ - \sin 70^\circ + \sin 10^\circ)$?
 (a) 1 (b) $\frac{1}{\sqrt{2}}$
 (c) $\frac{\sqrt{3}}{2}$ (d) 0
31. What is $\left(\frac{\sec 18^\circ}{\sec 144^\circ} + \frac{\operatorname{cosec} 18^\circ}{\operatorname{cosec} 144^\circ}\right)$ equal to?
 (a) $\sec 18^\circ$ (b) $\operatorname{cosec} 18^\circ$
 (c) $-\sec 18^\circ$ (d) $-\operatorname{cosec} 18^\circ$
32. One of the angle of a triangle is $\frac{1}{2}$ rad and the other is 99° . What is the third angle in radian measure?
 (a) $\frac{9\pi-10}{\pi}$ (b) $\frac{90\pi-100}{7\pi}$
 (c) $\frac{90\pi-10}{\pi}$ (d) None of these
33. Which one of the following is correct?
 (a) $\sin 1^\circ > \sin 1$ (b) $\sin 1^\circ < \sin 1$
 (c) $\sin 1^\circ = \sin 1$ (d) $\sin 1^\circ = \frac{\pi}{180} \sin 1$
34. If $\cos x \neq -1$, then what is $\frac{\sin x}{1+\cos x}$ equal to?
 (a) $-\cot \frac{x}{2}$ (b) $\cot \frac{x}{2}$
 (c) $\tan \frac{x}{2}$ (d) $-\tan \frac{x}{2}$

35. What is the value of $\frac{1 + \tan 15^\circ}{1 - \tan 15^\circ}$?
- (a) 1 (b) $\frac{1}{\sqrt{2}}$
 (c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{3}$
36. If angles A, B and C are in AP, then what is $\sin A + 2\sin B + \sin C$ equal to?
- (a) $4\sin B \cos^2 \left(\frac{A-C}{2} \right)$ (b) $4\sin B \cos^2 \left(\frac{A-C}{4} \right)$
 (c) $4\sin(2B) \cos^2 \left(\frac{A-C}{2} \right)$ (d) $4\sin(2B) \cos^2 \left(\frac{A-C}{4} \right)$
37. What is $\tan \left(7\frac{1}{2} \right)^\circ$ equal to?
- (a) $\sqrt{6} + \sqrt{3} - \sqrt{2} + 2$ (b) $\sqrt{6} + \sqrt{3} + \sqrt{2} + 2$
 (c) $\sqrt{6} - \sqrt{3} + \sqrt{2} - 2$ (d) $\sqrt{6} + \sqrt{3} + \sqrt{2} - 2$
38. The angle A lies in the third quadrant and it satisfies the equation $4(\sin^2 x + \cos x) = 1$. What is the measure of the angle A?
- (a) 225° (b) 240°
 (c) 210° (d) None of these
39. What is the value of $\frac{\cos 15^\circ + \cos 45^\circ}{\cos^3 15^\circ + \cos^3 45^\circ}$?
- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
 (c) $\frac{1}{3}$ (d) None of these

ANSWER KEY

1.	d	2.	b	3.	b	4.	d	5.	c	6.	a	7.	b	8.	a	9.	a	10.	c
11.	c	12.	b	13.	c	14.	c	15.	d	16.	b	17.	b	18.	c	19.	b	20.	d
21.	a	22.	d	23.	a	24.	a	25.	c	26.	b	27.	a	28.	d	29.	c	30.	d
31.	a	32.	d	33.	b	34.	c	35.	d	36.	b	37.	c	38.	c	39.	d		

Solutions

Sol.1. (d)

$$\theta = 1200^\circ = 1080 + 120^\circ$$

$$I. (\sec\theta + \tan\theta)^{-1} = \frac{1}{(\sec\theta + \tan\theta)}$$

$$= (\sec\theta - \tan\theta) = (-2 + \sqrt{3}) = \text{negative}$$

$$II. (\csc\theta - \cot\theta) = \frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} = \text{positive}$$

Sol.2. (b)

Sol.3. (b)

$$A + B + C = \frac{\pi}{2}$$

$$A + B = \frac{\pi}{2} - C$$

$$\tan(A + B) = \tan\left(\frac{\pi}{2} - C\right)$$

$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = \cot C = \frac{1}{\tan C}$$

$$= \tan A \tan C + \tan B \tan C + \tan A \tan B = 1$$

Sol.4. (d)

$$\cos 20^\circ + \cos 60^\circ + \cos 100^\circ + \cos 140^\circ$$

$$\cos 20^\circ + \cos 140^\circ + \cos 60^\circ + \cos 100^\circ$$

$$2\cos 80^\circ \cos 60^\circ + \cos 60^\circ + \cos 100^\circ$$

$$\cos 80^\circ + \cos 60^\circ + \cos 100^\circ = \cos 60^\circ = \frac{1}{2}$$

Sol.5. (c)

$$\cot\theta = 5/12 \text{ then } \sin\theta = 5/13 \text{ and } \cos\theta = 12/13$$

$$2\sin\theta + 3\cos\theta = -10/13 - 36/13 = -46/13$$

Sol.6. (a)

$$\cot\theta = 2\cos\theta$$

$$\sin\theta = 1/2 \text{ in second quadrant } \theta = 150^\circ$$

Sol.7. (b)

$$p = \sin(989^\circ)\cos(991^\circ)$$

$$p = -\cos 1^\circ \sin 1^\circ = \text{negative number}$$

Sol.8. (a)

Sol.9. (a)

$$\tan(-1575^\circ) = -\tan(1440^\circ + 135^\circ) = -\tan 135^\circ = 1$$

Sol.10. (c)

$$\cos^2\theta = 3\sqrt{3} \cot\theta - 5$$

$$1 + \cot^2\theta = 3\sqrt{3} \cot\theta - 5$$

$$\cot^2\theta - 3\sqrt{3} \cot\theta + 6 = 0$$

$$\cot\theta = \frac{3\sqrt{3} \pm \sqrt{27 - 24}}{2} = \frac{3\sqrt{3} \pm \sqrt{3}}{2}$$

$$\cot\theta = 2\sqrt{3} \text{ or } \sqrt{3}$$

$$\cot\theta = \sqrt{3} \Rightarrow \theta = \frac{\pi}{6}$$

Sol.11. (c)

$$\tan^2\theta = 2 \tan^2\phi + 1$$

$$\frac{1}{\tan^2\theta} = \frac{1}{2 \tan^2\phi + 1}$$

apply componendo and dividendo

$$\frac{1 - \tan^2\theta}{1 + \tan^2\theta} = \frac{-2 \tan^2\phi}{2 + 2 \tan^2\phi} = \frac{-\tan^2\phi}{1 + \tan^2\phi}$$

$$\frac{1 - \tan^2\theta}{1 + \tan^2\theta} = -\sin^2\phi = \frac{\cos 2\phi - 1}{2}$$

Sol.12. (b)

Sol.13. (c)

$$\sin^2x + \sin^2y = 1$$

$$\text{put } x=0^\circ \text{ and } y=90^\circ$$

$$\text{then } \cot(x+y) = \cot 90^\circ = 0$$

Sol.14. (c)

$$\sin x + \sin y = a \quad \dots\dots\dots(i)$$

$$\cos x + \cos y = b \quad \dots\dots\dots(ii)$$

$$(i)^2 + (ii)^2$$

$$1 + 1 + 2\sin x \sin y + 2\cos x \cos y = a^2 + b^2$$

$$2 + 2\cos(x-y) = a^2 + b^2$$

$$2\cos(x-y) = a^2 + b^2 - 2$$

$$\cos(x-y) = \frac{1}{2}[a^2 + b^2 - 2]$$

Sol.15. (d)

$$\left(1 + \cos 67\frac{1}{2}^\circ\right)\left(1 + \cos 112\frac{1}{2}^\circ\right)$$

$$\left(1 + \cos\left(90^\circ - 22\frac{1}{2}^\circ\right)\right)\left(1 + \cos\left(90^\circ + 22\frac{1}{2}^\circ\right)\right)$$

$$\left(1 + \sin 22\frac{1}{2}^\circ\right)\left(1 - \sin 22\frac{1}{2}^\circ\right)$$

$$\left(1 - \sin^2 22\frac{1}{2}^\circ\right) = 1 - \frac{1}{4}(2 - \sqrt{2}) = \frac{1}{2} + \frac{\sqrt{2}}{4}$$

Sol.16. (b)

Sol.17. (b)

$$\sin A = \frac{1}{\sqrt{5}}, \text{ then } \cos A = \frac{2}{\sqrt{5}}$$

$$\cos B = \frac{3}{\sqrt{10}}, \text{ then } \sin B = \frac{1}{\sqrt{10}}$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

after substituting these values

$$\cos(A+B) = \frac{1}{\sqrt{2}}$$

$$A+B = 45^\circ$$

Sol.18. (c)

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + 2\cos 4A}}} = \sqrt{2 + \sqrt{2 + \sqrt{2(1 + \cos 4A)}}}$$

$$\sqrt{2 + \sqrt{2 + \sqrt{2.2\cos^2 2A}}} = \sqrt{2 + \sqrt{2 + 2\cos 2A}}$$

$$\sqrt{2 + \sqrt{2(1 + \cos 2A)}} = \sqrt{2 + \sqrt{2.2\cos^2 A}}$$

$$\sqrt{2 + 2\cos A} = \sqrt{2(1 + \cos A)}$$

$$2\cos \frac{A}{2}$$

Sol.19. (b)

$$\sin 1110^\circ = \sin(1080^\circ + 30^\circ) = \sin 30^\circ = \frac{1}{2}$$

Sol.20. (d)

$$\sin 2A = \frac{4}{5} \Rightarrow \frac{2 \tan A}{1 + \tan^2 A} = \frac{4}{5}$$

$$10 \tan A = 4 + 4 \tan^2 A$$

$$4 \tan^2 A - 10 \tan A + 4 = 0$$

$$2 \tan^2 A - 5 \tan A + 2 = 0$$

$$(2 \tan A - 1)(\tan A - 2) = 0$$

$$\tan A = 2$$

Sol.21. (a)

$$\frac{\cos 10^\circ - \sin 10^\circ}{\cos 10^\circ + \sin 10^\circ} = \frac{1 - \tan 10^\circ}{1 + \tan 10^\circ}$$

$$= \tan(45^\circ - 10^\circ) = \tan 35^\circ$$

Sol.22. (d)

Sol.23. (a)

Sol.24. (a)

$$\left(\sin 22\frac{1}{2}^\circ + \cos 22\frac{1}{2}^\circ\right)^4 = (1 + \sin 45^\circ)^2$$

$$= \left(1 + \frac{1}{\sqrt{2}}\right)^2 = \frac{3 + 2\sqrt{2}}{2}$$

Sol.25. (c)

$$\sin\left(292\frac{1}{2}^\circ\right) = \sin \frac{585^\circ}{2}$$

$$= \sqrt{\frac{1 - \cos 585^\circ}{2}}$$

$$= \sqrt{\frac{1 + \cos 45^\circ}{2}} = \sqrt{\frac{1 + \frac{1}{\sqrt{2}}}{2}}$$

$$= \sqrt{\frac{\sqrt{2} + 1}{2\sqrt{2}}} = \frac{1}{2}\sqrt{2 + \sqrt{2}}$$

Sol.26. (b)

Consider

$$\frac{\sin\theta + \cos\theta - \tan\theta}{\sec\theta + \csc\theta - \cot\theta}$$

$$\text{Now, put value of } \theta = \frac{3\pi}{4}, \text{ we get}$$

$$\frac{\sin \frac{3\pi}{4} + \cos \frac{3\pi}{4} - \tan \frac{3\pi}{4}}{\sec \frac{3\pi}{4} + \csc \frac{3\pi}{4} - \cot \frac{3\pi}{4}}$$

$$= \frac{\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - 1}{-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + 1} = \frac{-1}{1} = -1$$

Sol.27. (a)

Consider $15^\circ \sin 75^\circ$

$$= \left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right) = \frac{3-1}{8} = \frac{2}{8} = \frac{1}{4}$$

Sol.28. (d)

Consider

$$\frac{1 + \sin\theta}{\cos\theta} = \frac{1 + \frac{2 \tan \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}}{\frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}}$$

$$= \frac{1 + \tan \frac{\theta}{2}}{1 - \tan \frac{\theta}{2}} \cdot \frac{\cos \frac{\theta}{2} + \sin \frac{\theta}{2}}{\cos \frac{\theta}{2} - \sin \frac{\theta}{2}}$$

Multiplied and divide by

$$\frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} + 2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} - 2 \sin^2 \frac{\theta}{2}}$$

$$= \frac{\sin \theta + 1 - \cos \theta}{\sin \theta - 1 + \cos \theta}$$

Sol.29. (c)

Sum of a number and its reciprocal is always greater than 2.

Sol.30. (d)

Consider $\sin 50^\circ - \sin 70^\circ + \sin 10^\circ$

$$= 2 \cos \frac{70^\circ + 50^\circ}{2} \cdot \sin \frac{50^\circ - 70^\circ}{2} + \sin 10^\circ$$

$$= -2 \cos 60^\circ \sin 10^\circ + \sin 10^\circ = \sin 10^\circ - \sin 10^\circ = 0$$

$$\left(\because \frac{1}{2} \right)$$

Sol.31. (a)

$$\begin{aligned} \text{Consider } & \left(\frac{\sec 18^\circ}{\sec 144^\circ} + \frac{\csc 18^\circ}{\csc 144^\circ} \right) \\ &= \frac{\sec 18^\circ}{\sec(180^\circ - 36^\circ)} + \frac{\csc 18^\circ}{\csc(180^\circ - 36^\circ)} \\ &= -\frac{\sec 18^\circ}{\sec 36^\circ} + \frac{\csc 36^\circ}{\csc 36^\circ} \end{aligned}$$

($\because$ sin, cosec are +ve in 2nd quadrant and sec is -ve in 2nd quadrant)

$$\begin{aligned} &= \frac{\sin 36^\circ}{\sin 18^\circ} - \frac{\cos 36^\circ}{\cos 18^\circ} = \\ &= \frac{\sin 36^\circ \cos 18^\circ - \cos 36^\circ \sin 18^\circ}{\sin 18^\circ \cos 18^\circ} \\ &= \frac{\sin(36^\circ - 18^\circ)}{\sin 18^\circ \cos 18^\circ} = \frac{\sin 18^\circ}{\sin 18^\circ \cos 18^\circ} \\ &= \sec 18^\circ \end{aligned}$$

Sol.32. (d)

$$\text{Let } \angle A = \frac{1}{2}$$

$$\text{radian, } \angle B = 99^\circ = \frac{99^\circ \times \pi}{180^\circ} = \frac{11\pi}{20}$$

We know that, $\angle A + \angle B + \angle C = \pi$

(by angle sum property of triangle)

$$\Rightarrow \frac{1}{2} + \frac{11\pi}{20} + \angle C = \pi$$

$$\Rightarrow \angle C = \pi - \frac{11\pi}{20} - \frac{1}{2} = \frac{9\pi - 10}{20}$$

Hence, the third angle in radian is $\frac{9\pi - 10}{20}$

Sol.33. (b)

Sol.34. (c)

$$\begin{aligned} \text{Consider } & \frac{\sin x}{1 + \cos x} = \frac{2 \sin x / 2 \cos x / 2}{1 + 2 \cos^2(x/2) - 1} \\ &= \frac{2 \sin x / 2 \cos x / 2}{2 \cos^2 x / 2} = \frac{\sin x / 2}{\cos x / 2} = \tan x / 2 \end{aligned}$$

Sol.35. (d)

$$\begin{aligned} \frac{1 + \tan 15^\circ}{1 - \tan 15^\circ} &= \frac{\tan 45^\circ + \tan 15^\circ}{1 - \tan 45^\circ \tan 15^\circ} (\because = 1) \\ &= \tan(45^\circ + 15^\circ) \\ &= \tan 60^\circ = \sqrt{3} \end{aligned}$$

Sol.36. (b)

A, B, C are in AP then $A + C = 2B$ given
 $\sin A + 2 \sin B + \sin C = \sin A + \sin C + 2 \sin B$

$$2 \sin B + 2 \sin \left(\frac{A+C}{2} \right) \cos \left(\frac{A-C}{2} \right)$$

$$2 \sin B + 2 \sin B \cos \left(\frac{A-C}{2} \right)$$

$$2 \sin B \left(1 + \cos \left(\frac{A-C}{2} \right) \right)$$

$$2 \sin B \left(2 \cos^2 \left(\frac{A-C}{4} \right) \right)$$

$$4 \sin B \cos^2 \left(\frac{A-C}{4} \right)$$

Sol.37. (c)

$$\tan \left(7 \frac{1}{2} \right) = \frac{\sin \left(7 \frac{1}{2} \right)}{\cos \left(7 \frac{1}{2} \right)}$$

Multiply and divide by $2 \sin \left(7 \frac{1}{2} \right)$; we get

$$\frac{2 \sin^2 \left(7 \frac{1}{2} \right)}{2 \sin \left(7 \frac{1}{2} \right) \cos \left(7 \frac{1}{2} \right)} = \frac{2 \sin^2 \left(15 \right)}{2 \sin \left(15 \right) \cos \left(15 \right)}$$

$$\begin{aligned} &= \frac{1 - \cos 15^\circ}{\sin 15^\circ} = \frac{1 - \left(\frac{\sqrt{3} + 1}{2\sqrt{2}} \right)}{\frac{\sqrt{3} - 1}{2\sqrt{2}}} = \sqrt{2}(\sqrt{3} + 1) - (2 + \sqrt{3}) \\ &= \sqrt{6} + \sqrt{2} - 2 - \sqrt{3} = \sqrt{6} - \sqrt{3} + \sqrt{2} - 2 \end{aligned}$$

Sol.38. (c)

Given equation is $4(\sin^2 x + \cos x) = 1$
 $\Rightarrow 4 \sin^2 x + 4 \cos x = 1$

$$\Rightarrow 4 \sin^2 x + 4 \cos x - 1 = 0$$

$$\Rightarrow 4(1 - \cos^2 x) + 4 \cos x - 1 = 0$$

$$\Rightarrow 4 - 4 \cos^2 x + 4 \cos x - 1 = 0$$

$$\Rightarrow -4 \cos^2 x + 4 \cos x + 3 = 0$$

$$\Rightarrow -4 \cos^2 x - 4 \cos x + 3 = 0$$

This is the quadratic in $\cos x$.

$$\Rightarrow 4 \cos^2 x - 6 \cos x + 2 \cos x - 3 = 0$$

$$\Rightarrow (2 \cos x - 3)(2 \cos x + 1) = 0$$

$$\cos x = \frac{3}{2} \text{ and } \cos x = -\frac{1}{2}$$

$$\cos x = \frac{3}{2} \text{ is not possible therefore } \cos x = -\frac{1}{2}$$

$$\Rightarrow \cos A = -\frac{1}{2} = \cos 210^\circ$$

$$\Rightarrow A = 210^\circ$$

Sol.39. (d)

$$\text{Consider } \frac{\cos 15^\circ + \cos 15^\circ}{\cos^3 15^\circ + \cos^3 15^\circ}$$

$$\frac{\cos 15^\circ + \cos 45^\circ}{(\cos 15^\circ + \cos 45^\circ)(\cos^2 45^\circ + \cos^2 15^\circ - \cos 45^\circ \cos 15^\circ)}$$

$$\left(\because (a+b)(a^2+b^2-ab) \right)$$

$$= \frac{1}{\cos^2 45^\circ + \cos^2 15^\circ - \cos 45^\circ \cos 15^\circ}$$

$$= \frac{1}{\frac{1}{2} + (\cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ)^2 - \frac{\cos 15^\circ}{\sqrt{2}}}$$

$$= \frac{1}{\frac{1}{2} + \left(\frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} \right) - \frac{1}{\sqrt{2}} \left(\frac{\sqrt{3}+1}{2\sqrt{2}} \right)}$$

$$= \frac{1}{\frac{1}{2} + \frac{3+1+2\sqrt{3}-\sqrt{3}+1}{8} - \frac{\sqrt{3}+1}{4}}$$

$$= \frac{1}{4+4+2\sqrt{3}-2\sqrt{3}-2} = \frac{8}{6} = \frac{4}{3}$$

NDA PYQ

1. If an angle α is divided into two parts A and B such that $A-B = x$ and $\tan A : \tan B = 2:1$, then what is $\sin x$ equal to?
 (a) $3\sin\alpha$ (b) $\frac{2\sin\alpha}{3}$
 (c) $\frac{\sin\alpha}{3}$ (d) $2\sin\alpha$
[NDA (I) - 2011]
 2. If $x = \sin\theta + \cos\theta$ and $y = \sin\theta \cdot \cos\theta$, then what is the value of $x^4 - 4x^2y - 2x^2 + 4y^2 + 4y + 1$?
 (a) 0 (b) 1
 (c) 2 (d) None of these
[NDA (I) - 2011]
 3. If $(1+\tan\theta)(1+\tan\phi)=2$, then what is $(\theta+\phi)$ equal to?
 (a) 30° (b) 45°
 (c) 60° (d) 90°
[NDA (II) - 2011]
 4. Which one of the following statements is correct?
 (a) $\sin^2 30^\circ, \sin^2 45^\circ, \sin^2 60^\circ$ are in GP
 (b) $\cos^2 30^\circ, \cos^2 45^\circ, \cos^2 60^\circ$ are in GP
 (c) $\cot^2 30^\circ, \cot^2 45^\circ, \cot^2 60^\circ$ are in AP
 (d) $\tan^2 30^\circ, \tan^2 45^\circ, \tan^2 60^\circ$ are in GP
[NDA-2011(I)]
 5. If, in general, the value of $\sin A$ is known, but the value of A is not known, then how many values of $\tan(A/2)$ can be calculated?
 (a) 1 (b) 2
 (c) 3 (d) 4
[NDA-2011(I)]
 6. What is the range of $f(x) = \cos 2x - \sin 2x$?
 (a) $[2, 4]$ (b) $[-1, 1]$
 (c) $[-\sqrt{2}, \sqrt{2}]$ (d) $(-\sqrt{2}, \sqrt{2})$
[NDA (I) - 2011]
 7. If $x = y \cos\left(\frac{2\pi}{3}\right) = z \cos\left(\frac{4\pi}{3}\right)$, then what is $xy + yz + zx$ equal to?
 (a) -1 (b) 0
 (c) 1 (d) 2
[NDA (II) - 2011]
 8. What is the value of $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$?
 (a) 1 (b) 2
 (c) 3 (d) 4
[NDA (II) - 2011]
 9. If $\tan \theta = \sqrt{m}$, where m is a non-square natural number m , then $\sec 2\theta$ is:
 (a) A negative number
 (b) A transcendental number
 (c) An irrational number
 (d) A rational number
[NDA (II) - 2011]
 10. If $\sin 3A = 1$, then how many distinct value can $\sin A$ assume?
 (a) 1 (b) 2
 (c) 3 (d) 4
[NDA (II) - 2011]
 11. If $\sin A + \sin B + \sin C = 3$, then what is $\cos A + \cos B + \cos C$ equal to?
 (a) -1 (b) 0
 (c) 1 (d) 3
[NDA (II) - 2011]
 12. If $\sin\theta = \cos^2\theta$, then what is $\cos^2\theta (1+\cos^2\theta)$ equal to?
 (a) 1 (b) 0
 (c) $\cos^2\theta$ (d) $2\sin\theta$
[NDA (II) - 2011]
 13. If $\tan A - \tan B = x$ and $\cot B - \cot A = y$, then what is $\cot(A-B)$ equal to?
 (a) $\frac{1}{y} - \frac{1}{x}$ (b) $\frac{1}{x} - \frac{1}{y}$
 (c) $\frac{1}{x} + \frac{1}{y}$ (d) $-\frac{1}{x} - \frac{1}{y}$
[NDA (II) - 2011]
 14. What is the maximum value of $3\cos x + 4\sin x + 5$?
 (a) 5 (b) 7
 (c) 10 (d) 12
[NDA (II) - 2011]
 15. If $\tan A = 1/2$ and $\tan B = 1/3$, then what is the value of $4A + 4B$?
 (a) $\pi/4$ (b) $\pi/2$
 (c) π (d) 2π
[NDA (II) - 2011]
 16. What is $\frac{\sin x}{1 + \cos x} + \frac{1 + \cos x}{\sin x}$ equal to?
 (a) $2\tan x$ (b) $2\operatorname{cosec} x$
 (c) $2\cos x$ (d) $2\sin x$
[NDA (II) - 2011]
 17. What is the value of $\tan 15^\circ \tan 195^\circ$?
 (a) $7-4\sqrt{3}$ (b) $7+4\sqrt{3}$
 (c) $7+2\sqrt{3}$ (d) $7+6\sqrt{3}$
[NDA (II) - 2011]
- Directions (For Next Three):**
 Let $\sin(A+B) = 1$ and $\sin(A-B) = \frac{1}{2}$ where $AB \in \left[0, \frac{\pi}{2}\right]$
18. What is the value of A ?
 (a) $\pi/6$ (b) $\pi/3$
 (c) $\pi/4$ (d) $\pi/8$
[NDA (I) - 2012]
 19. What is the value of $\tan(A+2B) \tan(2A+B)$?
 (a) -1 (b) 0
 (c) 1 (d) 2
[NDA (I) - 2012]
 20. What is the value of $\sin^2 A - \sin^2 B$?
 (a) 0 (b) $\frac{1}{2}$
 (c) 1 (d) 2
[NDA (I) - 2012]
 21. Consider the following statements:
 I. 1° in radian measure is less than 0.02 radians
 II. 1 radian in degree measure is greater than 45° .
 Which of the above statements is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
[NDA (I) - 2012]
 22. If $\tan\theta + \sec\theta = 4$, then what is the value of $\sin\theta$?
 (a) $8/17$ (b) $8/15$

- (c) 15/17 (d) 23/12 [NDA (I) - 2012]
23. What the value of $\sin A \cos A \tan A + \cos A \sin A \cot A$?
 (a) $\sin A$ (b) $\cos A$
 (c) $\tan A$ (d) 1 [NDA (I) - 2012]
24. What is the maximum value of $\sin 3\theta \cos 2\theta + \cos 3\theta \sin 2\theta$?
 (a) 1 (b) 2
 (c) 4 (d) 10 [NDA (I) - 2012]
25. If $\cot A \cot B = 2$ then what is the value of $\cos (A+B) \sec (A-B)$?
 (a) $1/3$ (b) $2/3$
 (c) 1 (d) -1 [NDA (I) - 2012]
26. What is the value of $\sin (1920^\circ)$?
 (a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{2}}$
 (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{3}$ [NDA (I) - 2012]
27. Which one of the following is positive in the third quadrant?
 (a) $\sin \theta$ (b) $\cos \theta$
 (c) $\tan \theta$ (d) $\sec \theta$ [NDA (I) - 2012]
28. If $\operatorname{cosec} \theta - \cot \theta = \frac{1}{\sqrt{3}}$, where $\theta \neq 0$, then what is the value of $\cos \theta$?
 (a) 0 (b) $\frac{\sqrt{3}}{2}$
 (c) $\frac{1}{2}$ (d) $\frac{1}{\sqrt{2}}$ [NDA (I) - 2012]
29. What is the maximum value of $\sin^2 x$?
 (a) -1 (b) 0
 (c) 1 (d) Infinity [NDA (I) - 2012]
30. What is the value of $\frac{\sin \theta}{\operatorname{cosec} \theta} + \frac{\cos \theta}{\sec \theta}$?
 (a) 1 (b) $1/2$
 (c) $1/3$ (d) 2 [NDA (I) - 2012]
31. What is $\tan \left(\frac{\pi}{12} \right)$ equal to?
 (a) $2-\sqrt{3}$ (b) $2+\sqrt{3}$
 (c) $\sqrt{3}-\sqrt{3}$ (d) $\sqrt{3}-\sqrt{2}$ [NDA(I)-2012]
32. If $\theta=18^\circ$, then what is the value of $4 \sin^2 \theta + 2 \sin \theta$?
 (a) -1 (b) 1
 (c) 0 (d) 2 [NDA(I)-2012]
33. If ABCD is a cyclic quadrilateral then what is $\sin A + \sin B - \sin C - \sin D$ equal to?
 (a) 0 (b) 1
 (c) 2 (d) $2(\sin A + \sin B)$ [NDA(I)-2012]
34. What is the value of $\sin 420^\circ \cdot \cos 390^\circ + \cos(-300^\circ) \sin(-330^\circ)$?
 (a) 0 (b) 1
 (c) 2 (d) -1 [NDA (I) - 2012]
35. If $\sin A = \frac{2}{\sqrt{5}}$ and $\cos B = \frac{1}{\sqrt{10}}$, where A and B are acute angles, then what is the value of $A+B$?
 (a) 135° (b) 90°
 (c) 75° (d) 60° [NDA (I) - 2012]
36. If $4 \sin^2 \theta = 1$, where $0 < \theta < 2\pi$, how many values does θ take?
 (a) 1 (b) 2
 (c) 4 (d) None of these [NDA (II) - 2012]
37. What is the value of $\sin 15^\circ$?
 (a) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ (b) $\frac{\sqrt{3}+1}{2\sqrt{2}}$
 (c) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$ (d) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$ [NDA (II) - 2012]
38. The line making an angle (-120°) with x-axis is situated in the:
 (a) First quadrant (b) Second quadrant
 (c) Third quadrant (d) Fourth quadrant [NDA (II) - 2012]
39. That is the value of $\tan (-585^\circ)$?
 (a) 1 (b) -1
 (c) $-\sqrt{3}$ (d) $-\sqrt{3}$ [NDA (II) - 2012]
40. If $\sec \alpha = \frac{13}{5}$, where $270^\circ < \alpha < 360^\circ$, then what is the value of $\sin \alpha$?
 (a) $5/13$ (b) $12/13$
 (c) $-12/13$ (d) $-13/12$ [NDA (II) - 2012]
41. Consider the following statements:
 I. The value of $\cos 46^\circ - \sin 46^\circ$ is positive
 II. The value of $\cos 44^\circ - \sin 44^\circ$ is negative
 Which of the above statements is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II [NDA (II) - 2012]
42. What is the value of $\sin 18^\circ \cos 36^\circ$?
 (a) 4 (b) 2
 (c) 1 (d) $1/4$ [NDA (II) - 2012]
43. The angle subtended at the centre of a circle of radius 3cm by an arc of length 1cm is:
 (a) $\frac{30^\circ}{\pi}$ (b) $\frac{60^\circ}{\pi}$
 (c) 60° (d) none of the above [NDA-2012(2)]
44. What is the value of $\tan 105^\circ$?
 (a) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$ (b) $\frac{\sqrt{3}+1}{1-\sqrt{3}}$
 (c) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$ (d) $\frac{\sqrt{3}+2}{\sqrt{3}-1}$ [NDA (I) - 2013]

45. If $\sin\theta + 2\cos\theta = 1$, then what is $2\sin\theta - \cos\theta$ equal to?
 (a) 0 (b) 1
 (c) 2 (d) 4

[NDA (I) - 2013]

46. What is $\frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$ equal to?

(a) $\sin x \cos x$ (b) $\tan x$
 (c) $\sin x$ (d) $\cos x$

[NDA (I) - 2013]

47. The expression $\frac{\cot x + \operatorname{cosec} x - 1}{\cot x - \operatorname{cosec} x + 1}$ is equal to:

(a) $\frac{\sin x}{1 - \cos x}$ (b) $\frac{1 - \cos x}{\sin x}$
 (c) $\frac{1 + \cos x}{\sin x}$ (d) $\frac{\sin x}{1 + \cos x}$

[NDA (I) - 2013]

48. What is the value of $(\sin^4\theta - \cos^4\theta + 1) \operatorname{cosec}^2\theta$.

(a) -2 (b) 0
 (c) 1 (d) 2

[NDA (I) - 2013]

49. If $\tan A = x + 1$ and $\tan B = x - 1$, then $x^2 \tan(A-B)$ has the value.

(a) 1 (b) x
 (c) 0 (d) 2

[NDA (I) - 2013]

50. If $A + B = 90^\circ$, then what is the value of $\sqrt{\sin A \sec B - \sin A \cos B}$?

(a) $\sin A$ (b) $\cos A$
 (c) $\tan A$ (d) 0

[NDA (I) - 2013]

51. What is $\tan^4 A - \sec^4 A + \tan^2 A + \sec^2 A$ equal to?

(a) 0 (b) 1
 (c) 2 (d) -1

[NDA (I) - 2013]

52. If $\operatorname{cosec} \theta + \cot \theta = c$, then what is $\cos \theta$ equal to?

(a) $\frac{c}{c^2 - 1}$ (b) $\frac{c}{c^2 + 1}$
 (c) $\frac{c^2 - 1}{c^2 + 1}$ (d) None of these

[NDA (I) - 2013]

53. If d is the number of degrees contained in an angle, m is the number of minutes and s is the number of seconds, then the value of $(s-m)/(m-d)$ is:

(a) 1 (b) 60
 (c) $1/60$ (d) none of the above

[NDA-2013(I)]

54. Consider the following:

I. $\tan\left(\frac{\pi}{6}\right)$ II. $\tan\left(\frac{3\pi}{4}\right)$
 III. $\tan\left(\frac{5\pi}{4}\right)$ IV. $\tan\left(\frac{2\pi}{3}\right)$

Which one of the following is the correct order?

(a) $I < IV < II < III$ (b) $IV < II < I < III$
 (c) $IV < II < III < I$ (d) $I < II < III < II$

[NDA (II) - 2013]

55. What is $(1 - \sin^2 \theta)(1 + \tan^2 \theta)$ equal to?

(a) $\sin^2 \theta$ (b) $\cos^2 \theta$
 (c) $\tan^2 \theta$ (d) 1

[NDA (II) - 2013]

56. What is $\sin^2 20^\circ + \sin^2 70^\circ$ equal to?

(a) 1 (b) 0
 (c) -1 (d) $\frac{1}{2}$

[NDA (II) - 2013]

57. If $\cos x = 1/3$. Then, what is $\sin x \cdot \cot x \operatorname{cosec} x \cdot \tan x$ equal to?

(a) $2/3$ (b) $3/2$
 (c) 2 (d) 1

[NDA (II) - 2013]

58. What is $\frac{\cot 54^\circ}{\tan 36^\circ} + \frac{\tan 20^\circ}{\cot 70^\circ}$ equal to?

(a) 0 (b) 1
 (c) 2 (d) 3

[NDA (II) - 2013]

59. What is $\tan 15^\circ$ equal to?

(a) $2 - \sqrt{3}$ (b) $2 + \sqrt{3}$
 (c) $1 - \sqrt{3}$ (d) $1 + \sqrt{3}$

[NDA (II) - 2013]

60. What is the angle (in circular measure) between the hour hand and the minute hand of a clock when the time is half past 4?

(a) $\pi/3$ (b) $\pi/4$
 (c) $\pi/6$ (d) none of the above

[NDA-2013(II)]

61. What is the value of $\frac{1 + \sin A}{1 - \sin A} - \frac{1 - \sin A}{1 + \sin A}$?

(a) $\sec A - \tan A$ (b) $2 \sec A \cdot \tan A$
 (c) $4 \sec A \cdot \tan A$ (d) $4 \operatorname{cosec} A \cdot \cot A$

[NDA (I) - 2014]

62. If $A + B + C = \pi$, then what is $\cos(A+B) + \cos C$ equal to?

(a) 0 (b) $2 \cos C$
 (c) $\cos C - \sin C$ (d) $2 \sin C$

[NDA (I) - 2014]

63. What is $\cos 20^\circ + \cos 100^\circ + \cos 140^\circ$ equal to?

(a) 2 (b) 1
 (c) $1/2$ (d) 0

[NDA (I) - 2014]

64. What is $\sin^2(3\pi) + \cos^2(4\pi) + \tan^2(5\pi)$ equal to?

(a) 0 (b) 1
 (c) 2 (d) 3

[NDA (I) - 2014]

65. Consider the following statements:

I. Value of $\sin \theta$ oscillates between -1 and 1
 II. Value of $\cos \theta$ oscillates between 0 to 1
 Which of the above statements is/are correct?

(a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

[NDA (I) - 2014]

66. Consider the following statements

I. $\sin|x| + \cos|x|$ is always positive
 II. $\sin(x^2) + \cos(x^2)$ is always positive
 Which of the above statement is/are correct?

(a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

[NDA (I) - 2014]

67. What is the value of $\frac{\cot 224^\circ - \cot 134^\circ}{\cot 226^\circ + \cot 316^\circ}$?

- (a) $-\operatorname{cosec} 88^\circ$ (b) $-\operatorname{cosec} 2^\circ$
(c) $-\operatorname{cosec} 44^\circ$ (d) $-\operatorname{cosec} 45^\circ$

[NDA (I) - 2014]

68. What is the value of $\cos 36^\circ$?

- (a) $\frac{\sqrt{5}-1}{4}$ (b) $\frac{\sqrt{5}+1}{4}$
(c) $\frac{\sqrt{10+2\sqrt{5}}}{4}$ (d) $\frac{\sqrt{10-2\sqrt{5}}}{4}$

[NDA (I) - 2014]

69. Consider the following statements:

I. $\ln \left(\sin^2 67^\circ \frac{1^\circ}{2} - \sin^2 22^\circ \frac{1^\circ}{2} \right) > 1$ for all positive integers $n \geq 2$.

II. If x is any positive real number, then $nx > 1$ for all positive integers $n \geq 2$.

Which of the above statements is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (I) - 2014]

70. Consider the following statements:

I. If 3θ is an acute angle such that $\sin 3\theta = \cos 2\theta$, then the measurement of θ in radian equals to $\pi/10$.

II. One radian is the angle subtended at the centre of a circle by an arc of the same circle whose length is equal to the diameter of that circle.

Which of the above statements is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (I) - 2014]

71. The complete solution of $3 \tan^2 x = 1$ is given by:

- (a) $x = n\pi \pm \frac{\pi}{3}$ (b) $x = n\pi + \frac{\pi}{3}$ only
(c) $x = n\pi \pm \frac{\pi}{6}$ (d) $x = n\pi + \frac{\pi}{6}$ only

Where $n \in \mathbb{Z}$

[NDA-2014(1)]

Directions (For Next Three):

Read the following information carefully and answer the questions given below.

Given, $16 \sin^5 x = p \sin 5x + q \sin 3x + r \sin x$

72. What is the value of p ?

- (a) 1 (b) 2
(c) -1 (d) -2

[NDA (II) - 2014]

73. What is the value of q ?

- (a) 3 (b) 5
(c) 10 (d) -5

[NDA (II) - 2014]

74. What is the value of r ?

- (a) 5 (b) 8
(c) 10 (d) -10

[NDA (II) - 2014]

75. What is $\frac{\cos 7x - \cos 3x}{\sin 7x - 2 \sin 5x + \sin 3x}$ equal to?

- (a) $\tan x$ (b) $\cot x$
(c) $\tan 2x$ (d) $\cot 2x$

[NDA (II) - 2014]

76. If $\frac{\sin(x+y)}{\sin(x-y)} = \frac{a+b}{a-b}$, then what is $\frac{\tan x}{\tan y}$ equal to?

- (a) b/a (b) a/b
(c) ab (d) 1

[NDA (II) - 2014]

77. What is $\sqrt{1 + \sin 2\theta}$ equal to?

- (a) $\cos \theta - \sin \theta$ (b) $\cos \theta + \sin \theta$
(c) $2 \cos \theta$ (d) $\cos \theta + 2 \sin \theta$

[NDA (II) - 2014]

78. If $\cot A = 2$ and $\cot B = 3$, then what is the value of $(A+B)$?

- (a) $\pi/6$ (b) π
(c) $\pi/2$ (d) $\pi/4$

[NDA (II) - 2014]

79. What is $\sin^2 66^\circ \frac{1^\circ}{2} - \sin^2 23^\circ \frac{1^\circ}{2}$ equal to?

- (a) $\sin 47^\circ$ (b) $\cos 47^\circ$
(c) $2 \sin 47^\circ$ (d) $2 \cos 47^\circ$

[NDA (II) - 2014]

80. If $\sin A \sin(60^\circ - A) \sin(60^\circ + A) = k \sin 3A$, then what is k equal to?

- (a) $1/4$ (b) $1/2$
(c) 1 (d) 4

[NDA (II) - 2014]

81. Which one of the following is one of the solutions of the equation $\tan 2\theta \tan \theta = 1$?

- (a) $\pi/12$ (b) $\pi/6$
(c) $\pi/4$ (d) $\pi/3$

[NDA (II) - 2014]

82. The line $y = \sqrt{3}$ meets the graph $y = \tan x$, where $x \in \left(0, \frac{\pi}{2}\right)$, in

- k points. What is k equal to?
(a) one (b) two
(c) three (d) infinity

[NDA(II)-2014]

Directions (For Next Two)

Let α be the root of the equation $25 \cos^2 \theta + 5 \cos \theta - 12 = 0$,

where $\frac{\pi}{2} < \alpha < \pi$.

83. What is the $\tan \alpha$ equal to?

- (a) $-3/4$ (b) $3/4$
(c) $-4/5$ (d) $-4/5$

[NDA (I) - 2015]

84. What is $\sin 2\alpha$ equal to?

- (a) $24/25$ (b) $-24/25$
(c) $-5/12$ (d) $-21/25$

[NDA (I) - 2015]

85. $(1 - \sin A + \cos A)^2$ is equal to:

- (a) $2(1 - \cos A)(1 + \sin A)$ (b) $2(1 - \sin A)(1 + \cos A)$
(c) $2(1 - \cos A)(1 - \sin A)$ (d) None of these

[NDA (I) - 2015]

86. What is $\frac{\cos \theta}{1 - \tan \theta} + \frac{\sin \theta}{1 - \cot \theta}$ equal to?

- (a) $\sin \theta - \cos \theta$ (b) $\sin \theta + \cos \theta$
(c) $2 \sin \theta$ (d) $2 \cos \theta$

[NDA (I) - 2015]

87. Let θ be a positive angle. If the number of degrees in θ is divided by the number of radians in θ , then an irrational

number $\frac{180}{\pi}$ results. If the number of degrees in θ is

multiplied by the number of radians in θ , then an irrational number $\frac{125\pi}{9}$ results. The angle θ must be equal to:

- (a) 30° (b) 45°

(c) 50°

(d) 60°

[NDA (I) - 2015]

88. If $\sin x + \sin y = a$ and $\cos x + \cos y = b$, then $\tan^2$

$\left(\frac{x+y}{2}\right) + \tan^2\left(\frac{x-y}{2}\right)$ is equal to:

(a) $\frac{a^4 + b^4 + 4b^2}{a^2b^2 + b^4}$

(b) $\frac{a^4 - b^4 + 4b^2}{a^2b^2 + b^4}$

(c) $\frac{a^4 - b^4 + 4a^2}{a^2b^2 + a^4}$

(d) None of these

[NDA (II) - 2015]

89. If $p = \tan\left(-\frac{11\pi}{6}\right)$, $q = \tan\left(\frac{21\pi}{4}\right)$, $r = \cot\left(\frac{283\pi}{6}\right)$ then which of

the following is/are correct?

1. The value of $p \times r$ is 2

2. p, q and r are in G.P.

Select the correct answer using the code given below:

(a) Only 1

(b) Only 2

(c) Both 1 and 2

(d) Neither 1 nor 2

[NDA (II) - 2015]

90. The value of $\sin^2 5^\circ + \sin^2 10^\circ + \sin^2 15^\circ + \sin^2 20^\circ + \dots + \sin^2 90^\circ$ is:

(a) 7

(b) 8

(c) 9

(d) $19/2$

[NDA (II) - 2015]

91. On simplifying

$\frac{\sin^3 A + \sin 3A}{\sin A} + \frac{\cos^3 A - \cos 3A}{\cos A}$ we get

(a) $\sin 3A$

(b) $\cos 3A$

(c) $\sin A + \cos A$

(d) 3

[NDA (II) - 2015]

92. If $A = (\cos 12^\circ - \cos 36^\circ)(\sin 96^\circ + \sin 24^\circ)$ and $B = (\sin 60^\circ - \sin 12^\circ)(\cos 48^\circ - \cos 72^\circ)$, then what is A/B equal to?

(a) -1

(b) 0

(c) 1

(d) 2

[NDA (I) - 2016]

93. $\sin A + 2\sin 2A + \sin 3A$ is equal to which of the following?

1. $4\sin 2A \cos^2\left(\frac{A}{2}\right)$

2. $2\sin 2A \left(\sin \frac{A}{2} + \cos \frac{A}{2}\right)^2$

3. $8 \sin A \cos A \cos^2\left(\frac{A}{2}\right)$

Select the correct answer using the codes given below:

(a) Both 1 and 2

(b) Both 2 and 3

(c) Both 1 and 3

(d) 1, 2 and 3

[NDA (II) - 2016]

94. If $\sin \theta_1 + \sin \theta_2 + \sin \theta_3 + \sin \theta_4 = 4$, then what is the value of $\cos \theta_1 + \cos \theta_2 + \cos \theta_3 + \cos \theta_4$?

(a) 0

(b) 1

(c) 2

(d) 4

[NDA (II) - 2016]

95. If $x \cos \theta + y \sin \theta = z$, then what is the value of $(x \sin \theta - y \cos \theta)^2$?

(a) $x^2 + y^2 - z^2$

(b) $x^2 - y^2 - z^2$

(c) $x^2 - y^2 + z^2$

(d) $x^2 + y^2 + z^2$

[NDA (II) - 2016]

96. If $x = \sin 70^\circ$, $\sin 50^\circ$ and $y = \cos 60^\circ$, $\cos 80^\circ$, then what is xy equal to?

(a) $1/16$

(b) $1/8$

(c) $1/4$

(d) $1/2$

[NDA (II) - 2016]

97. What is the value of:

$\left(1 + \cos \frac{\pi}{8}\right)\left(1 + \cos \frac{3\pi}{8}\right)\left(1 + \cos \frac{5\pi}{8}\right)\left(1 + \cos \frac{7\pi}{8}\right)$?

(a) $\frac{1}{2}$

(b) $\frac{1}{2} + \frac{1}{2\sqrt{2}}$

(c) $\frac{1}{2} - \frac{1}{2\sqrt{2}}$

(d) $\frac{1}{8}$

[NDA (II) - 2016]

98. If $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$, then what is the value of $\sin 81^\circ$.

(a) $\frac{\sqrt{3+\sqrt{5}} + \sqrt{5-\sqrt{5}}}{4}$

(b) $\frac{\sqrt{3+\sqrt{5}} + \sqrt{5+\sqrt{5}}}{4}$

(c) $\frac{\sqrt{3-\sqrt{5}} + \sqrt{5-\sqrt{5}}}{4}$

(d) $\frac{\sqrt{3+\sqrt{5}} - \sqrt{5-\sqrt{5}}}{4}$

[NDA (II) - 2016]

99. What is $\frac{1 - \tan 2^\circ \cot 62^\circ}{\tan 152^\circ - \cot 88^\circ}$ equal to?

(a) $\sqrt{3}$

(b) $-\sqrt{3}$

(c) $\sqrt{2} - 1$

(d) $1 - \sqrt{2}$

[NDA (II) - 2016]

100. If $\sin A = 3/5$, where $450^\circ < A < 540^\circ$, then $\cos A/2$ is equal to:

(a) $\frac{1}{\sqrt{10}}$

(b) $-\sqrt{\frac{3}{10}}$

(c) $\frac{\sqrt{3}}{\sqrt{10}}$

(d) None of these

[NDA (I) - 2017]

101. The expression $\frac{\sin \alpha + \sin \beta}{\cos \alpha + \cos \beta}$ is equal to:

(a) $\tan\left(\frac{\alpha + \beta}{2}\right)$

(b) $\cot\left(\frac{\alpha + \beta}{2}\right)$

(c) $\sin\left(\frac{\alpha + \beta}{2}\right)$

(d) $\cos\left(\frac{\alpha + \beta}{2}\right)$

[NDA (I) - 2017]

102. What is the value of $\tan 18^\circ$?

(a) $\frac{\sqrt{5}-1}{\sqrt{10+2\sqrt{5}}}$

(b) $\frac{\sqrt{5}-1}{\sqrt{10+\sqrt{5}}}$

(c) $\frac{\sqrt{10+2\sqrt{5}}}{\sqrt{5}-1}$

(d) $\frac{\sqrt{10+\sqrt{5}}}{\sqrt{5}-1}$

[NDA (I) - 2017]

103. If $\tan(\alpha + \beta) = 2$ and $\tan(\alpha - \beta) = 1$, then $\tan(2\alpha)$ is equal to:

(a) -3

(b) -2

(c) -1/3

(d) 1

[NDA (I) - 2017]

104. What is $\frac{1}{\sin 10^\circ} - \frac{\sqrt{3}}{\cos 10^\circ}$ equal to?

(a) 0

(b) 1

(c) 2

(d) 4

[NDA (I) - 2017]

105. The maximum value of:

$\sin\left(x + \frac{\pi}{6}\right) + \cos\left(x + \frac{\pi}{6}\right)$ in the interval $\left(0, \frac{\pi}{2}\right)$ is attained

at:

- (a) $\pi/12$ (b) $\pi/6$
(c) $\pi/3$ (d) $\pi/2$

[NDA (I) - 2017]

106. If $k = \sin\left(\frac{\pi}{18}\right)\sin\left(\frac{5\pi}{18}\right)\sin\left(\frac{7\pi}{18}\right)$, then what is the value of K?

- (a) $1/2$ (b) $1/4$
(c) $1/8$ (d) $1/16$

[NDA (I) - 2017]

107. If $\sin\theta = 3\sin(\theta+2\alpha)$, then the value of $\tan(\theta+\alpha) + 2\tan\alpha$ is equal to:

- (a) -1 (b) 0
(c) 1 (d) 2

[NDA (I) - 2017]

108. If $\sec\theta - \operatorname{cosec}\theta = \frac{4}{3}$, then what is $(\sin\theta - \cos\theta)$ equal to?

- (a) Only -2 (b) Only $1/2$
(c) Both -2 and $1/2$ (d) Neither $1/2$ nor -2

[NDA (I) - 2017]

109. The value of $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$ is equal to:

- (a) -1 (b) 0
(c) 1 (d) 4

[NDA (II) - 2017]

110. The value of $\sqrt{3}\operatorname{cosec} 20^\circ - \sec 20^\circ$ is equal to:

- (a) 4 (b) 2
(c) 1 (d) -4

[NDA (II) - 2017]

111. Angle α is divided into two parts A and B such that $A - B = x$ and $\tan A : \tan B = p : q$. The value of $\sin x$ is equal to:

- (a) $\frac{(p+q)\sin\alpha}{p-q}$ (b) $\frac{p\sin\alpha}{p+q}$
(c) $\frac{p\sin\alpha}{p-q}$ (d) $\frac{(p-q)\sin\alpha}{p+q}$

[NDA (II) - 2017]

112. $\sqrt{1+\sin A} = -\left(\sin\frac{A}{2} + \cos\frac{A}{2}\right)$ is true if:

- (a) $\frac{3\pi}{2} < A < \frac{5\pi}{2}$ (b) $\frac{\pi}{2} < A < \frac{3\pi}{2}$
(c) $\frac{3\pi}{2} < A < \frac{7\pi}{2}$ (d) $0 < A < \frac{3\pi}{2}$

[NDA (II) - 2017]

113. The maximum value of

$\sin\left(x + \frac{\pi}{5}\right) + \cos\left(x + \frac{\pi}{5}\right)$, where $x \in \left(0, \frac{\pi}{2}\right)$, is attained

at:

- (a) $\pi/20$ (b) $\pi/15$
(c) $\pi/10$ (d) $\pi/2$

[NDA (I) - 2018]

114. If $\cos\alpha + \cos\beta + \cos\gamma = 0$, where $0 < \alpha \leq \frac{\pi}{2}$, $0 < \beta \leq \frac{\pi}{2}$, $0 < \gamma \leq \frac{\pi}{2}$, then what is the value of $\sin\alpha + \sin\beta + \sin\gamma$?

- (a) 0 (b) 3
(c) $\frac{5\sqrt{2}}{2}$ (d) $\frac{3\sqrt{2}}{2}$

[NDA (I) - 2018]

115. Suppose $\cos A$ is given. If only one value of $\cos(A/2)$ is possible, then A must be:

- (a) An odd multiple of 90°
(b) A multiple of 90°
(c) An odd multiple of 180°
(d) A multiple of 180°

[NDA (I) - 2018]

116. If $\sin\alpha + \sin\beta = 0 = \cos\alpha + \cos\beta$, where $0 < \beta < \alpha < 2\pi$, then which one of the following is correct?

- (a) $\alpha = \pi - \beta$ (b) $\alpha = \pi + \beta$
(c) $\alpha = 2\pi - \beta$ (d) $2\alpha = \pi + 2\beta$

[NDA (I) - 2018]

117. If $\frac{\sin(x+y)}{\sin(x-y)} = \frac{a+b}{a-b}$, then what is $\frac{\tan x}{\tan y}$ equal to?

- (a) $\frac{a}{b}$ (b) $\frac{b}{a}$
(c) $\frac{a+b}{a-b}$ (d) $\frac{a-b}{a+b}$

[NDA (I) - 2018]

118. What is $\sin 105^\circ + \cos 105^\circ$ equal to:

- (a) $\sin 50^\circ$ (b) $\cos 50^\circ$
(c) $1/\sqrt{2}$ (d) 0

[NDA (I) - 2018]

119. What is $\frac{\sin 5x - \sin 3x}{\cos 5x + \cos 3x}$ equal to?

- (a) $\sin x$ (b) $\cos x$
(c) $\tan x$ (d) $\cot x$

[NDA (I) - 2018]

120. If $\sin x = \frac{1}{\sqrt{5}}$, $\sin y = \frac{1}{\sqrt{10}}$, where $0 < x < \frac{\pi}{2}$, $0 < y < \frac{\pi}{2}$, then what is $(x+y)$ equal to?

- (a) π (b) $\pi/2$
(c) $\pi/4$ (d) 0

[NDA (I) - 2018]

121. If $A = \sin^2\theta + \cos^4\theta$, the for real θ , which one of the following is correct?

- (a) $1 \leq A \leq 2$ (b) $\frac{3}{4} \leq A \leq 1$
(c) $\frac{13}{16} \leq A \leq 1$ (d) $\frac{3}{4} \leq A \leq \frac{13}{16}$

[NDA (II) - 2018]

122. If $\theta = \frac{\pi}{8}$, then what is the value of $(2\cos\theta+1)^{10} (2\cos\theta-1)^{10}$

$(2\cos\theta-1)^{10} (2\cos4\theta-1)^{10}$?

- (a) 0 (b) 1
(c) 2 (d) 4

[NDA (II) - 2018]

123. What is/are the solutions of the trigonometric equation $\operatorname{cosec} x + \cot x = \sqrt{3}$, where $0 < x < 2\pi$?

- (a) $\frac{5\pi}{3}$ only (b) $\frac{\pi}{3}$ only
(c) π only (d) $\pi, \frac{\pi}{3}, \frac{5\pi}{3}$

[NDA (II) - 2018]

124. A is an angle in the fourth quadrant. If satisfies the trigonometric equation $3(3 - \tan^2 A - \cot A)^2 = 1$. Which one of the following is a value of A?

(a) 300° (b) 315°
(c) 330° (d) 345°

[NDA (II) - 2018]

125. If $\sec(\theta - \alpha)$, $\sec\theta$ and $\sec(\theta + \alpha)$ are in AP, where $\cos \alpha \neq 1$, then what is the value of $\sin^2\theta + \cos\alpha$?

(a) 0 (b) 1
(c) -1 (d) $1/2$

[NDA (II) - 2018]

126. What is $\frac{2 \tan \theta}{1 + \tan^2 \theta}$ equal to?

(a) $\cos 2\theta$ (b) $\tan \theta$
(c) $\sin 2\theta$ (d) $\operatorname{cosec} 2\theta$

[NDA (II) - 2018]

127. If $\sin\theta = -\frac{1}{2}$ and $\tan\theta = \frac{1}{\sqrt{3}}$ then in which quadrant does θ lie?

(a) First (b) Second
(c) Third (d) Fourth

[NDA (I) - 2019]

128. If $\sin\theta + \cos\theta = \sqrt{2} \cos\theta$ then what $\cos\theta - \sin\theta$ equal to

(a) $-\sqrt{2} \cos\theta$ (b) $-\sqrt{2} \sin\theta$
(c) $\sqrt{2} \sin\theta$ (d) $2 \sin\theta$

[NDA (I) - 2019]

129. In a circle of diameter 44 cm, the length of chord is 22cm. What is the length of minor arc of the chord?

(a) $\frac{484}{21} \text{ cm}$ (b) $\frac{242}{21} \text{ cm}$
(c) $\frac{121}{21} \text{ cm}$ (d) $\frac{44}{7} \text{ cm}$

[NDA (I) - 2019]

130. What is $(1 + \tan\alpha \tan\beta)^2 + (\tan\alpha - \tan\beta)^2 - \sec^2\alpha \sec^2\beta$ equal to ?

(a) 0 (b) 1
(c) 2 (d) 4

[NDA (I) - 2019]

131. If $p = \operatorname{cosec}\theta - \cot\theta$ and $q = (\operatorname{cosec}\theta + \cot\theta)^{-1}$ then which one of the following is correct ?

(a) $pq = 1$ (b) $p = q$
(c) $p + q = 1$ (d) $p + q = 0$

[NDA (I) - 2019]

132. What is the value of

$$\frac{\sin 34^\circ \cos 236^\circ - \sin 56^\circ \sin 124^\circ}{\cos 28^\circ \cos 88^\circ + \cos 178^\circ \sin 208^\circ}$$

(a) -2 (b) -1
(c) 2 (d) 1

[NDA (I) - 2019]

133. $\tan 54^\circ$ can be expressed as

(a) $\frac{\sin 9^\circ + \cos 9^\circ}{\sin 9^\circ - \cos 9^\circ}$ (b) $\frac{\sin 9^\circ - \cos 9^\circ}{\sin 9^\circ + \cos 9^\circ}$
(c) $\frac{\cos 9^\circ + \sin 9^\circ}{\cos 9^\circ - \sin 9^\circ}$ (d) $\frac{\sin 36^\circ}{\cos 36^\circ}$

[NDA (I) - 2019]

Consider the following (for next three) items:

$$\text{If } p = X \cos\theta - Y \sin\theta, q = X \sin\theta + Y \cos\theta \text{ and } p^2 + 4pq + q^2 = AX^2 + BY^2, 0 \leq \theta \leq \frac{\pi}{2}$$

134. What is the value of θ ?

(a) $\pi/2$ (b) $\pi/3$
(c) $\pi/4$ (d) $\pi/6$

[NDA (I) - 2019]

135. What is the value of A?

(a) 4 (b) 3
(c) 2 (d) 1

[NDA (I) - 2019]

136. What is the value of B?

(a) -1 (b) 0
(c) 1 (d) 2

[NDA (I) - 2019]

Consider the following (for next three) items:

It given that $\cos(\theta - \alpha) = a$, $\cos(\theta - \beta) = b$.

137. What is $\cos(\alpha - \beta)$ equal to?

(a) $ab + \sqrt{1-a^2} \sqrt{1-b^2}$ (b) $ab - \sqrt{1-a^2} \sqrt{1-b^2}$
(c) $a\sqrt{1-b^2} - b\sqrt{1-a^2}$ (d) $a\sqrt{1-b^2} + b\sqrt{1-a^2}$

[NDA (I) - 2019]

138. What is $\sin^2(\alpha - \beta) + 2ab \cos(\alpha - \beta)$ equal to

(a) $a^2 + b^2$ (b) $a^2 - b^2$
(c) $b^2 - a^2$ (d) $-(a^2 + b^2)$

[NDA (I) - 2019]

139. If $\sin\alpha + \cos\alpha = p$, then what is $\cos^2(2\alpha) =$

(a) p^2 (b) $p^2 - 1$
(c) $p^2(2 - p^2)$ (d) $p^2 + 1$

[NDA (I) - 2019]

140. If $\tan\theta = (1/2)$, $\tan\phi = (1/3)$, then what is the value of $(\theta + \phi)$?

(a) 0 (b) $\pi/6$
(c) $\pi/4$ (d) $\pi/2$

[NDA (I) - 2019]

141. If $\cos A = 3/4$, then what is the value of $\sin\left(\frac{A}{2}\right) \sin\left(\frac{3A}{2}\right)$?

(a) $5/8$ (b) $5/16$
(c) $5/24$ (d) $7/32$

[NDA (I) - 2019]

142. What is the value of $\tan 75^\circ + \cot 75^\circ$?

(a) 2 (b) 4
(c) $2\sqrt{3}$ (d) $4\sqrt{3}$

[NDA (I) - 2019]

143. What is the value of $\cos 46^\circ \cos 47^\circ \cos 48^\circ \cos 49^\circ \cos 50^\circ \dots \cos 135^\circ$

(a) -1 (b) 0
(c) 1 (d) Greater than 1

[NDA (I) - 2019]

144. If $\sin 2\theta = \cos 3\theta$, where $0 \leq \theta \leq \frac{\pi}{2}$, then what is $\sin\theta$ equal to?

(a) $\frac{\sqrt{5}+1}{4}$ (b) $\frac{\sqrt{5}-1}{4}$
(c) $\frac{\sqrt{5}+1}{16}$ (d) $\frac{\sqrt{5}-1}{16}$

[NDA (I) - 2019]

145. What is the least value of $25 \operatorname{cosec}^2 x + 36 \sec^2 x$?

(a) 1 (b) 11
(c) 120 (d) 121

[NDA-2019(1)]

146. What is the minimum value of $\frac{a^2}{\cos^2 x} + \frac{b^2}{\sin^2 x}$ where $a > 0$ and $b < 0$?

(a) $(a+b)^2$ (b) $(a-b)^2$
(c) a^2+b^2 (d) $|a^2+b^2|$

[NDA-2019(2)]

147. If $\cos \sec \theta = \frac{29}{21}$, where $0 < \theta < 90^\circ$, then what is the value of $4 \sec \theta + 4 \tan \theta$?

(a) 5 (b) 10
(c) 15 (d) 20

[NDA (II) - 2019]

148. Consider the following statements:

1. $\cos \theta + \sec \theta$ can never be equal to 1.5

2. $\tan \theta + \cot \theta$ can never be less than 2

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2019]

149. What is the length of the chord of a unit circle which subtends an angle θ at the centre?

(a) $\sin\left(\frac{\theta}{2}\right)$ (b) $\cos\left(\frac{\theta}{2}\right)$
(c) $2\sin\left(\frac{\theta}{2}\right)$ (d) $2\cos\left(\frac{\theta}{2}\right)$

[NDA (II) - 2019]

150. If $\tan A - \tan B = x$, and $\cot B - \cot A = y$, then what is the value of $\cot(A-B)$?

(a) $\frac{1}{x} + \frac{1}{y}$ (b) $\frac{1}{y} - \frac{1}{x}$
(c) $\frac{xy}{x+y}$ (d) $1 + \frac{1}{xy}$

[NDA (II) - 2019]

151. What is $\sin(\alpha + \beta) - 2\sin\alpha\cos\beta + \sin(\alpha - \beta) =$

(a) 0 (b) $2\sin\alpha$
(c) $2\sin\beta$ (d) $\sin\alpha + \sin\beta$

[NDA (II) - 2019]

152. If $2\tan A = 3\tan B = 1$, then what is $\tan(A-B)$ equal to?

(a) $1/5$ (b) $1/6$
(c) $1/7$ (d) $1/9$

[NDA (II) - 2019]

153. What is $\cos 80^\circ + \cos 40^\circ - \cos 20^\circ$ equal to?

(a) 2 (b) 1
(c) 0 (d) -19

[NDA (II) - 2019]

154. What is $\cot(A/2) - \tan(A/2)$ equal to?

(a) $\tan A$ (b) $\cot A$
(c) $2\tan A$ (d) $2\cot A$

[NDA (II) - 2019]

155. What is $\cot \theta + \operatorname{cosec} \theta$ equal to?

(a) $\tan\left(\frac{\theta}{2}\right)$ (b) $\cot\left(\frac{\theta}{2}\right)$
(c) $2\tan\left(\frac{\theta}{2}\right)$ (d) $2\cot\left(\frac{\theta}{2}\right)$

[NDA (II) - 2019]

156. What is $\tan 25^\circ \tan 15^\circ + \tan 15^\circ \tan 50^\circ + \tan 25^\circ \tan 50^\circ$ equal to?

(a) 0 (b) 1
(c) 2 (d) 4

[NDA (II) - 2019]

Direction (For next three):

Read the following information and answer the given items
Let $a \sin^2 x + b \cos^2 x = c$; $b \sin^2 y + a \cos^2 y = d$; and $p \tan x = q \tan y$.

157. What is $\tan^2 x$ equal to?

(a) $\frac{c-b}{a-c}$ (b) $\frac{a-c}{c-b}$
(c) $\frac{c-a}{c-b}$ (d) $\frac{c-b}{c-a}$

[NDA 2020]

158. What is $\frac{d-a}{b-d}$ equal to?

(a) $\sin^2 y$ (b) $\cos^2 y$
(c) $\tan^2 y$ (d) $\cot^2 y$

159. What is $\frac{p^2}{q^2}$ equal to?

(a) $\frac{(b-c)(b-d)}{(a-d)(a-c)}$ (b) $\frac{(a-d)(c-a)}{(b-c)(d-b)}$
(c) $\frac{(d-a)(c-a)}{(b-c)(d-b)}$ (d) $\frac{(b-c)(b-d)}{(c-a)(a-d)}$

[NDA 2020]

Direction (For next three):

Read the following information and answer the given items:

Let $t_n = \sin^n \theta + \cos^n \theta$

160. What is $\frac{t_3 - t_5}{t_5 - t_7}$ equal to?

(a) $\frac{t_1}{t_3}$ (b) $\frac{t_3}{t_5}$
(c) $\frac{t_5}{t_7}$ (d) $\frac{t_1}{t_7}$

[NDA 2020]

161. What is $t_1^2 - t_2$ equal to?

(a) $\cos 2\theta$ (b) $\sin 2\theta$
(c) $2 \cos \theta$ (d) $2 \sin \theta$

[NDA 2020]

162. What is the value of t_{10} where $\theta = 45^\circ$?

(a) 1 (b) $1/4$
(c) $1/16$ (d) $1/32$

[NDA 2020]

Direction (For next three):

Read the following information and answer the given items

Let $\alpha = \beta = 15^\circ$

163. What is the value of $\sin \alpha + \cos \beta$

(a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{2\sqrt{2}}$
(c) $\frac{\sqrt{3}}{2\sqrt{2}}$ (d) $\frac{\sqrt{3}}{\sqrt{2}}$

[NDA 2020]

164. What is the value of $\sin 7\alpha - \cos 7\beta$

(a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{2\sqrt{2}}$

(c) $\frac{\sqrt{3}}{2\sqrt{2}}$

(d) $\frac{\sqrt{3}}{\sqrt{2}}$

[NDA 2020]

165. What is $\sin(\alpha + 1^\circ) + \cos(\beta + 1^\circ)$ equal to?

(a) $\sqrt{3} \cos 1^\circ + \sin 1^\circ$

(b) $\sqrt{3} \cos 1^\circ - \frac{1}{2} \sin 1^\circ$

(c) $\frac{1}{\sqrt{2}} (\sqrt{3} \cos 1^\circ + \sin 1^\circ)$

(d) $\frac{1}{2} (\sqrt{3} \cos 1^\circ + \sin 1^\circ)$

[NDA 2020]

166. If $\sin x + \sin y = \cos y - \cos x$, where $0 < x < y < \frac{\pi}{2}$, then

what is $\tan\left(\frac{x-y}{2}\right)$ equal to?

(a) 0

(b) $\frac{1}{2}$

(c) 1

(d) 2

[NDA 2020]

167. If $\cot \alpha$ and $\cot \beta$ are the roots of equation $x^2 - 3x + 2 = 0$, then what is $\cot(\alpha + \beta)$ equal to?

(a) $\frac{1}{2}$

(b) $\frac{1}{3}$

(c) 2

(d) 3

[NDA 2020]

Direction (For next two):

Read the following information and answer the given items

Let $\frac{\tan 3A}{\tan A} = K$, where $\tan A \neq 0$ and $K \neq \frac{1}{3}$

168. What is $\tan^2 A$ equal to?

(a) $\frac{K+3}{3K-1}$

(b) $\frac{K-3}{3K-1}$

(c) $\frac{3K-3}{K-3}$

(d) $\frac{K+3}{3K+1}$

[NDA 2020]

169. For real value of $\tan A$, K cannot lie between

(a) $\frac{1}{3}$ and 3

(b) $\frac{1}{2}$ and 2

(c) $\frac{1}{5}$ and 5

(d) $\frac{1}{7}$ and 7

[NDA 2020]

170. If $\tan \theta = \frac{\cos 17^\circ - \sin 17^\circ}{\cos 17^\circ + \sin 17^\circ}$, then what is the value of θ ?

(a) 0°

(b) 28°

(c) 38°

(d) 52°

[NDA 2020]

171. A and B are positive acute angles such that $\cos 2B = 3\sin^2 A$ and $3\sin 2A = 2\sin 2B$. What is the value of $(A + 2B)$?

(a) $\frac{\pi}{6}$

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$

(d) $\frac{\pi}{2}$

[NDA 2020]

172. What is $\sin 3x + \cos 3x + 4\sin^3 x - 3\sin x + 3\cos x - 4\cos^3 x$ equal to?

(a) 0

(b) 1

(c) $2 \sin 2x$

(d) $4 \cos 4x$

[NDA 2020]

173. The value of ordinate of the graph of $y = 2 + \cos x$ lie in the interval?

(a) $[0, 1]$

(b) $[0, 3]$

(c) $[-1, 1]$

(d) $[1, 3]$

[NDA 2020]

174. What is the value of $8\cos 10^\circ \cos 20^\circ \cos 40^\circ$?

(a) $\tan 10^\circ$

(b) $\cot 10^\circ$

(c) $\operatorname{cosec} 10^\circ$

(d) $\sec 10^\circ$

[NDA 2020]

175. What is the value of $\cos 48^\circ - \cos 12^\circ$?

(a) $\frac{\sqrt{5}-1}{4}$

(b) $\frac{1-\sqrt{5}}{4}$

(c) $\frac{\sqrt{5}+1}{4}$

(d) $\frac{1-\sqrt{5}}{8}$

[NDA 2020]

176. If $\tan A = \frac{1}{7}$, then what is $\cos 2A$ equal to?

(a) $\frac{24}{25}$

(b) $\frac{18}{25}$

(c) $\frac{12}{25}$

(d) $\frac{6}{25}$

[NDA (I) - 2021]

177. What is the value of $\tan 31^\circ \tan 33^\circ \tan 35^\circ \dots \tan 57^\circ \tan 59^\circ$ the following

(a) -1

(b) 0

(c) 1

(d) 2

[NDA (I) - 2021]

178. What is the value of the following $(\sin 24^\circ + \cos 66^\circ)(\sin 24^\circ - \cos 66^\circ)$?

(a) -1

(b) 0

(c) 1

(d) 2

[NDA (I) - 2021]

179. What is $(1 + \cot \theta - \operatorname{cosec} \theta)(1 + \tan \theta + \sec \theta)$ equal to?

(a) 1

(b) 2

(c) 3

(d) 4

[NDA (I) - 2021]

180. What is $\frac{1 + \tan^2 \theta}{1 + \cot^2 \theta} - \left(\frac{1 - \tan \theta}{1 - \cot \theta}\right)^2$ equal to?

(a) 0

(b) 1

(c) $2 \tan \theta$

(d) $2 \cot \theta$

[NDA (I) - 2021]

181. What is the interior angle of a regular octagon of side length 2 cm?

(a) $\frac{\pi}{2}$

(b) $\frac{3\pi}{4}$

(c) $\frac{3\pi}{5}$

(d) $\frac{3\pi}{8}$

[NDA (I) - 2021]

182. If $7\sin \theta + 24\cos \theta = 25$, then what is the value of $(\sin \theta + \cos \theta)$?

(a) 1

(b) $\frac{26}{25}$

(c) $\frac{6}{5}$

(d) $\frac{31}{25}$

[NDA (I) - 2021]

183. If $3\cos \theta = 4\sin \theta$, then what is the value of $\tan(45^\circ + \theta)$?

(a) 10

(b) 7

(c) $\frac{7}{2}$

(d) $\frac{7}{4}$

[NDA (I) - 2021]

184. The number of integer values of k , for which the equation $2\sin x = 2k + 1$ has a solution, is

(a) zero

(b) one

(c) two

(d) four

[NDA (I) - 2021]

185. What is the value of the following: $\cot 2x \cdot \cot 4x - \cot 4x \cdot \cot 6x - \cot 6x \cdot \cot 2x$?

(a) -1

(b) 0

(c) 1

(d) 2

[NDA (I) - 2021]

186. If $\tan x = -\frac{3}{4}$ and x is in second quadrant, then what is the value of $\sin x \cdot \cos x$?

(a) 6/25
(c) -6/25

(b) 12/25
(d) -12/25

[NDA (I) - 2021]

187. What is the value of $\operatorname{cosec}\left(\frac{7\pi}{6}\right)\sec\left(\frac{5\pi}{3}\right)$?

(a) 4/3
(c) -4

(b) 4
(d) $-\frac{4}{\sqrt{3}}$

[NDA (I) - 2021]

188. Let $4\sin^2 x = 3$, here $0 \leq x \leq \pi$. What is $\tan 3x$ equal to?

(a) -2
(c) 0

(b) -1
(d) 1

[NDA (II) - 2021]

189. Let the equation $\sec x \cdot \operatorname{cosec} x = p$ have a solution, where p is a positive real number. What should be the smallest value of p ?

(a) $\frac{1}{2}$
(b) 1
(c) 2

(d) Minimum does not exist

[NDA (II) - 2021]

190. If $\sin\theta$ and $\cos\theta$ are the roots of the equation $ax^2 + bx + c = 0$, then which one of the following is correct?

(a) $a^2 + b^2 - 2ac = 0$

(b) $-a^2 + b^2 + 2ac = 0$

(c) $a^2 - b^2 + 2ac = 0$

(d) $a^2 + b^2 + 2ac = 0$

[NDA (II) - 2021]

191. If $\sin(A+B) = 1$ and $2\sin(A-B) = 1$, where $0 < A, B < \frac{\pi}{2}$, then what is $\tan A : \tan B$ equal to?

(a) 1 : 2

(b) 2 : 1

(c) 1 : 3

(d) 3 : 1

[NDA (II) - 2021]

192. Let $\sin x + \sin y = \cos x + \cos y$ for all $x, y \in \mathbb{R}$. What is $\tan\left(\frac{x}{2} + \frac{y}{2}\right)$ equal to?

(a) 1
(c) $\sqrt{2}$

(b) 2
(d) $2\sqrt{2}$

[NDA (II) - 2021]

193. If $\tan\theta = -\frac{5}{12}$, then what can be the value of $\sin\theta$?

(a) $\frac{5}{13}$ but cannot be $-\frac{5}{13}$

(b) $-\frac{5}{13}$ but cannot be $\frac{5}{13}$

(c) $\frac{5}{13}$ or $-\frac{5}{13}$

(d) None of the above

[NDA (I) - 2022]

194. What is the value of $\cos^4 \frac{7\pi}{8} + \cos^4 \frac{5\pi}{8}$?

(a) 3/2
(c) 3/8

(b) 3/4
(d) 3/16

[NDA (I) - 2022]

195. What is $\sin^2\left(\frac{\pi}{4} + \theta\right) - \sin^2\left(\frac{\pi}{4} - \theta\right)$ equal to?

(a) $\sin 2\theta$
(c) $2\sin\theta$

(b) $\cos 2\theta$
(d) $2\cos\theta$

[NDA (I) - 2022]

196. If $\tan\alpha + \tan\beta = 1 - \tan\alpha \cdot \tan\beta$, where $\tan\alpha \cdot \tan\beta \neq 1$, then which of the following is one of the value of $(\alpha + \beta)$?

(a) $\pi/6$
(c) $\pi/3$

(b) $\pi/4$
(d) $\pi/2$

[NDA (I) - 2022]

197. If $(1 + \tan\theta)(1 + \tan 9\theta) = 2$, then what is the value of $\tan(10\theta)$?

(a) 0
(c) 2

(b) 1
(d) infinite

[NDA (I) - 2022]

198. What is the value of $\sin 0^\circ + \sin 10^\circ + \sin 20^\circ + \sin 30^\circ + \dots + \sin 360^\circ$?

(a) -1
(c) 1

(b) 0
(d) 2

[NDA (I) - 2022]

199. What is value of $\cot^2 15^\circ + \tan^2 15^\circ$?

(a) 12
(c) $8\sqrt{3}$

(b) 14
(d) 4

[NDA (I) - 2022]

200. If $\alpha + \beta = \frac{\pi}{4}$ and $2\tan\alpha = 1$, then what is $\tan 2\beta$ equal to?

(a) 1/3
(c) 3/4

(b) 2/3
(d) 3/5

[NDA (I) - 2022]

201. If $\tan(45^\circ + \theta) = 1 + \sin 2\theta$, where $-\pi/4 < \theta < \pi/4$, then what is the value of $\cos 2\theta$?

(a) 0
(c) 1

(b) 1/2
(d) 2

[NDA (I) - 2022]

202. Let $\sin 2\theta = \cos 3\theta$ where θ is acute angle. What is the value of $1 + 4\sin\theta$? (given that $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$)

(a) $\sqrt{3}$
(c) $\sqrt{5}$

(b) 2
(d) 3

[NDA (I) - 2022]

203. What is the value of $\operatorname{cosec}\left(-\frac{73\pi}{3}\right)$?

(a) $\frac{2}{\sqrt{3}}$
(c) 2

(b) $-\frac{2}{\sqrt{3}}$
(d) -2

[NDA 2022 (II)]

204. What is the value of $\cos\left(\frac{5\pi}{17}\right) + \cos\left(\frac{7\pi}{17}\right) + 2\cos\left(\frac{11\pi}{17}\right)\cos\left(\frac{\pi}{17}\right)$?

(a) 0
(b) 1
(c) $4\cos\left(\frac{6\pi}{17}\right)\cos\left(\frac{\pi}{17}\right)$
(d) $4\cos\left(\frac{11\pi}{17}\right)\cos\left(\frac{\pi}{17}\right)$

[NDA 2022 (II)]

205. What is the value of $\tan\left(\frac{3\pi}{8}\right)$?

(a) $\sqrt{2} - 1$
(c) $1 - \sqrt{2}$

(b) $\sqrt{2} + 1$
(d) $-(\sqrt{2} + 1)$

[NDA 2022 (II)]

206. What is $\cos 36^\circ - \cos 72^\circ$ equal to?

(a) $\frac{\sqrt{5}}{2}$

(b) $-\frac{\sqrt{5}}{2}$

(c) $\frac{1}{2}$

(d) $-\frac{1}{2}$

[NDA 2022 (II)]

207. If $\sec x = \frac{25}{24}$ and x lies in the fourth quadrant, then what is the value of $\tan x + \sin x$?

(a) $-\frac{625}{168}$

(b) $-\frac{343}{600}$

(c) $\frac{625}{168}$

(d) $\frac{343}{600}$

[NDA 2022 (II)]

208. What is the value of $\tan^2 165^\circ + \cot^2 165^\circ$?

(a) 7

(b) 14

(c) $4\sqrt{3}$

(d) $8\sqrt{3}$

[NDA 2022 (II)]

209. What is the value of $\sin\left(2n\pi + \frac{5\pi}{6}\right)\sin\left(2n\pi - \frac{5\pi}{6}\right)$, where

$n \in \mathbb{Z}$?

(a) $-1/4$

(b) $-3/4$

(c) $1/4$

(d) $3/4$

[NDA 2022 (II)]

210. If $1 + 2(\sin x + \cos x)(\sin x - \cos x) = 0$ where $0 < x < 360^\circ$, then how many values does x take?

(a) only one value

(b) only two values

(c) only three values

(d) four values

[NDA 2022 (II)]

211. What is the maximum value of $3(\sin x - \cos x) + 4(\cos^3 x - \sin^3 x)$?

(a) 1

(b) $\sqrt{2}$

(c) $\sqrt{3}$

(d) 2

[NDA 2022 (II)]

212. What is the diameter of a circle inscribed in a regular polygon of 12 sides, each of length 1 cm?

(a) $1 + \sqrt{2} \text{ cm}$

(b) $2 + \sqrt{2} \text{ cm}$

(c) $2 + \sqrt{3} \text{ cm}$

(d) $3 + \sqrt{3} \text{ cm}$

[NDA 2022 (II)]

Consider the following for the next two (02) items that follow:

Given that:

$$\sin x + \cos x + \tan x + \cot x + \sec x + \csc x = 7$$

213. The given equation can be reduced to:

(a) $\sin^2 2x - 44 \sin 2x + 36 = 0$

(b) $\sin^2 2x + 44 \sin 2x - 36 = 0$

(c) $\sin^2 2x - 22 \sin 2x + 18 = 0$

(d) $\sin^2 2x + 22 \sin 2x - 18 = 0$

[NDA - 2023 (1)]

214. If $\sin 2x = a - b\sqrt{c}$, where a and b are natural numbers and c is prime number, then what is the value of $a - b + 2c$?

(a) 0

(b) 14

(c) 21

(d) 28

[NDA - 2023 (1)]

Consider the following for the next two (02) items that follow:

Let $\sin \beta$ be the GM of $\sin \alpha$ and $\cos \alpha$; $\tan \gamma$ be the AM of $\sin \alpha$ and $\cos \alpha$.

215. What is $\cos 2\beta$ equal to?

(a) $(\cos \alpha - \sin \alpha)^2$

(b) $(\cos \alpha + \sin \alpha)^2$

(c) $(\cos \alpha - \sin \alpha)^3$

(d) $\frac{(\cos \alpha - \sin \alpha)^2}{2}$

[NDA - 2023 (1)]

216. What is the value of $\sec 2\gamma$?

(a) $\frac{3 - \sin 2\alpha}{5 + 2 \sin 2\alpha}$

(b) $\frac{5 + \sin 2\alpha}{3 - \sin 2\alpha}$

(c) $\frac{3 - 2 \sin 2\alpha}{4 + \sin 2\alpha}$

(d) $\frac{3 - \sin 2\alpha}{4 + 3 \sin 2\alpha}$

[NDA - 2023 (1)]

217. If $\tan(\pi \cos \theta) = \cot(\pi \sin \theta)$, $0 < \theta < \pi/2$; then what is the value of $8 \sin^2\left(\theta + \frac{\pi}{4}\right)$?

(a) 16

(b) 2

(c) 1

(d) $1/2$

[NDA-2023 (2)]

218. If $\tan \alpha = 1/7$, $\sin \beta = \frac{1}{\sqrt{10}}$; $0 < \alpha, \beta < \pi/2$, then what is the

value of $\cos(\alpha + 2\beta)$?

(a) $-1/2$

(b) $-1/\sqrt{2}$

(c) $1/\sqrt{2}$

(d) $1/2$

[NDA-2023 (2)]

Consider the following for the next (02) items that follow

Let $p = \cos\left(\frac{\pi}{5}\right)\cos\left(\frac{2\pi}{5}\right)$ and $q = \cos\left(\frac{4\pi}{5}\right)\cos\left(\frac{8\pi}{5}\right)$

219. What is the value of $p + q$?

(a) $-1/2$

(b) $-1/4$

(c) 0

(d) $1/2$

[NDA-2023 (2)]

220. What is the value of pq ?

(a) $-1/16$

(b) $-1/4$

(c) $1/4$

(d) $1/16$

[NDA-2023 (2)]

Consider the following for the next (02) items that follow

Let $p = \frac{1}{3} - \frac{\tan 3x}{\tan x}$ and $q = 1 - 3 \tan^2 x$, $0 < x < \pi$, $x \neq \pi/2$

221. What is pq equal to?

(a) 1

(b) 2

(c) $8/3$

(d) $-8/3$

[NDA-2023 (2)]

222. For how many values of x does $1/p$ become zero?

(a) no value

(b) only one value

(c) only two values

(d) only three values

[NDA-2023 (2)]

Consider the following for the next (02) items that follow

Let $\sin x + \sin y = \sqrt{3}(\cos y - \cos x)$; $x + y = \pi/2$, $0 < x, y < \pi/2$.

223. What is a value of $\sin 3x + \sin 3y$?

(a) -1

(b) 0

(c) 1

(d) 3

[NDA-2023 (2)]

224. What is a value of $\cos^3 x + \cos^3 y$?

(a) $\frac{3\sqrt{3}}{8}$

(b) $\frac{3\sqrt{6}}{8}$

(c) $\frac{3\sqrt{6}}{4}$

(d) 1

[NDA-2023 (2)]

225. What is the value of $\sin 10^\circ \cdot \sin 50^\circ + \sin 50^\circ \cdot \sin 250^\circ + \sin 250^\circ \cdot \sin 10^\circ$ equal to?

- (a) $-\frac{1}{4}$ (b) $-\frac{3}{4}$
(c) $\frac{3\sin 10^\circ}{4}$ (d) $-\frac{3\cos 10^\circ}{4}$

[NDA-2024 (1)]

226. If $f(\theta) = \frac{1}{1 + \tan \theta}$ and $\alpha + \beta = \frac{5\pi}{4}$, then what is the value of $f(\alpha) f(\beta)$?

- (a) $-\frac{1}{2}$ (b) $\frac{1}{2}$
(c) 1 (d) 2

[NDA-2024 (1)]

227. If $\tan \alpha$ and $\tan \beta$ are the roots of the equation $x^2 - 6x + 8 = 0$, then what is the value of $\cos(2\alpha + 2\beta)$?

- (a) $\frac{13}{75}$ (b) $\frac{13}{85}$
(c) $\frac{17}{85}$ (d) $\frac{19}{85}$

[NDA-2024 (1)]

228. What is the value of $\tan 65^\circ + 2\tan 45^\circ - 2\tan 40^\circ - \tan 25^\circ$?

- (a) 0 (b) 1
(c) 2 (d) 4

[NDA-2024 (1)]

229. What is $\frac{\sqrt{3}\cos 10^\circ - \sin 10^\circ}{\sin 25^\circ \cos 25^\circ}$ equal to?

- (a) 1 (b) $\sqrt{3}$
(c) 2 (d) 4

[NDA-2024 (1)]

230. What is $(\sin 9^\circ - \cos 9^\circ)$ equal to?

- (a) $\frac{\sqrt{5}-\sqrt{5}}{2}$ (b) $\frac{\sqrt{5}-\sqrt{3}}{2}$
(c) $\frac{\sqrt{5}-\sqrt{5}}{2}$ (d) $\frac{\sqrt{5}-\sqrt{5}}{4}$

[NDA-2024 (1)]

231. What is the number of solutions of $(\sin \theta - \cos \theta)^2 = 2$ where $-\pi < \theta < \pi$?

- (a) Only one (b) Only two
(c) Four (d) No solution

[NDA-2024 (1)]

232. What is the maximum value of $a\cos x + b\sin x + c$?

- (a) $\sqrt{a^2 + b^2} + c$ (b) $\sqrt{a^2 + b^2} + c$
(c) $\sqrt{a^2 + b^2} - c$ (d) $\sqrt{a^2 + b^2}$

[NDA-2024 (2)]

233. Consider the following numbers

- I. $\tan 22.5^\circ$
II. $\cot 22.5^\circ$
III. $\tan 22.5^\circ - \cot 22.5^\circ$
How many of the above are irrational numbers?

- (a) none (b) only one
(c) only two (d) All three

[NDA-2024 (2)]

234. If $\frac{x}{\cos \theta} = \frac{y}{\cos\left(\frac{2\pi}{3} - \theta\right)} = \frac{z}{\cos\left(\frac{2\pi}{3} + \theta\right)}$ then what is $x + y + z$ equal to?

- (a) -1 (b) 0

- (c) 1 (d) 3

[NDA-2024 (2)]

235. If $p \tan(\theta - 30^\circ) = q \tan(\theta + 120^\circ)$, then what is $(p + q)/(p - q)$ equal to?

- (a) $\sin 2\theta$ (b) $\cos 2\theta$
(c) $2\sin 2\theta$ (d) $2\cos 2\theta$

[NDA-2024 (2)]

236. What is the number of solutions of the equation $\cot 2x \cdot \cot 3x = 1$ for $0 < x < \pi$?

- (a) only one (b) only two
(c) only five (d) More than five

[NDA-2024 (2)]

237. What is the general solution of $\cos^{100} x - \sin^{100} x = 1$?

- (a) $n\pi$ (b) $(2n + 1)\pi$
(c) $2n\pi$ (d) $(2n + 1)\pi/2$

where n is an integer.

[NDA-2024 (2)]

238. What is $\sin 12^\circ \sin 48^\circ$ equal to?

- (a) $\frac{\sqrt{5}-1}{4}$ (b) $\frac{\sqrt{5}+1}{4}$
(c) $\frac{\sqrt{5}-1}{8}$ (d) $\frac{\sqrt{5}+1}{8}$

[NDA-2024 (2)]

239. What is the value of $\frac{\cos 17^\circ - \sin 17^\circ}{\cos 17^\circ + \sin 17^\circ}$ equal to?

- (a) $\tan 17^\circ$ (b) $\cot 34^\circ$
(c) $\tan 62^\circ$ (d) $\cot 62^\circ$

[NDA-2024 (2)]

Consider the following for the three (03) items that follow:

Let $p = \sin 35^\circ$, $q = \sin 25^\circ$, and $r = \sin(-95^\circ)$

240. What is $(p + q + r)$ equal to?

- (a) -1 (b) 0
(c) $2\sin 5^\circ$ (d) $2\cos 5^\circ$

[NDA-2025 (1)]

241. What is $(pq + qr + rp)$ equal to?

- (a) $-3/4$ (b) 0
(c) $1/4$ (d) $3/4$

[NDA-2025 (1)]

242. What is $(p^2 + q^2 + r^2)$ equal to?

- (a) $1/2$ (b) 1
(c) $3/2$ (d) 2

[NDA-2025 (1)]

Consider the following for the two (02) items that follow:

Let $p = |\sin \alpha - \sin(\alpha - 90^\circ)|$

243. What is the minimum value of p ?

- (a) 0 (b) $1/2$
(c) $1/\sqrt{2}$ (d) 1

[NDA-2025 (1)]

244. What is the maximum value of p ?

- (a) 1 (b) $\sqrt{2}$
(c) $\sqrt{3}$ (d) 2

[NDA-2025 (1)]

Consider the following for the three (03) items that follow:

245. Let $p = \tan 2\alpha - \tan \alpha$ and $q = \cot \alpha - \cot 2\alpha$

What is p/q equal to?

- (a) $-\tan \alpha \cdot \tan 2\alpha$ (b) $-\cot \alpha \cdot \cot 2\alpha$
(c) $\tan \alpha \cdot \tan 2\alpha$ (d) $\cot \alpha \cdot \cot 2\alpha$

[NDA-2025 (1)]

246. What is $(p + q)$ equal to?

- (a) $\sec 4\alpha$ (b) $\operatorname{cosec} 4\alpha$
(c) $2\sec 4\alpha$ (d) $2\operatorname{cosec} 4\alpha$

[NDA-2025 (1)]

247. What is $\tan^2 \alpha$ equal to?

- (a) $\frac{pq}{(p+q)}$ (b) $\frac{(p+2q)}{p}$
(c) $\frac{p}{(p+2q)}$ (d) $\frac{p}{(2p+q)}$

[NDA-2025 (1)]

Consider the following for the two (02) items that follow:

Let $2\sin \alpha + \cos \alpha = 2$, where $0 < \alpha < 90^\circ$

248. What is $\tan \alpha$ equal to?

- (a) $\frac{1}{2}$ (b) 1
(c) $\frac{3}{4}$ (d) 2

[NDA-2025 (1)]

249. What is $2\sin 2\alpha + \cos 2\alpha$ equal to?

- (a) $\frac{11}{10}$ (b) $\frac{11}{5}$
(c) $\frac{12}{5}$ (d) $\frac{13}{5}$

[NDA-2025 (1)]

Direction for next two:

Let $\sin A + \sin B = p$ and $\cos A + \cos B = q$

250. What is $\frac{p}{q}$ equal to?

- (a) $\tan\left(\frac{A-B}{2}\right)$ (b) $\cot\left(\frac{A-B}{2}\right)$
(c) $\tan\left(\frac{A+B}{2}\right)$ (d) $\cot\left(\frac{A+B}{2}\right)$

[NDA-2025 (2)]

251. What is $\frac{p^2 - q^2}{p^2 + q^2}$ equal to?

- (a) $\cos(A+B)$ (b) $\cos(A-B)$
(c) $\cos\left(\frac{\pi}{2} - A - B\right)$ (d) $\cos(\pi - A - B)$

[NDA-2025 (2)]

Direction for next two:

Let $p = \operatorname{cosec} 20^\circ$ and $q = \operatorname{cosec} 70^\circ$

252. What is $\left(\frac{\sqrt{3}p - q}{4}\right)$ equal to?

- (a) -1 (b) 0
(c) 1 (d) 2

[NDA-2025 (2)]

253. What is $\frac{p^2 + q^2}{p^2 q^2}$ equal to?

- (a) $\frac{1}{2}$ (b) 1
(c) $\frac{3}{2}$ (d) 2

[NDA-2025 (2)]

Direction for next two:

Let $\cos(2x+3y) = \frac{1}{2}$ and $\cos(3x+2y) = \frac{\sqrt{3}}{2}$

Where $-\pi < (2x+3y) < \pi, -\pi < (3x+2y) < \pi$

254. How many values does $(x+y)$ have?

- (a) Two (b) Three
(c) Four (d) More than four

[NDA-2025 (2)]

255. How many values does $(y-x)$ have?

- (a) Two (b) Three
(c) Four (d) More than four

[NDA-2025 (2)]

256. How many values of θ , where $-\pi < \theta < \pi$, satisfy both the equations $\cot \theta = -\sqrt{3}$ and $\operatorname{cosec} \theta = -2$ simultaneously?

- (a) 4 (b) 2
(c) 1 (d) None

[NDA-2025 (2)]

257. If $x + \frac{1}{x} = 2 \cos \theta$, then what is $x^3 + \frac{1}{x^3}$ equal to?

- (a) $\cos^3 \theta$ (b) $\cos 3\theta$
(c) $2 \cos 3\theta$ (d) $3 \cos 3\theta$

[NDA-2025 (2)]

258. If $0 \leq \theta \leq \frac{\pi}{2}$, then what is the number of values of x satisfying the equation $\tan x + \sec x = 2 \cos x$?

- (a) 0 (b) 1
(c) 2 (d) 3

[NDA-2025 (2)]

ANSWER KEY

1.	c	2.	a	3.	b	4.	d	5.	b	6.	c	7.	b	8.	d	9.	a	10.	b
11.	b	12.	a	13.	c	14.	c	15.	c	16.	b	17.	a	18.	b	19.	c	20.	b
21.	c	22.	c	23.	d	24.	a	25.	a	26.	c	27.	c	28.	c	29.	c	30.	a
31.	a	32.	b	33.	a	34.	b	35.	a	36.	c	37.	a	38.	c	39.	b	40.	c
41.	d	42.	d	43.	b	44.	b	45.	c	46.	d	47.	c	48.	d	49.	d	50.	b
51.	a	52.	c	53.	c	54.	b	55.	d	56.	a	57.	d	58.	c	59.	a	60.	b
61.	c	62.	a	63.	d	64.	b	65.	a	66.	d	67.	b	68.	b	69.	a	70.	a
71.	c	72.	a	73.	d	74.	c	75.	b	76.	b	77.	b	78.	d	79.	b	80.	a
81.	b	82.	a	83.	a	84.	b	85.	b	86.	b	87.	c	88.	b	89.	b	90.	d
91.	d	92.	c	93.	c	94.	a	95.	a	96.	a	97.	d	98.	a	99.	b	100.	d
101.	a	102.	a	103.	a	104.	d	105.	a	106.	c	107.	b	108.	b	109.	d	110.	a
111.	d	112.	c	113.	a	114.	b	115.	c	116.	b	117.	a	118.	c	119.	c	120.	c
121.	b	122.	b	123.	b	124.	a	125.	a	126.	c	127.	c	128.	c	129.	a	130.	a
131.	b	132.	a	133.	c	134.	c	135.	b	136.	a	137.	a	138.	a	139.	c	140.	c
141.	b	142.	b	143.	b	144.	b	145.	d	146.	a	147.	b	148.	a	149.	c	150.	a
151.	a	152.	c	153.	c	154.	d	155.	b	156.	b	157.	a	158.	c	159.	b	160.	a
161.	b	162.	c	163.	d	164.	d	165.	c	166.	c	167.	b	168.	b	169.	a	170.	b
171.	d	172.	a	173.	d	174.	b	175.	b	176.	a	177.	c	178.	b	179.	b	180.	a
181.	b	182.	d	183.	b	184.	c	185.	c	186.	d	187.	c	188.	c	189.	c	190.	c
191.	d	192.	a	193.	c	194.	b	195.	a	196.	b	197.	b	198.	b	199.	b	200.	c
201.	c	202.	c	203.	b	204.	a	205.	b	206.	c	207.	b	208.	b	209.	a	210.	d
211.	b	212.	c	213.	a	214.	d	215.	a	216.	b	217.	c	218.	c	219.	c	220.	a
221.	d	222.	c	223.	b	224.	b	225.	b	226.	b	227.	b	228.	c	229.	d	230.	a
231.	b	232.	b	233.	c	234.	b	235.	d	236.	c	237.	a	238.	c	239.	d	240.	b
241.	a	242.	c	243.	a	244.	b	245.	c	246.	d	247.	c	248.	c	249.	b	250.	c
251.	d	252.	c	253.	b	254.	c	255.	c	256.	c	257.	c	258.	b				

Solutions

Sol.1. (c)

Since, angle α is divided into two parts A and B.

$$\therefore \alpha = A + B$$

and $x = A - B$ (given)

on solving (1) and (2) we get,

$$A = \frac{\alpha + x}{2}, B = \frac{\alpha - x}{2}$$

$$\text{Now consider } \frac{\tan A}{\tan B} = \frac{2}{1}$$

$$\Rightarrow \frac{\tan\left(\frac{\alpha + x}{2}\right)}{\tan\left(\frac{\alpha - x}{2}\right)} = \frac{2}{1}$$

$$\Rightarrow \frac{\sin\left(\frac{\alpha + x}{2}\right)\cos\left(\frac{\alpha - x}{2}\right)}{\tan\left(\frac{\alpha + x}{2}\right)\sin\left(\frac{\alpha - x}{2}\right)} = \frac{2}{1}$$

Multiply and divide by 2,

$$\Rightarrow \frac{2\sin\left(\frac{\alpha + x}{2}\right)\cos\left(\frac{\alpha - x}{2}\right)}{2\cos\left(\frac{\alpha + x}{2}\right)\sin\left(\frac{\alpha - x}{2}\right)} = 2$$

$$\Rightarrow \frac{\sin \alpha + \sin x}{\sin \alpha - \sin x} = 2$$

$$\Rightarrow \sin \alpha + \sin x = 2 \sin \alpha - 2 \sin x$$

$$\Rightarrow 3 \sin x = \sin \alpha$$

$$\Rightarrow \sin x = \frac{\sin \alpha}{3}$$

Sol.2. (a)

$$\text{Let } x = \sin \theta + \cos \theta$$

$$\text{and } y = \sin \theta \cdot \cos \theta$$

$$\text{Now, } x^4 - 4x^2y - 2x^2 + 4y^2 + 4y + 1$$

$$= (\sin \theta + \cos \theta)^4 - 4(\sin \theta + \cos \theta)^2 y -$$

$$2(\sin \theta + \cos \theta)^2 + 4y^2 + 4y + 1$$

$$= (\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta)^2 -$$

$$4(\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta) y$$

$$- 2(\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta) + 4y^2 + 4y + 1$$

$$= 1 + 4y^2 + 4y - 4y - 8y^2 - 2 - 4y + 4y^2 + 4y + 1 = 0$$

$$3 \sin^3 A - 4 \sin^3 A = 1$$

$$4 \sin^3 A - 3 \sin A + 1 = 0$$

$$2\left(\sin A - \frac{1}{2}\right)\left(\sin A - \frac{1}{2}\right)(\sin A + 1) = 0$$

two distinct values 30° and 270° .

Sol.11. (b)

$$\text{Let } \sin A + \sin B + \sin C = 3$$

$$\Rightarrow \sin A = \sin B = \sin C = 1$$

$$(\because \text{max value of } \sin \text{ is } 1)$$

$$\therefore \cos A = \sqrt{1 - \sin^2 A} = \sqrt{1 - 1} = 0$$

$$\text{Similarly, } \cos B = 0 = C$$

$$\text{Hence, } \cos A + \cos B + \cos C = 0 + 0 + 0 = 0$$

Sol.12. (a)

$$\text{Let } \sin \theta = \cos^2 \theta$$

$$\Rightarrow \sin^2 \theta = \cos^4 \theta$$

By solving quadratic equation we will get two values of $\tan(A/2)$

Sol.6. (c)

$$-\sqrt{a^2 + b^2} \leq a \sin \theta + b \cos \theta \leq \sqrt{a^2 + b^2}$$

$$-\sqrt{2} \leq \cos 2\theta - \sin 2\theta \leq \sqrt{2}$$

Sol.7. (b)

$$\text{Let } x = y \cos\left(\frac{2\pi}{3}\right) = z \cos\left(\frac{4\pi}{3}\right)$$

$$\Rightarrow x = y \cos\left(\pi - \frac{2\pi}{3}\right) = -y \cos \frac{\pi}{3} = \frac{-y}{2} \quad \dots(1)$$

And

$$\Rightarrow x = z \cos\left(\pi - \frac{\pi}{3}\right) = -z \cos \frac{\pi}{3} = \frac{-z}{2} \quad \dots(2)$$

from (1) and (2)

$$\frac{-y}{2} = \frac{-z}{2} \Rightarrow y = z$$

Thus,

$$xy + yz + zx = zx + z^2 + xz = 2xz + z^2$$

$$= -y, (y) + y^2 = -y^2 + y^2 = 0$$

Sol.8. (d)

Given expression is

$$\tan 9^\circ - \tan 27^\circ - \tan(90^\circ - 27^\circ) + \tan(90^\circ - 9^\circ)$$

$$\tan 9^\circ - \tan 27^\circ - \cot 27^\circ + \cot 9^\circ$$

$$= (\tan 9^\circ + \cot 9^\circ) - (\tan 27^\circ + \cot 27^\circ)$$

$$= \frac{1}{\sin 9^\circ \sin 9^\circ} - \frac{1}{\sin 27^\circ \sin 27^\circ}$$

$$= \frac{2}{\sin 18^\circ} - \frac{2}{\sin 54^\circ}$$

$$= \frac{2}{\sin 18^\circ} - \frac{2}{\sin(90^\circ - 36^\circ)}$$

$$= \frac{2}{\sin 18^\circ} - \frac{2}{\sin 36^\circ}$$

$$= 2\left[\frac{4}{\sqrt{5}-1} - \frac{4}{\sqrt{5}+1}\right] = 8 \times \frac{2}{4} = 4$$

Sol.9. (a)

$$\sec 2\theta = \frac{1}{\cos 2\theta} = \frac{1 + \tan^2 \theta}{1 - \tan^2 \theta} = \frac{1 + m}{1 - m}$$

Sol.10. (b)

$$\sin 3A = 1 \text{ it is clear that one angle is } 30^\circ$$

$$3 \sin A - 4 \sin^3 A = 1$$

$$4 \sin^3 A - 3 \sin A + 1 = 0$$

$$2\left(\sin A - \frac{1}{2}\right)\left(\sin A - \frac{1}{2}\right)(\sin A + 1) = 0$$

two distinct values 30° and 270° .

Sol.11. (b)

$$\text{Let } \sin A + \sin B + \sin C = 3$$

$$\Rightarrow \sin A = \sin B = \sin C = 1$$

$$(\because \text{max value of } \sin \text{ is } 1)$$

$$\therefore \cos A = \sqrt{1 - \sin^2 A} = \sqrt{1 - 1} = 0$$

$$\text{Similarly, } \cos B = 0 = C$$

$$\text{Hence, } \cos A + \cos B + \cos C = 0 + 0 + 0 = 0$$

Sol.12. (a)

$$\text{Let } \sin \theta = \cos^2 \theta$$

$$\Rightarrow \sin^2 \theta = \cos^4 \theta$$

Consider

$$\cos^2 \theta (1 + \cos^2 \theta) = \cos^2 \theta + \cos^4 \theta$$

$$= \cos^2 \theta + \sin^2 \theta \text{ (using 1)}$$

$$= 1$$

Sol.13. (c)

$$\text{Let } \tan A - \tan B = x$$

$$\text{and } \cot B - \cot A = y$$

$$\Rightarrow \frac{1}{\tan B} - \frac{1}{\tan A} = y$$

$$\Rightarrow \frac{\tan A - \tan B}{\tan A \tan B} = y \Rightarrow \frac{x}{\tan A \tan B} = y$$

$$\text{Consider } \cot(A - B) = \left(\frac{1}{\tan(A - B)}\right)$$

$$= \frac{1 + \tan A \tan B}{\tan A - \tan B} = \frac{1 + \frac{x}{y}}{x} = \frac{y + x}{xy} = \frac{1}{x} + \frac{1}{y}$$

Sol.14. (c)

Maximum value of

$$3 \cos x + 4 \sin x + 5 =$$

$$\sqrt{(4)^2 + (3)^2} + 5 = \sqrt{16 + 9} + 5$$

$$\sqrt{25} + 5 = 5 + 5 = 10$$

Sol.15. (c)

$$\text{Let } \tan A = \frac{1}{2}, \tan B = \frac{1}{3}$$

We know,

$$\tan(A + B) = 1$$

$$\Rightarrow A + B = \tan^{-1}(1) = \frac{\pi}{4}$$

Multiply by 4 on both side,

$$4(A + B) = \frac{\pi}{4} \times 4 \Rightarrow 4A + 4B = \pi$$

Sol.16. (b)

$$\frac{\sin x}{1 + \cos x} + \frac{1 + \cos x}{\sin x} = \frac{\sin^2 x + 1 + \cos^2 x + 2 \cos x}{(1 + \cos x)(\sin x)}$$

$$\frac{2 + 2 \cos x}{(1 + \cos x)(\sin x)} = \frac{2(1 + \cos x)}{(1 + \cos x)(\sin x)}$$

$$= \frac{2}{\sin x} = 2 \csc x$$

Sol.17. (a)

$$\text{Consider } \tan 15^\circ \tan 195^\circ$$

$$= \tan 15^\circ \tan(180^\circ + 15^\circ)$$

$$= \tan 15^\circ \tan 15^\circ$$

$$(\because \tan(\theta + \theta) = \tan \theta)$$

$$= (\tan 15^\circ)^2$$

$$= (2 - \sqrt{3})^2 \quad (\because \tan 15^\circ = 2 - \sqrt{3})$$

Sol.18. (b)

$$A + B = 90^\circ$$

$$A - B = 30^\circ$$

$$A = 60^\circ \text{ and } B = 30^\circ$$

Sol.19. (c)

by substituting the values of A and B
 $\tan 120^\circ \tan 150^\circ = 1$

Sol.20. (b)

$$\sin^2 A - \sin^2 B = \sin^2 60^\circ - \sin^2 30^\circ$$

$$\sin^2 A - \sin^2 B = \sin^2 60^\circ - \sin^2 30^\circ = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$$

Sol.21. (c)

(1) $1^\circ = \left(\frac{\pi}{180}\right)$ radian = 0.01746 radian which is less than 0.02 radian = 0.01746 radian which is less than 0.02 radian.

(2) $1 \text{ radian} = \left(\frac{180}{\pi}\right)^\circ = 57^\circ 16' 22''$ approx. which is greater than 45°

Hence, both statements are true

Sol.22. (c)

$$\tan \theta + \sec \theta = 4$$

$$\Rightarrow \frac{\sin \theta}{\cos \theta} + \frac{1}{\cos \theta} = 4$$

$$\Rightarrow 1 + \sin \theta = 4 \cos \theta$$

Squaring on both side,

$$(1 + \sin \theta)^2 = 16 \cos^2 \theta = 16(1 - \sin^2 \theta)$$

$$(1 + \sin \theta)^2 = 16(1 - \sin^2 \theta)(1 + \sin \theta)$$

$$1 + \sin \theta = 16 - 16 \sin \theta$$

$$17 \sin \theta = 16 - 1$$

$$\sin \theta = \frac{15}{17}$$

Sol.23. (d)

Given expression

$$= \sin A \cos A \cdot \frac{\sin A}{\cos A} + \cos A \sin A \cdot \frac{\cos A}{\sin A}$$

$$\sin^2 A + \cos^2 A = 1$$

Sol.24. (a)

$$\text{Consider } \sin 3\theta \cos 2\theta + \cos 3\theta \sin 2\theta$$

$$= \sin(3\theta + 2\theta) = \sin(5\theta)$$

We know, $-1 < \sin \theta < 1$

Hence, maximum value of given expression is 1.

Sol.25. (a)

$$\text{Consider } \cos(A+B) \cdot \sec(A-B)$$

$$= \frac{\cos(A+B)}{\cos(A-B)} = \frac{\cos A \cos B - \sin A \sin B}{\cos A \cos B + \sin A \sin B}$$

Divide Nr and Dr. by $\sin A \sin B$

$$\frac{\cot A \cot B - 1}{\cot A \cot B + 1} = \frac{2-1}{2+1} = \frac{1}{3}$$

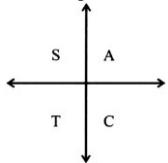
Sol.26. (c)

$$\sin 1920^\circ = \sin(1800^\circ + 120^\circ) = \sin 120^\circ$$

$$= \sin(90^\circ + 30^\circ) = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

Sol.27. (c)

Tan θ is positive in third quadrant



Sol.28. (c)

$$\text{Consider, } \operatorname{cosec} \theta - \cot \theta = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta} = \frac{1}{\sqrt{3}} \Rightarrow \frac{1 - \cos \theta}{\sin \theta} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{1 - (1 - 2 \sin^2 \frac{\theta}{2})}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} = \frac{1}{\sqrt{3}} \Rightarrow \frac{2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} = \frac{1}{\sqrt{3}} \Rightarrow \tan \frac{\theta}{2} = \tan 30^\circ$$

$$\Rightarrow \theta = 60^\circ$$

$$\therefore \cos \theta = \cos 60^\circ = \frac{1}{2}$$

Sol.29.

Maximum value of $\sin^2 x$ is 1

Sol.30. (a)

$$\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sec \theta} = \frac{\sin \theta}{1} + \frac{\cos \theta}{1} = \sin^2 \theta + \cos^2 \theta = 1$$

Sol.31. (a)

$$\tan 15^\circ = 2 - \sqrt{3}$$

Sol.32. (b)

$$4 \sin^2 18^\circ + 2 \sin 18^\circ$$

$$= 4 \left(\frac{\sqrt{5}-1}{4} \right)^2 + 2 \left(\frac{\sqrt{5}-1}{4} \right) = 1$$

Sol.33. (a)

For cyclic quadrilateral

$$A + C = 180^\circ \text{ and } B + D = 180^\circ$$

$$\sin A + \sin B + \sin C + \sin D$$

$$= \sin A + \sin B - \sin(180^\circ - A) - \sin(180^\circ - B)$$

$$= \sin A + \sin B - \sin A - \sin B = 0$$

Sol.34. (b)

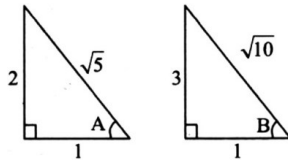
$$\sin 420^\circ \cdot \cos 390^\circ + \cos(-300^\circ) \sin(-330^\circ)$$

$$\sin(360^\circ + 60^\circ) \cdot \cos(360^\circ + 30^\circ) - \cos(360^\circ - 60^\circ) \sin(360^\circ - 30^\circ)$$

$$\sin(60^\circ) \cdot \cos(30^\circ) + \cos(60^\circ) \sin(30^\circ) = 1$$

Sol.35. (a)

$$\text{Given, } \sin A = \frac{2}{\sqrt{5}} \text{ and } \cos B = \frac{2}{\sqrt{10}}$$



$$\therefore \cos A = \frac{1}{\sqrt{5}} \quad \sin B = \frac{2}{\sqrt{10}}$$

$$\therefore \sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$= \left(\frac{2}{\sqrt{5}} \right) \left(\frac{1}{\sqrt{10}} \right) + \left(\frac{1}{\sqrt{5}} \right) \left(\frac{2}{\sqrt{10}} \right)$$

$$= \frac{2}{\sqrt{50}} + \frac{2}{\sqrt{50}} = \frac{4}{\sqrt{50}} = \frac{4}{5\sqrt{2}} = \frac{2}{5\sqrt{2}}$$

$$\sin(A+B) = \frac{1}{\sqrt{2}} \Rightarrow \sin(90^\circ + 45^\circ) = \frac{1}{\sqrt{2}}$$

$$(\therefore 90^\circ + 45^\circ) = \cos 45^\circ$$

$$\therefore A = 90^\circ \text{ and } B = 45^\circ$$

$$\therefore A + B = 90^\circ + 45^\circ = 135^\circ$$

$$= 4 + 3 - 4\sqrt{2} = 7 - 4\sqrt{2}$$

Sol.36. (c)

$$4 \sin^2 \theta = 1$$

$$\Rightarrow \sin^2 \theta = \frac{1}{4} \Rightarrow \sin \theta = \pm \frac{1}{2} \Rightarrow \theta = \pm \frac{\pi}{6}$$

Hence θ take 4 values. (one value for each quadrant)

Sol.37. (a)

$$\sin 15^\circ = \sin[45^\circ - 30^\circ]$$

$$= \sin 45^\circ \cdot \cos 30^\circ - \cos 45^\circ \sin 30^\circ$$

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2}$$

$$= \frac{\sqrt{3}-1}{2\sqrt{2}} = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

Sol.38. (c)

-120 is lie in third quadrant.

Sol.39. (b)

$$\tan(-585^\circ) = -\tan 585^\circ$$

$$= -\tan[540^\circ + 45^\circ] = -\tan\left(3\pi + \frac{\pi}{4}\right)$$

$$= \frac{\tan 3\pi + \tan \frac{\pi}{4}}{1 - \tan 3\pi \tan \frac{\pi}{4}} = \frac{0+1}{1-0 \times 1} = -1$$

Sol.40. (c)

We have $\sec \alpha = 13/5$

$$\text{Since } \frac{3\pi}{2} < \alpha < 2\pi$$

$$\therefore \sin \alpha < 0$$

$$\text{Now, } \sin \alpha = -\sqrt{1 - \frac{1}{\sec^2 \alpha}}$$

$$= -\sqrt{1 - \frac{25}{169}} = -\sqrt{\frac{144}{169}} = -\frac{12}{13}$$

Sol.41. (d)

Before 45° sine is smaller than cosine and after 45° sine is greater than cosine.

Sol.42. (d)

$$\text{We know that } \sin 18^\circ = \frac{\sqrt{5}-1}{4}$$

$$\text{Now, } \cos 36^\circ = 1 - 2 \sin^2 18^\circ$$

$$= 1 - \frac{2}{16} (\sqrt{5}-1)^2 = 1 - \frac{(3-\sqrt{5})}{4}$$

$$\text{Now, } \cos 36^\circ = 1 - 2 \sin^2 18^\circ$$

$$= 1 - \frac{2}{16} (\sqrt{5}-1)^2 = 1 - \frac{(3-\sqrt{5})}{4}$$

$$\cos 36^\circ = \frac{\sqrt{5}+1}{4}$$

$$\text{Now, } \sin 18^\circ \cos 36^\circ = \frac{(\sqrt{5}-1)^2 - (1)^2}{16} = \frac{4}{16} = \frac{1}{4}$$

Sol.43. (a)

$$\text{arc} = \frac{\pi \theta}{180} \Rightarrow 1 = \frac{3\pi \theta}{180} \Rightarrow \theta = \frac{60}{\pi}$$

Sol.44. (b)

$$\tan 105^\circ = -\tan(90^\circ + 15^\circ) = -\tan 15^\circ$$

Sol.45. (c)

$$\text{Let } \sin \theta + 2 \cos \theta = 1 \dots (i)$$

$$\text{Consider } 2 \sin \theta - \cos \theta = \alpha \text{ (let)} \dots (ii)$$

Squaring and adding

$$\sin^2 \theta + 4 \cos^2 \theta + 4 \sin \theta \cos \theta +$$

$$4 \sin^2 \theta + \cos^2 \theta - 4 \sin \theta \cos \theta = 1 + \alpha^2$$

$$\Rightarrow (\sin^2 \theta + \cos^2 \theta) + 4(\cos^2 \theta + \sin^2 \theta) = 1 + \alpha^2$$

$$\Rightarrow 1 + 4 = 1 + \alpha^2 \Rightarrow \alpha^2 = 4 \Rightarrow \alpha = 2$$

Sol.46. (d)

$$1 - \tan^2 \frac{x}{2} + \frac{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}{2}$$

$$1 + \tan^2 \frac{x}{2} + \frac{\cot^2 \frac{x}{2} + \cot^2 \frac{x}{2}}{2}$$

$$= \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2} = \cos x$$

$$(\therefore \quad = \cos^2 x - \sin^2 x)$$

Sol.47. (c)

$$\frac{\cot x + \operatorname{cosec} x - (\cos^2 x - \cot^2 x)}{(\cos^2 x - \cot^2 x)} \\ = \frac{\cot x + \operatorname{cosec} x - [(\cos^2 x - \cot^2 x)(\cos^2 x + \cot^2 x)]}{\cot x - \cos^2 x + 1}$$

$$= \frac{(\cot x + \operatorname{cosec} x)(1 + \cot x - \cos^2 x)}{(\cot x - \cos^2 x + 1)}$$

$$= \cot x + \cos^2 x \\ = \frac{\cos x}{\sin x} + \frac{1}{\sin x} = \frac{1 + \cos x}{\sin x}$$

Sol.48. (d)

$$\text{Consider } (\sin^4 \theta - \cos^4 + 1) \cos^2 \theta$$

$$= [(\sin^2 \theta)^2 - (\cos^2 \theta)^2 + 1] \cos^2 \theta$$

$$= [(\sin^2 \theta - \cos^2 \theta)(\sin^2 \theta + \cos^2 \theta) + 1] \cos^2 \theta$$

$$= [1 - \cos^2 \theta + \sin^2 \theta] \cos^2 \theta$$

$$= 2 \sin^2 \theta \times \frac{1}{\sin^2 \theta} = 2$$

Sol.49. (d)

$$x^2 \tan(A - B)$$

$$= x^2 \left(\frac{\tan A - \tan B}{1 + \tan A \tan B} \right) = x^2 \left(\frac{(x+1) - (x-1)}{1 + (x^2 - 1)} \right)$$

$$= x^2 \left(\frac{2}{x^2} \right) = 2$$

Sol.50. (b)

$$\text{Let } A+B=90^\circ$$

$$\text{Consider } \sqrt{\sin A \sec B - \sin A \cos B}$$

$$= \sqrt{\sin A \sec(90^\circ - A) - \sin A \cos(90^\circ - A)}$$

$$= \sqrt{\sin A \sec A - \sin A \sin A} = \sqrt{1 - \sin^2 A} = \cos A$$

Sol.51. (a)

$$\text{Consider } \tan^4 A - \sec^4 A + \tan^2 A + \sec^2 A$$

$$= (\tan^2 A)^2 - (\sec^2 A)^2 + \tan^2 A + \sec^2 A$$

$$= (\tan^2 A + \sec^2 A)^2 - (\tan^2 A - \sec^2 A)$$

$$= -(\tan^2 A + \sec^2 A)^2 + \tan^2 A + \sec^2 A = 0$$

Sol.52. (c)

$$\text{Let } \cos^2 \theta + \cot^2 \theta = c$$

$$\Rightarrow \frac{1}{\sin^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} = c \Rightarrow \frac{1 + \cos^2 \theta}{\sin^2 \theta} = c$$

$$\frac{1 + \left(2 \cos^2 \frac{\theta}{2} - 1\right)}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} = c \Rightarrow \frac{2 \cos^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} = c$$

$$\Rightarrow \cot \frac{\theta}{2} = c \Rightarrow \cos \theta = \frac{1 - \frac{1}{c^2}}{1 - \frac{1}{c^2}} = \frac{c^2 - 1}{c^2 + 1}$$

$$\therefore \frac{1 - \tan^2 \left(\frac{\theta}{2}\right)}{1 + \tan^2 \left(\frac{\theta}{2}\right)}$$

$$\Rightarrow \tan \frac{\theta}{2} = \frac{1}{c}$$

Sol.53. (c)

$$(s-m)/(m-d)$$

$$m = 60s \text{ and } d = 60m$$

$$(s-60s)/(m-60m) = (-59s)/(-59m)$$

$$= s/m = s/60s = 1/60$$

Sol.54. (b)

$$1. \tan \left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}} \quad 2. \tan \left(\frac{3\pi}{4}\right) = -1$$

$$3. \tan \left(\frac{5\pi}{4}\right) = 1 \quad 4. \tan \left(\frac{2\pi}{3}\right) = -\sqrt{3}$$

$$-\sqrt{3} < 1 < \frac{1}{\sqrt{3}} < 1$$

$$\text{Hence, } 4 < 2 < 1 < 3$$

Sol.55. (d)

$$(1 - \sin^2 \theta)(1 + \tan^2 \theta) = \cos^2 \theta \cdot \sec^2 \theta$$

$$= \cos^2 \theta \cdot \frac{1}{\cos^2 \theta} = 1$$

Sol.56. (a)

$$\sin^2 20^\circ + \sin^2 70^\circ = \sin^2 20^\circ + \sin^2 (90^\circ - 20^\circ)$$

$$= \sin^2 20^\circ + \cos^2 20^\circ = 1$$

Sol.57. (d)

$$\sin x \cdot \cot x \cdot \cos^2 x \cdot \tan x$$

$$= (\sin x \cdot \cos^2 x) \cdot (\cot x \cdot \tan x)$$

$$= 1 \times 1 = 1$$

Sol.58. (c)

$$\frac{\cot 54^\circ}{\tan 36^\circ} + \frac{\cot 20^\circ}{\tan 70^\circ} = \frac{\cot(90^\circ - 36^\circ)}{\tan 36^\circ} + \frac{\tan(90^\circ - 70^\circ)}{\tan 70^\circ}$$

$$= \frac{\tan 36^\circ}{\tan 36^\circ} + \frac{\cot 70^\circ}{\cot 70^\circ} = 1 + 1 = 2$$

Sol.59. (a)

$$\tan 15^\circ = \sqrt{\frac{1 - \cos 30^\circ}{1 + \cos 30^\circ}} = \sqrt{\frac{1 - \frac{\sqrt{3}}{2}}{1 + \frac{\sqrt{3}}{2}}} = \sqrt{\frac{2 - \sqrt{3}}{2 + \sqrt{3}}}$$

$$= \sqrt{\frac{2 - \sqrt{3}}{2 + \sqrt{3}} \times \frac{2 - \sqrt{3}}{2 - \sqrt{3}}} = \sqrt{\frac{(2 - \sqrt{3})^2}{1}} = 2 - \sqrt{3}$$

Sol.60. (b)

$$\text{Half past means its 4:30}$$

$$\theta = \frac{|11m - 60h|}{2} = \frac{11(30) - 60(4)}{2} = 45^\circ$$

Sol.61. (c)

$$\frac{1 + \sin A}{1 - \sin A} - \frac{1 - \sin A}{1 + \sin A}$$

$$= \frac{(1 + \sin A)^2 - (1 - \sin A)^2}{(1 - \sin A)(1 + \sin A)} = \frac{4 \sin A}{\cos^2 A}$$

$$= \frac{4 \sin A}{\cos A} \cdot \frac{1}{\cos A} = 4 \sec A \cdot \tan A$$

Sol.62. (a)

$$A + B + C = \pi$$

$$A + B = \pi - C$$

$$\cos(A + B) = \cos(\pi - C)$$

$$\cos(A + B) = -\cos C$$

$$\cos(A + B) + \cos C = 0$$

Sol.63. (d)

$$\cos 20^\circ + \cos 100^\circ + \cos 140^\circ$$

$$= (\cos 140^\circ + \cos 20^\circ) + \cos 100^\circ$$

$$= 2 \cos \left(\frac{160^\circ}{2}\right) \cdot \cos \left(\frac{120^\circ}{2}\right) + \cos 100^\circ$$

$$= 2 \cdot \cos 80^\circ \cdot \frac{1}{2} + \cos 100^\circ$$

$$= 2 \cos \left(\frac{180^\circ}{2}\right) \cdot \cos \left(\frac{20^\circ}{2}\right)$$

$$= 2 \cos 90^\circ \cdot \cos 10^\circ$$

$$= 2 \times 0 \times \cos 10^\circ = 0$$

Sol.64. (b)

$$\sin^2(3\pi) + \cos^2(4\pi) + \tan^2(5\pi)$$

$$= \sin^2(3\pi) + \cos^2(\pi + 3\pi) + \tan^2(5\pi)$$

$$= (\sin^2(3\pi) + \cos^2(3\pi) + \tan^2(2 \times 2\pi + \pi))$$

$$= 1 + \tan^2 \pi = \sec^2 \pi = 1$$

Sol.65. (a)

$$\sin \theta \in [-1, 1]; Q \in \mathbb{R}, \text{ the value of } \sin \theta \text{ lie between } -1 \text{ to } 1.$$

$$\cos \theta \in [-1, 1]; Q \in \mathbb{R} \text{ the value of } \sin \theta \text{ lies between } 1 \text{ to } 1.$$

Sol.66. (d)

Statement 1:

$$f_1(x) = \sin|x| + \cos|x|, \text{ the value of } \sin|x|$$

$$\text{and } \cos|x| \text{ depends on its angle.}$$

$$\sin|x| + \cos|x| \text{ is not always positive.}$$

Statement 2: $f_2(x) = \sin(x^2) = \cos(x^2)$, the

value of x^2 between any value which lies in the interval $\left(\pi, \frac{3\pi}{2}\right)$, then value of

$$f_2(x) = \sin(x^2) + \cos(x^2) \text{ is always negative}$$

Sol.67. (b)

$$\frac{\cot 224^\circ - \cot 134^\circ}{\cot 226^\circ + \cot 316^\circ}$$

$$\frac{\cot(180^\circ - 44^\circ) - \cot(180^\circ - 46^\circ)}{\cot(180^\circ + 44^\circ) + \cot(270^\circ + 46^\circ)}$$

$$\frac{\cot 44^\circ + \cot 46^\circ}{\cot 46^\circ - \tan 46^\circ} = \frac{\cot 46^\circ + \cot 44^\circ}{\cot 44^\circ - \tan 46^\circ}$$

$$\frac{\sin(46^\circ + 44^\circ)}{\cot(44^\circ - 46^\circ)} = -\operatorname{cosec} 2^\circ$$

$$\frac{\cot(44^\circ - 46^\circ)}{\cot(44^\circ - 46^\circ)} = -\operatorname{cosec} 2^\circ$$

Sol.68. (b)

$$\cos 36^\circ = \frac{\sqrt{5} + 1}{2}$$

Sol.69. (a)

Statement 1:

$$\text{Given } n \left(\sin^2 67^\circ \frac{1}{2} - \sin^2 22^\circ \frac{1}{2} \right)$$

$$= 2n \cdot \sin 45^\circ \times \frac{1}{2} = n \cdot \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\therefore \frac{n}{\sqrt{2}} > 1 \text{ for all positive integers } n \geq 2.$$

$$\therefore \text{Statement 1 is true}$$

Sol.70. (a)

statement : 1

$$\sin 3\theta = \cos 2\theta$$

$$3\theta = \frac{\pi}{2} - 2\theta$$

$$5\theta = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{10}$$

Statement: 2

One radian is the angle subtended at the centre of a circle by an arc of the same circle whose length is equal to radius of that circle.
Hence, statement I is correct.

Sol.71. (c)

$$3 \tan^2 x = 1$$

$$\tan^2 x = 1/3$$

$$\alpha = \pi/6$$

general solution of $\tan^2 x$ is

$$x = n\pi \pm \alpha = n\pi \pm \frac{\pi}{6}$$

Sol.72. (a)

$$16 \sin^5 x = 16 (\sin^2 x)^2 \cdot \sin x$$

$$-16 \left(\frac{1 - \cos 2x}{2} \right)^2 \cdot \sin x$$

$$= 4 (1 + \cos^2 2x - 2 \cos 2x) \cdot \sin x$$

$$= 4 \left(1 + \frac{1 + \cos 4x}{2} - 2 \cos 2x \right) \cdot \sin x$$

$$= \frac{4}{2} (3 + \cos 4x - 4 \cos 2x) \cdot \sin x$$

$$[\because \cos B = \sin(A+B) + \sin(A-B)]$$

$$= 6 \sin x + \sin 5x - \sin 3x - 4 \sin 3x + 4 \sin x$$

$$= \sin 5x - 5 \sin 3x + 10 \sin x$$

$$p = 1$$

Sol.73. (d)

By above solution

Sol.74. (c)

By above solution

Sol.75. (b)

$$\cos 7x - \cos 3x$$

$$\sin 7x - 2 \sin 5x + \sin 3x$$

$$-2 \sin \frac{7x+3x}{2} \cdot \sin \frac{7x-3x}{2}$$

$$2 \sin \frac{7x+3x}{2} \cdot \cos \frac{7x-3x}{2} - 2 \sin 5x$$

$$\left(\because D = 2 \sin \left(\frac{C+D}{2} \right) \cdot \cos \left(\frac{C-D}{2} \right) \right)$$

$$\text{and } \cos C - \cos D = -2 \sin \left(\frac{C+D}{2} \right) \sin \left(\frac{C-D}{2} \right)$$

$$= \frac{-2 \sin 5x \cdot \sin 2x}{2 \sin 5x \cos 2x - 2 \sin 5x}$$

$$= \frac{-2 \sin 5x \cdot \sin 2x}{2 \sin 5x \cos 2x - 2 \sin 5x}$$

$$= \frac{-2 \sin 5x \cdot \sin 2x}{-2 \sin 5x [1 - \cos 2x]}$$

$$= \frac{\sin 2x}{1 - 1 + 2 \sin^2 x} = \cot x$$

Sol.76. (b)

$$\frac{\sin(x+y)}{\sin(x-y)} = \frac{a+b}{a-b}$$

$$\frac{\sin(x+y) + (x-y)}{\sin(x+y) - (x-y)} = \frac{(a+b) + (a-b)}{(a+b) - (a-b)}$$

$$\therefore \frac{\tan x}{\tan y} = \frac{a}{b}$$

Sol.77. (b)

$$\text{consider, } \sqrt{1 + \sin 2\theta}$$

$$= \sqrt{\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta}$$

$$= \sqrt{(\sin \theta + \cos \theta)^2} = \sin \theta + \cos \theta$$

Sol.78. (d)

$\cot A = 2$ and $\cot B = 3$

$$\cot(A+B) = \frac{\cot A \cot B - 1}{\cot A + \cot B} = \frac{6-1}{2+3} = \frac{5}{5} = 1$$

$$\cot(A+B) = \cot\left(\frac{\pi}{4}\right) \Rightarrow A+B = \frac{\pi}{4}$$

Sol.79. (b)

$$\sin^2 66^\circ - \sin^2 23^\circ$$

$$= \left[\sin\left(90^\circ - 23^\circ\right) \right]^2 - \sin^2 23^\circ$$

$$= \cos^2\left(23^\circ\right) = \sin^2 23^\circ$$

$$= \cos 2\left(23^\circ\right) = \cos 47^\circ$$

$$(\because \cos^2 A - \sin^2 A)$$

$$= \cos\left[2 \times \left(\frac{47^\circ}{2}\right)\right] = \cos 47^\circ$$

Sol.80. (a)

$$\sin A \cdot \sin(60^\circ + A) = k \sin 3A$$

$$\Rightarrow \sin A \cdot \frac{\sin 3A}{4 \sin A} = k \cdot \sin 3A$$

$$\left[\because \sin(60^\circ - A) = \frac{\sin 3A}{4 \sin A} \right]$$

$$\Rightarrow \frac{\sin 3A}{4} = k \cdot \sin 3A$$

$$\therefore k = \frac{1}{4}$$

Sol.81. (b)

$$\tan 2\theta \cdot \tan \theta = 1$$

$$\Rightarrow \frac{2 \tan \theta}{1 - \tan^2 \theta} \cdot \tan \theta = 1$$

$$\Rightarrow 2 \tan^2 \theta = 1 - \tan^2 \theta \Rightarrow 3 \tan^2 \theta = 1$$

$$\Rightarrow \tan^2 \theta = \frac{1}{3} \Rightarrow \tan \theta = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \tan^2 \theta = \tan^2 \theta (30^\circ) = \tan^2 \left(\frac{\pi}{6}\right)$$

$$\therefore \theta = \frac{\pi}{6}$$

Sol.82. (a)

$$\tan x = \sqrt{3}$$

$x = 60^\circ$ only one point between 0 to 90°

Sol.83. (a)

Here α is the root of equation

$$25 \cos^2 \theta + 5 \cos \theta - 12 = 0$$

$$\Rightarrow 25 \cos^2 \alpha + 5 \cos \alpha - 12 = 0$$

$$\Rightarrow 25 \cos^2 \alpha + 20 \cos \alpha - 15 \cos \alpha - 12 = 0$$

$$\Rightarrow 5 \cos \alpha (5 \cos \alpha + 4) - 3 (5 \cos \alpha + 4) = 0$$

$$(5 \cos \alpha - 3)(5 \cos \alpha + 4) = 0$$

$$\cos \alpha = \frac{3}{5} \text{ or } \cos \alpha = -\frac{4}{5}$$

$$\frac{\pi}{2} < \alpha < \pi$$

$$\therefore \cos \alpha = -\frac{4}{5}$$

($\because$ in 2nd quadrant, $\cos \alpha$ value is negative)

$$\text{Now } \sin \alpha = \sqrt{1 - \cos^2 \alpha} = \sqrt{1 - \frac{16}{25}}$$

$$\therefore \sin \alpha = \frac{3}{5}$$

$$\therefore \tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \frac{3}{5} \times \frac{-5}{4} = -\frac{3}{4}$$

Option (a) is correct

Sol.84. (b)

$$\sin 2\alpha = 2 \sin \alpha \cdot \cos \alpha$$

$$= 2 \left(\frac{3}{5} \right) \left(-\frac{4}{5} \right)$$

$$= \frac{6}{5} \times \frac{-4}{5} = -\frac{24}{25}$$

$\therefore$ Option (b) is correct.

Sol.85. (b)

$$(1 - \sin A + \cos A)^2$$

$$= 1 + \sin^2 A + \cos^2 A - 2 \sin A$$

$$- 2 \sin A \cdot \cos A + 2 \cos A$$

$$= 2(1 - \sin A) + 2 \cos A(1 - \sin A)$$

$$= 2(1 + \sin A)(1 - \sin A)$$

$$= 2(1 + \cos A)(1 - \sin A)$$

$\therefore$ Option (b) is correct

Sol.86. (b)

$$\frac{\cos \theta}{1 - \tan \theta} + \frac{\sin \theta}{1 - \cot \theta}$$

$$= \frac{\cos \theta}{1 - \frac{\sin \theta}{\cos \theta}} + \frac{\sin \theta}{1 - \frac{\cos \theta}{\sin \theta}}$$

$$= \frac{\cos^2 \theta}{\cos \theta - \sin \theta} + \frac{\sin^2 \theta}{\sin \theta - \cos \theta}$$

$$= \frac{\cos^2 \theta - \sin^2 \theta}{\cos \theta - \sin \theta} = \frac{(\cos \theta + \sin \theta)(\cos \theta - \sin \theta)}{\cos \theta - \sin \theta}$$

$$= \cos \theta + \sin \theta$$

$\therefore$ Option (b) is correct

Sol.87. (c)

From going by the options, option (a), $\theta = 30^\circ$, as we know that $180^\circ = \pi$ radian

$$\therefore 30^\circ = \frac{30\pi}{180} \text{ radian}$$

Now according to question,

$$\frac{30^\circ \times 180^\circ}{30^\circ \pi} = \frac{180}{\pi}$$

Now number of degree in θ is multiplied by number of radians in θ .

$$\therefore 30^\circ \times \frac{30\pi}{180} = \frac{900\pi}{180} = \frac{10\pi}{2} = 5\pi \neq \frac{125\pi}{9}$$

From option (b), $\theta = 45^\circ$

$$\therefore 45^\circ = \frac{45\pi}{180} \text{ radian}$$

Now according to question,

$$\frac{45^\circ \times 180^\circ}{45^\circ \pi} = \frac{180}{\pi}$$

Now number of degree in θ is multiplied by number of radian in θ

$$\therefore 45^\circ \times \frac{45\pi}{180} = \frac{45^\circ \pi}{4} \neq \frac{125\pi}{9}$$

From option (c),

$$\theta = 50^\circ$$

As we know that $180^\circ = \pi$ radian

$$\therefore 50^\circ = \frac{50\pi}{180} \text{ radian}$$

Now according to question

$$\frac{50^\circ \times 180^\circ}{50^\circ \pi} = \frac{180}{\pi}$$

Now number of degree in ' θ ' is multiplied by number of radian in θ .

$$\therefore 50^\circ \times \frac{50\pi}{180} = \frac{2500\pi}{180} = \frac{125\pi}{9}$$

$\therefore$ Option (c) is correct.

Sol.88. (b)

$$\sin x + \sin y = a$$

$$\Rightarrow 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) = a$$

$$\cos x + \cos y = b$$

$$\Rightarrow 2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) = b$$

Dividing eq (1) & (2)

$$\tan\left(\frac{x+y}{2}\right) = \frac{a}{b}$$

Squaring eq(1) & (2) and adding

$$4 \cos^2\left(\frac{x-y}{2}\right) = a^2 + b^2$$

$$\sec^2\left(\frac{x-y}{2}\right) = \frac{4}{a^2 + b^2}$$

$$\text{Again, } \tan^2\left(\frac{x+y}{2}\right) + \tan^2\left(\frac{x-y}{2}\right)$$

$$= \left(\frac{a}{b}\right)^2 + \sec^2\left(\frac{x-y}{2}\right) - 1$$

$$= \frac{a^2}{b^2} + \frac{4}{a^2 + b^2} - 1 = \frac{a^4 - b^4 + 4b^2}{a^2 b^2 + b^4}$$

Sol.89. (b)

$$p = \tan\left(-\frac{11\pi}{6}\right)$$

$$p = \tan\left(2\pi - \frac{\pi}{6}\right)$$

$$p = \tan\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}}$$

$$q = \tan\left(\frac{21\pi}{4}\right) = \tan\left(6\pi + \frac{3\pi}{4}\right)$$

$$q = \tan\left(\frac{3\pi}{4}\right) = + \tan\left(\frac{\pi}{4}\right) = 1$$

$$r = \cot\left(\frac{283\pi}{6}\right) = + \cot\left(46\pi + \frac{7\pi}{6}\right) = \cot\left(\pi + \frac{\pi}{6}\right)$$

$$r = \cot + \frac{\pi}{6} = \sqrt{3}$$

$$p \times r = \frac{1}{\sqrt{3}} \times \sqrt{3} = 1$$

$\therefore$ Statement (1) is incorrect

$$\text{Also } \frac{p}{q} = \frac{q}{r} = \frac{1}{\sqrt{3}}$$

So p, r are in G.P.

$\therefore$ Statement (2) is correct

Sol.90. (d)

$$\sin^2 5^\circ + \sin^2 10^\circ + \sin^2$$

$$15^\circ + \dots + \sin^2 75^\circ +$$

$$\sin^2 80^\circ + \sin^2 85^\circ + \sin^2 90^\circ$$

$$\Rightarrow \sin^2 + \sin^2 10^\circ + \sin^2 15^\circ + \dots \sin^2 (90-15^\circ) +$$

$$\sin^2 (90-10^\circ) + \sin^2 (90-5^\circ) + 1$$

$$\Rightarrow \sin^2 5^\circ + \sin^2 10^\circ + \sin^2 15^\circ +$$

$$\dots + \cos^2 15^\circ +$$

$$\dots + \cos^2 15^\circ + \cos^2 10^\circ + \cos^2 5^\circ + 1$$

$$\Rightarrow (1+1+1+\dots 8 \text{ times}) + \sin^2 45^\circ + 1$$

$$\Rightarrow 8 + \frac{1}{2} + 1 = \frac{19}{2}$$

Sol.91. (d)

$$\frac{\sin^3 A + \sin 3A}{\sin A} + \frac{\cos^3 A - \cos 3A}{\cos A}$$

$$\Rightarrow \frac{\sin^3 A + \sin 3A - 4 \sin^3 A}{\sin A}$$

$$+ \frac{\cos^3 A - [4 \cos^3 A - 3 \cos A]}{\cos A}$$

$$\Rightarrow \frac{3 \sin A - 3 \sin^3 A}{\sin A} + \frac{(-3 \cos^3 A + 3 \cos A)}{\cos A}$$

$$= 3 - 3 \sin^2 A - 3 \cos^2 A + 3$$

$$= 6 - 3(\cos^2 A + \sin^2 A) = 6 - 3(1) = 3$$

Sol.92. (b)

$$\Rightarrow \frac{A}{B} = \frac{[2 \sin 24^\circ \sin 12^\circ][2 \sin 60^\circ \cos 36^\circ]}{[2 \cos 36^\circ \sin 24^\circ][2 \sin 60^\circ \sin 12^\circ]}$$

$$\Rightarrow \frac{A}{B} = 1$$

Sol.93. (c)

$$\sin A + 2 \sin 2A + \sin 3A$$

$$= 2 \sin 2A + \sin A + \sin 3A$$

$$= 2 \sin 2A + 2 \sin 2A \cos A$$

$$= 2 \sin 2A (1 + \cos A)$$

now apply half angle formula in $1 + \cos A$

1,3 statements are correct.

Sol.94. (a)

$$\sin \theta_1 + \sin \theta_2 + \sin \theta_3 + \sin \theta_4 = 4 \dots (1)$$

Since max. value of $\sin \theta = 1$

We have four terms in LHS of eq.(1).

$\Rightarrow$ Every term should be =1

$$\Rightarrow \sin \theta_1 = \sin \theta_2 = \sin \theta_3 = \sin \theta_4 = 1$$

$$\Rightarrow \theta_1 = \theta_2 = \theta_3 = \theta_4 = 90^\circ$$

Now,

$$\cos \theta_1 + \cos \theta_2 + \cos \theta_3 + \cos \theta_4 = (\cos 90^\circ) \times 4 = 0$$

Sol.95. (a)

$$\text{Here, } z = x \cos \theta + y \sin \theta$$

$$z^2 = x^2 \cos^2 \theta + y^2 \sin^2 \theta + 2xy \sin \theta \cos \theta$$

$$\Rightarrow 2xy \sin \theta \cos \theta = z^2 - x^2 \cos^2 \theta - y^2 \sin^2 \theta$$

$$\text{Let, } L = (x \sin \theta - \cos \theta)^2$$

$$\Rightarrow L = x^2 \sin^2 \theta + y^2 \cos^2 \theta - 2xy \sin \theta \cos \theta$$

$$\Rightarrow L = x^2 \sin^2 \theta + y^2 \cos^2 \theta - [z^2 - x^2 \cos^2 \theta - y^2 \sin^2 \theta]$$

$$\Rightarrow L = x^2 [\sin^2 \theta + \cos^2 \theta] + y^2 [\sin^2 \theta + \cos^2 \theta] - z^2$$

$$\Rightarrow L = x^2 + y^2 - z^2$$

Sol.96. (a)

$$x = \sin 70^\circ \cdot \sin 50^\circ \text{ and } y = \cos 60^\circ \cdot \cos 80^\circ$$

$$xy = \frac{1}{2} \cdot \sin (90-20) \cdot \sin (90-40) \cdot \cos 80$$

$$\Rightarrow xy = \frac{1}{2} \cdot \cos 20^\circ \cdot \cos 40^\circ \cdot \cos 80$$

$$(\because \cos(-x) = \cos x)$$

$$\Rightarrow xy = \frac{1}{2} \cdot \cos 20^\circ \cdot \cos (60-20)^\circ \cdot \cos (60+20)^\circ$$

$$\Rightarrow xy = \frac{1}{2} \left[\frac{1}{4} \cos 3(20^\circ) \right] = \frac{1}{2} \times \frac{1}{2} \times \cos 60^\circ = \frac{1}{16}$$

$$\left[\because \cos (60-\theta) \cdot \cos (60+\theta) = \frac{1}{4} \cos 3\theta \right]$$

Sol.97. (d)

$$\left[1 + \cos \frac{\pi}{8} \right] \left[1 + \cos \frac{3\pi}{8} \right] \left[1 + \cos \frac{5\pi}{8} \right] \left[1 + \cos \frac{7\pi}{8} \right]$$

we have,

$$\cos \frac{7\pi}{8} = \cos \left[\pi - \frac{\pi}{8} \right] = -\cos \frac{\pi}{8}$$

$$\text{and } \cos \frac{5\pi}{8} = \cos \left[\pi - \frac{3\pi}{8} \right] = -\cos \frac{3\pi}{8}$$

$$\therefore \left[1 + \cos \frac{\pi}{8} \right] \left[1 + \cos \frac{3\pi}{8} \right] \left[1 - \cos \frac{\pi}{8} \right] \left[1 - \cos \frac{3\pi}{8} \right]$$

$$= \left[1 - \cos^2 \frac{\pi}{8} \right] \left[1 - \cos^2 \frac{3\pi}{8} \right] = \sin^2 \frac{\pi}{8} \cdot \sin^2 \frac{3\pi}{8}$$

$$= \frac{1}{4} \left[2 \sin^2 \frac{\pi}{8} \cdot 2 \sin^2 \frac{3\pi}{8} \right]$$

$$= \frac{1}{4} \left[\left(1 - \cos \frac{\pi}{4} \right) \left(1 - \cos \frac{3\pi}{4} \right) \right]$$

$$\left(\because 2 \sin^2 \frac{\theta}{2} \right)$$

$$= \frac{1}{4} \left[\left(1 - \frac{1}{\sqrt{2}} \right) \left(1 + \frac{1}{\sqrt{2}} \right) \right] = \frac{1}{8}$$

Sol.98. (a)

$$\therefore \dots = \frac{\sqrt{5}-1}{4}$$

$$x^2 = 4^2 - (\sqrt{5}-1)^2$$

$$\Rightarrow x = \sqrt{10+2\sqrt{5}}$$

$$\Rightarrow \cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4}$$

$$\Rightarrow 2 \cos^2 9^\circ - 1 = \frac{\sqrt{10+2\sqrt{5}}}{4}$$

$$\cos^2 9^\circ = \frac{\sqrt{10+2\sqrt{5}} + 4}{8}$$

$$\Rightarrow \sin^2 81^\circ = \frac{4 + \sqrt{10+2\sqrt{5}}}{8}$$

After squaring all the options available, we come to a conclusion that option (a) is correct.

Sol.99. (b)

$$L = \frac{1 - \tan 2^\circ \cot 62^\circ}{\tan 152^\circ - \cot 88^\circ} = \frac{1 - \tan 2^\circ \cot (90-28)^\circ}{\tan (180-28)^\circ - \cot (90-2)^\circ}$$

$$\Rightarrow L = \frac{1 - \tan 2^\circ \cot 28^\circ}{-\tan 28^\circ - \tan 2^\circ} = \left[\frac{1 - \tan 2^\circ \tan 28^\circ}{\tan 2^\circ + \tan 28^\circ} \right]$$

$$\Rightarrow L = \frac{1}{\tan (2+28)^\circ} = \frac{1}{\tan 30^\circ} = -\sqrt{3}$$

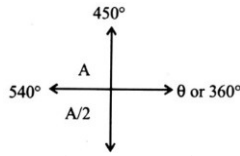
$$\left[\because \cot B = \frac{\tan A + \tan B}{1 - \tan A \tan B} \right]$$

Sol.100. (d)

$$\sin A = \frac{3}{5}; 450^\circ < A < 540^\circ$$

$$\Rightarrow 225^\circ < \frac{A}{2} < 270^\circ$$

$$\cos A = \frac{-4}{5} \quad (\because Q2)$$



$$\therefore \cos \frac{A}{2} = \frac{-1}{\sqrt{10}} \quad (\because Q3)$$

Sol.101. (a)

$$\frac{\sin \alpha + \sin \beta}{\cos \alpha + \cos \beta} = \frac{2 \sin \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)}{2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)}$$

$$= \tan \left(\frac{\alpha + \beta}{2} \right)$$

Sol.102. (a)

$$\tan 18^\circ = \frac{\sin 18^\circ}{\cos 18^\circ} = \frac{\frac{\sqrt{5}-1}{4}}{\frac{10+2\sqrt{5}}{4}} = \frac{\sqrt{5}-1}{\sqrt{10+2\sqrt{5}}}$$

Sol.103. (a)

$$\tan(\alpha + \beta) = 2$$

$$\tan(\alpha - \beta) = 1$$

$$\tan 2\alpha = \tan[(\alpha + \beta) + (\alpha - \beta)]$$

$$= \frac{\tan(\alpha + \beta) + \tan(\alpha - \beta)}{1 - \tan(\alpha + \beta) \tan(\alpha - \beta)}$$

$$= \frac{2+1}{1-2 \cdot 1} = \frac{3}{-1} = -3$$

Sol.104. (d)

$$\frac{1}{\sin 10^\circ} - \frac{\sqrt{3}}{\cos 10^\circ} = \frac{\cos 10^\circ - \sqrt{3} \sin 10^\circ}{\sin 10^\circ \cos 10^\circ}$$

$$= \frac{2 \times 2 \left[\frac{1}{2} \cos 10^\circ - \frac{\sqrt{3}}{2} \sin 10^\circ \right]}{2 \sin 10^\circ \cos 10^\circ}$$

$$= \frac{4(\cos 60^\circ \cos 10^\circ - \sin 60^\circ \sin 10^\circ)}{\sin 20^\circ}$$

$$= \frac{4 \cos(60^\circ + 10^\circ)}{\sin 20^\circ} = 4 \cdot \frac{\sin 20^\circ}{\sin 20^\circ} = 4$$

Sol.105. (a)

The maximum value of $\sin A + \cos A$ is at 45° so $x = 15^\circ$

Sol.106. (c)

$$K = \sin \left(\frac{\pi}{18} \right) \sin \left(\frac{5\pi}{18} \right) \sin \left(\frac{7\pi}{18} \right)$$

We know,

$$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

$$K = \frac{1}{2} \cdot \sin \frac{\pi}{18} \left[2 \sin \frac{5\pi}{18} \sin \frac{7\pi}{18} \right]$$

$$= \frac{1}{2} \cdot \sin \frac{\pi}{18} \left[\cos \frac{2\pi}{18} - \cos \frac{12\pi}{18} \right]$$

$$= \frac{1}{2} \cdot \frac{1}{2} \left[2 \sin \frac{\pi}{18} \cos \frac{2\pi}{18} - 2 \sin \frac{\pi}{18} \cos \frac{2\pi}{3} \right]$$

$$= \frac{1}{4} \left[\sin \left(\frac{3\pi}{18} \right) + \sin \left(\frac{-\pi}{18} \right) - 2 \sin \frac{\pi}{18} \cos \left(\pi - \frac{\pi}{3} \right) \right]$$

$$= \frac{1}{4} \left[\sin \frac{\pi}{6} - \sin \frac{\pi}{18} + 2 \sin \frac{\pi}{18} \cdot \cos \frac{\pi}{3} \right]$$

$$= \frac{1}{4} \left[\sin \frac{\pi}{6} - \sin \frac{\pi}{18} + 2 \sin \frac{\pi}{18} \cdot \frac{1}{2} \right]$$

$$= \frac{1}{4} \sin \frac{\pi}{6} = \frac{1}{4} \times \frac{1}{2} \times \frac{1}{8}$$

Sol.107. (b)

$$\text{Given, } \sin \theta = 3 \sin(\theta + 2\alpha)$$

$$\Rightarrow \frac{\sin(\theta + 2\alpha)}{\sin \theta} = \frac{1}{3}$$

Apply componendo and dividendo rule

$$\Rightarrow \frac{\sin(\theta + 2\alpha) + \sin \theta}{\sin(\theta + 2\alpha) - \sin \theta} = \frac{1+3}{1-3}$$

$$\Rightarrow \frac{2 \sin(\theta + \alpha) \cos \alpha}{2 \cos(\theta + \alpha) \sin \alpha} = \frac{4}{-2} = -2$$

$$\Rightarrow \frac{\tan(\theta + \alpha)}{\tan \alpha} = -2$$

$$\Rightarrow \tan(\theta + \alpha) = -2 \tan \alpha \Rightarrow$$

$$\tan(\theta + \alpha) + 2 \tan \alpha = 0$$

Sol.108. (b)

$$\sec \theta - \cos \theta = \frac{4}{3} \Rightarrow \frac{1}{\cos \theta} - \frac{1}{\sin \theta} = \frac{4}{3}$$

$$\Rightarrow \frac{\sin \theta - \cos \theta}{\sin \theta \cos \theta} = \frac{4}{3} \quad \dots(i)$$

$$\Rightarrow \frac{(\sin \theta - \cos \theta)^2}{(\sin \theta \cos \theta)^2} = \frac{16}{9}$$

$$\Rightarrow \frac{\sin^2 \theta + \cos^2 \theta - 2 \sin \theta \cos \theta}{\sin^2 \theta \cos^2 \theta} = \frac{16}{9}$$

$$\Rightarrow \frac{1 - 2 \sin \theta \cos \theta}{\sin^2 \theta \cos^2 \theta} = \frac{16}{9}$$

$$\text{Let } \sin \theta \cos \theta = x \Rightarrow \frac{1-2x}{x^2} = \frac{16}{9}$$

$$\Rightarrow 16x^2 + 18x - 9 = 0$$

$$\Rightarrow (8x - 3)(2x + 3) = 0$$

$$\Rightarrow x = \frac{3}{8}, x = \frac{-3}{2}$$

$$\therefore \sin \theta \cos \theta = \frac{3}{8}$$

$$\text{From (i), } \sin \theta \cos \theta = x \Rightarrow \frac{1-2x}{x^2} = \frac{16}{9}$$

$$\Rightarrow 16x^2 + 18x - 9 = 0$$

$$\Rightarrow (8x - 3)(2x + 3) = 0$$

$$\Rightarrow x = \frac{3}{8}, x = \frac{-3}{2}$$

$$\therefore \sin \theta \cos \theta = \frac{3}{8}$$

$$\text{From (i) } \sin \theta - \cos \theta = \frac{4}{3} \times \frac{3}{8} = \frac{4}{8} = \frac{1}{2}$$

Sol.109. (d)

$$\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$$

$$= \tan 9^\circ - \tan 27^\circ - \tan(90^\circ - 27^\circ) + \tan(90^\circ - 9^\circ)$$

$$= \tan 9^\circ - \tan 27^\circ - \cot 27^\circ + \cot 9^\circ$$

$$= \tan 9^\circ + \cot 9^\circ - (\tan 27^\circ + \cot 27^\circ)$$

We know,

$$\tan \theta + \cot \theta = \frac{1}{\sin \theta \cos \theta} = \frac{2}{2 \sin \theta \cos \theta} = \frac{2}{\sin 2\theta}$$

$$\therefore \tan 9^\circ + \cot 9^\circ - (\tan 27^\circ + \cot 27^\circ)$$

$$= \frac{2}{\sin 18^\circ} - \frac{2}{\sin 54^\circ} = 2 \left(\frac{\sin 54^\circ - \sin 18^\circ}{\sin 18^\circ \sin 54^\circ} \right)$$

We also know, $\sin C - \sin D$

$$= 2 \cos \frac{(C+D)}{2} \sin \frac{(C-D)}{2}$$

$$\therefore 2 \left(\frac{\sin 54^\circ - \sin 18^\circ}{\sin 18^\circ \sin 54^\circ} \right) = 2.2 \frac{\cos 36^\circ \sin 18^\circ}{\sin 18^\circ \sin 54^\circ}$$

$$= 2.2 \cdot \frac{\sin 54^\circ - \sin 18^\circ}{\sin 18^\circ \sin 54^\circ} = 4$$

Sol.110. (a)

$$\sqrt{3} \csc 20^\circ - \sec 20^\circ = \frac{\sqrt{3}}{\sin 20^\circ} - \frac{1}{\cos 20^\circ}$$

$$= \frac{\sqrt{3} \cos 20^\circ - \sin 20^\circ}{\sin 20^\circ \cos 20^\circ}$$

$$= \frac{\left(\frac{\sqrt{3}}{2} \cos 20^\circ - \frac{1}{2} \sin 20^\circ \right)}{\sin 20^\circ \cos 20^\circ}$$

$$= \frac{2(\cos 30^\circ \cos 20^\circ - \sin 30^\circ \sin 20^\circ)}{\left(\frac{2 \sin 20^\circ \cos 20^\circ}{2} \right)}$$

$$= \frac{4[\cos(30^\circ + 20^\circ)]}{\sin 40^\circ}$$

$$= \frac{4 \cos 50^\circ}{\sin 40^\circ} = \frac{4 \cos(90^\circ - 40^\circ)}{\sin 40^\circ} = \frac{4 \sin 40^\circ}{\sin 40^\circ} = 4$$

Sol.111. (d)

$$A - B = x \text{ and } \tan A : \tan B = p : q$$

$$\text{Also, given } \alpha = A + B$$

$$A - B = x \text{ and } \alpha = A + B$$

$$\Rightarrow A = \frac{x + \alpha}{2} \text{ and } B = \frac{\alpha - x}{2}$$

$$\text{Now, } \frac{\tan A}{\tan B} = \frac{\tan \left(\frac{x + \alpha}{2} \right)}{\tan \left(\frac{\alpha - x}{2} \right)} = \frac{p}{q} \quad (\text{Given})$$

$$\Rightarrow \frac{2 \sin \left(\frac{\alpha + x}{2} \right) \cos \left(\frac{\alpha - x}{2} \right)}{2 \cos \left(\frac{\alpha + x}{2} \right) \sin \left(\frac{\alpha - x}{2} \right)} = \frac{p}{q}$$

$$\Rightarrow \frac{2 \sin \alpha}{2 \sin x} = \frac{p + q}{p - q}$$

$$\Rightarrow \frac{\sin \alpha}{\sin x} = \frac{p + q}{p - q} \Rightarrow \sin x = \frac{(\sin \alpha)(p - q)}{p + q}$$

$$\Rightarrow \frac{2 \sin \alpha}{2 \sin x} = \frac{p + q}{p - q}$$

$$\Rightarrow \frac{\sin \alpha}{\sin x} = \frac{p + q}{p - q} \Rightarrow \sin x = \frac{(\sin \alpha)(p - q)}{p + q}$$

Sol.112. (c)

$$\sqrt{1 + \sin A} = - \left(\sin \frac{A}{2} + \cos \frac{A}{2} \right)$$

$$\text{We know, } 1 + \sin A = \left(\cos \frac{A}{2} + \sin \frac{A}{2} \right)^2$$

$$\therefore \sqrt{1 + \sin A} = \left| \cos \frac{A}{2} + \sin \frac{A}{2} \right|$$

$$\therefore \sqrt{1 + \sin A} = - \left(\sin \frac{A}{2} + \cos \frac{A}{2} \right) \text{ when}$$

$$\frac{3\pi}{4} < \frac{A}{2} < \frac{7\pi}{4}$$

$$\Rightarrow \frac{3\pi}{2} < A < \frac{7\pi}{2}$$

Sol.113. (a)
 $\sin A + \cos A$ has maximum value at 45°

$$x + \frac{\pi}{5} = \frac{\pi}{4} \Rightarrow x = \frac{\pi}{20}$$

Sol.114. (b)
 $\cos \alpha + \cos \beta + \cos \gamma = 0 \dots (1)$

$$\text{Given, } 0 < \alpha \leq \frac{\pi}{2}, 0 < \gamma \leq \frac{\pi}{2}$$

(i) is satisfied when $\alpha = \frac{\pi}{2}, \beta = \frac{\pi}{2}$ and $\gamma = \frac{\pi}{2}$.

$$\therefore \sin \alpha + \sin \beta + \sin \gamma = \sin \frac{\pi}{2} + \sin \frac{\pi}{2} + \sin \frac{\pi}{2}$$

$$= 1 + 1 + 1 = 3$$

Sol.115. (c)

Given, $\cos \frac{A}{2}$ has only one value

$$\text{We know, } \cos A = 2 \cos^2 \frac{A}{2} - 1$$

$$\Rightarrow 2 \cos^2 \frac{A}{2} = \cos A + 1 \Rightarrow \cos \frac{A}{2} = \sqrt{\frac{\cos A + 1}{2}}$$

$$\text{Since, } \cos \frac{A}{2} \text{ is single value, } \frac{\cos A + 1}{2} = 0$$

$$\Rightarrow \cos A = -1$$

So, A is an odd multiple of 180°

Sol.116. (b)

$$\sin \alpha + \sin \beta = 0 = \cos \alpha + \cos \beta$$

$$\sin \alpha + \sin \beta = 0$$

$$\Rightarrow \sin \alpha = -\sin \beta$$

$$\Rightarrow \sin \alpha + \sin(\pi + \beta)$$

$$\Rightarrow \alpha = \pi + \beta$$

Sol.117. (a)

$$\frac{\sin(x+y)}{\sin(x-y)} = \frac{a+b}{a-b}$$

Applying componendo and dividendo, we get

$$\frac{\sin(x+y) + \sin(x-y)}{\sin(x-y) - \sin(x-y)} = \frac{a+b+a-b}{a+b-a-b}$$

$$\Rightarrow \frac{2 \sin(x+y+x-y) \cos(x+y-x+y)}{2 \cos(x+y+x-y) \sin(x+y-x+y)} = \frac{2a}{2b}$$

$$\Rightarrow \frac{\sin x \cdot \cos y}{\cos x \cdot \sin y} = \frac{a}{b}$$

$$\Rightarrow \frac{\tan x}{\tan y} = \frac{a}{b}$$

Sol.118. (c)

$$\sin 105^\circ + \cos 105^\circ$$

$$= \sin(60^\circ + 45^\circ) + \cos(60^\circ + 45^\circ)$$

$$= (\sin 60^\circ \cdot \sin 45^\circ + \cos 60^\circ \cdot \cos 45^\circ)$$

$$+ (\cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ)$$

$$= \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} + \frac{1}{2} \cdot \frac{1}{\sqrt{2}} + \frac{1}{2} \cdot \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$$

Sol.119. (c)

$$\frac{\sin 5x - \sin 3x}{\cos 5x + \cos 3x}$$

$$\therefore \frac{\sin 5x - \sin 3x}{\cos 5x + \cos 3x} = \frac{2 \cos \left(\frac{5x+3x}{2} \right) \sin \left(\frac{5x-3x}{2} \right)}{2 \cos \left(\frac{5x+3x}{2} \right) \cos \left(\frac{5x-3x}{2} \right)}$$

$$= \frac{\sin(2x)}{\cos(2x)} = \frac{\sin x}{\cos x} = \tan x$$

Sol.120. (c)

$$\sin x = \frac{1}{\sqrt{5}}, \sin y = \frac{1}{\sqrt{10}}, 0 < x < \frac{\pi}{2}, 0 < y < \frac{\pi}{2}$$

$$\cos x = \sqrt{1 - \sin^2 x} \quad \cos y = \sqrt{1 - \sin^2 y}$$

$$= \frac{\sqrt{4}}{\sqrt{5}} = \frac{2}{\sqrt{5}} = \frac{\sqrt{9}}{\sqrt{10}} = \frac{3}{\sqrt{10}}$$

$$\sin(x+y) = \sin x \cos y + \cos x \sin y$$

$$= \frac{1}{\sqrt{5}} \cdot \frac{3}{\sqrt{10}} + \frac{2}{\sqrt{5}} \cdot \frac{1}{\sqrt{10}}$$

$$= \frac{5}{\sqrt{5} \cdot \sqrt{10}} = \frac{\sqrt{5} \cdot \sqrt{5}}{\sqrt{5} \cdot \sqrt{10}} = \frac{\sqrt{5}}{\sqrt{10}} = \frac{\sqrt{1}}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\therefore x + y = \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) = \frac{\pi}{4}$$

Sol.121. (b)

$$A = \sin^2 \theta + \cos^4 \theta$$

$$= \sin^2 \theta + (1 - \sin^2 \theta)^2$$

$$= 1 + \sin^4 \theta - \sin^2 \theta$$

$$= 1 - \sin^2 \theta (1 - \sin^2 \theta)$$

$$= 1 - \sin^2 \theta \cdot \cos^2 \theta$$

$$= \frac{4 - 4 \sin^2 \theta \cdot \cos^2 \theta}{4} = \frac{4 - \sin^2(2\theta)}{4}$$

$$\alpha \therefore A = \frac{4-0}{4} \text{ or } \frac{4-0}{4} \Rightarrow \frac{3}{4} \leq A \leq 1$$

Sol.122. (b)

$$[(2 \cos \theta + 1)(2 \cos \theta - 1)]^{10}$$

$$[(2 \cos 2\theta - 1)]^{10} [(2 \cos 4\theta - 1)]^{10}$$

$$\text{When, } \theta = \frac{\pi}{8} \text{ (given)}$$

$$= \left(4 \cos^2 \frac{\pi}{8} - 1 \right)^{10} \left(2 \times \frac{1}{\sqrt{2}} - 1 \right)^{10} (-1)^{10}$$

$$= \left(4 \times \frac{2+\sqrt{2}}{4} - 1 \right)^{10} (\sqrt{2} - 1)^{10}$$

$$= [(\sqrt{2} + 1)(\sqrt{2} - 1)]^{10} = 1^{10} = 1$$

Sol.123. (b)

Given equation:

$$\operatorname{cosec} x + \cot x = \sqrt{3}$$

$$\operatorname{cosec} x + \cot x = \frac{1}{\sqrt{3}}$$

$$\Rightarrow 2 \operatorname{cosec} x = \sqrt{3} + \frac{1}{\sqrt{3}} = \frac{4}{\sqrt{3}}$$

$$\Rightarrow \operatorname{cosec} x = \frac{2}{\sqrt{3}}$$

$$\Rightarrow \sin x = \frac{\sqrt{3}}{2}$$

$$\text{Possible values of } x = \frac{\pi}{3}, \frac{2\pi}{3}$$

$\therefore$ Option (b) is correct.

Sol.124. (a)

Checking through options

$$300^\circ - 60^\circ$$

$$\text{So, } 3[3 - \tan^2(-60^\circ) - \cot(-60^\circ)]^2$$

$$= \left[3 - 3 + \frac{1}{\sqrt{3}} \right]^2 = 3 \times \frac{1}{3} = 1$$

Sol.125. (a)

$\sec(\theta - \alpha)$, $\sec \theta$, $\sec(\theta + \alpha)$ are in AP

$$\Rightarrow 2 \sec \theta = \sec(\theta - \alpha) + \sec(\theta + \alpha)$$

$$\Rightarrow \frac{2}{\cos \theta} = \frac{1}{\cos(\theta - \alpha)} + \frac{1}{\cos(\theta + \alpha)}$$

$$\Rightarrow \frac{2}{\cos \theta} = \frac{\cos(\theta - \alpha) + \cos(\theta + \alpha)}{\cos(\theta - \alpha) \cos(\theta + \alpha)}$$

$$\Rightarrow \frac{2}{\cos \theta} = \frac{2 \cos \theta \cos \alpha}{\cos(\theta - \alpha) \cos(\theta + \alpha)}$$

$$\Rightarrow 2 \cos(\theta - \alpha) \cos(\theta + \alpha) = 2 \cos^2 \theta \cos \alpha$$

$$\Rightarrow \cos 2\theta + \cos 2\alpha = 2 \cos^2 \theta \cos \alpha$$

$$\Rightarrow 2 \cos^2 \theta + 2 \cos^2 \alpha - 2 = 2 \cos^2 \theta \cos \alpha$$

$$\Rightarrow \cos^2 \theta + \cos^2 \alpha - 1 = \cos^2 \theta \cos \alpha$$

$$\Rightarrow \cos^2 \theta (1 - \cos \alpha) + \cos^2 \alpha - 1 = 0$$

$$\Rightarrow \cos^2 \theta (1 - \cos \alpha) - (1 - \cos \alpha)(1 + \cos \alpha) = 0$$

$$\Rightarrow \cos^2 \theta - (1 + \cos \alpha) = 0$$

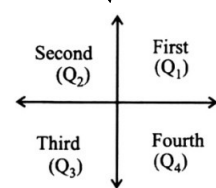
$$\Rightarrow -\sin^2 \theta - \cos \alpha = 0$$

$$\Rightarrow \sin^2 \theta + \cos \alpha = 0$$

Sol.126. (c)

Sol.127. (c)

$$\sin \theta = \frac{1}{2}, \tan \theta = \frac{1}{\sqrt{3}}$$



$\sin \theta$ is negative, $\tan \theta$ is positive lies in third quadrant.

Sol.128. (c)

$$\sin \theta + \cos \theta = \sqrt{2} \dots (1)$$

$$\text{Let } \cos \theta - \sin \theta = P$$

$$(1)^2 + (2)^2 \Rightarrow \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta + \cos^2 \theta + \sin^2 \theta$$

$$- 2 \sin \theta \cos \theta = 2 \cos^2 \theta + P^2$$

$$\Rightarrow 2 = 2 \cos^2 \theta + P^2$$

$$\Rightarrow P^2 = 2(1 - \cos^2 \theta) = 2 \sin^2 \theta$$

$$\Rightarrow P = \sqrt{2} \sin \theta$$

Sol.129. (a)

$$\text{arc length} = r \times \theta = 22 \times \frac{\pi}{3} = \frac{484}{21}$$

Sol.130. (a)

$$(1 + \tan \alpha \tan \alpha)^2$$

$$(\tan \alpha - \tan \beta)^2 - \sec^2 \alpha \sec^2 \beta$$

$$= 1 + \tan^2 \alpha - \tan^2 \beta + 2 \tan \alpha \tan \beta + \tan^2 \beta - \sec^2 \beta$$

$$= 2 \tan \alpha \tan \beta - \sec^2 \beta$$

$$= 2 + \tan^2 \alpha \tan^2 \beta + \tan^2 \beta - \sec^2 \beta$$

$$= (1 + \tan^2 \alpha)(1 + \tan^2 \beta) - \sec^2 \alpha \sec^2 \beta$$

$$= \sec^2 \alpha \sec^2 \beta - \sec^2 \alpha \sec^2 \beta = 0$$

Sol.131. (b)

$$p = \csc \theta - \cot \theta$$

$$q = (\csc \theta + \cot \theta)^{-1}$$

$$\Rightarrow \frac{1}{q} = \csc \theta + \cot \theta$$

$$\text{We know, } \csc^2 \theta - \cot^2 \theta = 1$$

$$\Rightarrow \left(\frac{1}{q}\right)(p) = 1$$

$$\Rightarrow p = q$$

Sol.132. (a)

$$\frac{\sin 34^\circ \cos 236^\circ - \sin 56^\circ \sin 124^\circ}{\cos 28^\circ \cos 88^\circ + \cos 178^\circ \sin 208^\circ} = \frac{\sin 34^\circ (-\cos 56^\circ) - \sin 56^\circ \cos 34^\circ}{\cos 28^\circ \sin 2^\circ + \cos 2^\circ \sin 28^\circ}$$

$$= \frac{-\sin 34^\circ \cos 56^\circ - \sin 56^\circ \cos 34^\circ}{\sin(28^\circ + 2^\circ)} = \frac{-\sin(34^\circ + 56^\circ)}{\sin 30^\circ} = \frac{-\sin 90^\circ}{\sin 30^\circ} = \frac{-1}{\frac{1}{2}} = -2$$

Sol.133. (c)

$$\tan 54^\circ = \tan(45^\circ + 9^\circ)$$

$$= \frac{\tan 45^\circ + \tan 9^\circ}{1 - \tan 45^\circ \cdot \tan 9^\circ} = \frac{1 + \tan 9^\circ}{1 - \tan 9^\circ}$$

$$1 + \frac{\sin 9^\circ}{\cos 9^\circ} = \frac{\cos 9^\circ + \sin 9^\circ}{\cos 9^\circ - \sin 9^\circ}$$

$$1 - \frac{\sin 9^\circ}{\cos 9^\circ} = \frac{\cos 9^\circ - \sin 9^\circ}{\cos 9^\circ + \sin 9^\circ}$$

Sol.134. (c)

$$p = x \cos \theta - y \sin \theta$$

$$q = x \sin \theta - y \cos \theta$$

$$\text{Given, } p^2 + 4pq + q^2 = Ax^2 + By^2$$

$$\text{Let us take } \theta = \frac{\pi}{4}$$

$$p = x \cos \frac{\pi}{4} - y \cos \frac{\pi}{4} = \frac{x-y}{\sqrt{2}}$$

$$q = x \sin \frac{\pi}{4} - y \cos \frac{\pi}{4} = \frac{x+y}{\sqrt{2}}$$

$$pq = \frac{x^2 - y^2}{2} \Rightarrow 2pq = x^2 - y^2$$

$$\Rightarrow 4pq = 2x^2 - 2y^2 \quad \dots\dots\dots(1)$$

Now

$$p^2 + q^2 = x^2 \cos^2 \theta + 2xy \sin \theta \cos \theta + y^2 \sin^2 \theta = x^2 + y^2$$

From (1), (2),

$$p^2 + q^2 + 4pq = x^2 + y^2 + 2x^2 - 2y^2 \quad \dots\dots\dots(2)$$

$$= 3x^2 - y^2$$

Comparing this with the given form, we get

$$\theta = \frac{\pi}{4}, A = 3, B = -1$$

Sol.135. (b)

By above solution

Sol.136. (a)

By above solution

Sol.137. (a)

$$\text{Given, } \cos(\theta - \alpha) = a \Rightarrow \sin(\theta - \alpha) = \sqrt{1 - a^2}$$

$$\cos(\theta - \beta) = b \Rightarrow \sin(\theta - \beta) = \sqrt{1 - b^2}$$

$$\therefore \cos(\alpha - \beta) = \cos(\theta - \beta) - (\theta - \alpha)$$

$$= (b)(a) + \sqrt{1 - b^2} \sqrt{1 - a^2}$$

$$= ab + \sqrt{1 - a^2} \sqrt{1 - b^2}$$

Sol.138. (a)

$$\sin^2(\alpha - \beta) + 2ab \cos(\alpha - \beta) = 1 - \cos^2(\alpha - \beta) + 2ab \cos(\alpha - \beta)$$

$$(\alpha - \beta) = 1 - \cos(\alpha - \beta) [\cos(\alpha - \beta) - 2ab]$$

$$= 1 - (ab + \sqrt{1 - a^2} \sqrt{1 - b^2})$$

$$= 1 - (\sqrt{1 - a^2} \sqrt{1 - b^2} + ab) (\sqrt{1 - a^2} \sqrt{1 - b^2} - ab)$$

$$= 1 - \left[(\sqrt{1 - a^2} \sqrt{1 - b^2})^2 - (ab)^2 \right]$$

$$= 1 - [(1 - a^2)(1 - b^2) - a^2 b^2]$$

$$= 1 - (1 - b^2 - a^2 + a^2 b^2 - a^2 b^2)$$

$$= 1 - 1 + b^2 + a^2 = a^2 + b^2$$

Sol.139. (c)

$$\sin \alpha + \cos \alpha = p$$

$$\Rightarrow (\sin \alpha + \cos \alpha)^2 = p^2$$

$$\Rightarrow \sin^2 \alpha + \cos^2 \alpha + 2 \sin \alpha \cos \alpha = p^2$$

$$\Rightarrow 1 + \sin 2\alpha = p^2$$

$$\Rightarrow \sin 2\alpha = p^2 - 1$$

$$\cos^2 2\alpha = 1 - \sin^2 2\alpha = 1 - (p^2 - 1)^2$$

$$= 1 - (p^4 + 1 - 2p^2) = -p^4 + 2p^2$$

$$= p^2(2 - p^2)$$

Sol.140. (c)

$$\tan \theta = \frac{1}{2} \quad \text{and} \quad \tan \phi = \frac{1}{3}$$

$$\tan(\theta + \phi) = \frac{\tan \theta + \tan \phi}{1 - \tan \theta \tan \phi}$$

$$= \frac{\frac{1}{2} + \frac{1}{3}}{1 - \left(\frac{1}{2}\right)\left(\frac{1}{3}\right)} = \frac{\frac{5}{6}}{\frac{5}{6}} = 1$$

$$\therefore \theta + \phi - \tan^{-1}(1) = \frac{\pi}{4}$$

Sol.141. (b)

$$\cos A = \frac{3}{4}$$

$$\sin\left(\frac{A}{2}\right) \cdot \sin\left(\frac{3A}{2}\right) = \frac{1}{2} \left[2 \sin \frac{A}{2} \sin \frac{3A}{2} \right]$$

$$= \frac{1}{2} \left[\cos\left(\frac{3A}{2} - \frac{A}{2}\right) - \cos\left(\frac{3A}{2} + \frac{A}{2}\right) \right]$$

$$= \frac{1}{2} [\cos A - \cos 2A] = \frac{1}{2} [\cos A - (2 \cos^2 A - 1)]$$

$$= \frac{1}{2} [\cos A - 2 \cos^2 A + 1]$$

$$= \frac{1}{2} \left[\left(\frac{3}{4}\right) - 2\left(\frac{3}{4}\right)^2 + 1 \right]$$

$$= \frac{1}{2} \left[\frac{3}{4} - \frac{18}{16} + 1 \right] = \frac{1}{2} \left[\frac{12 - 18 + 16}{16} \right]$$

$$= \frac{1}{2} \left[\frac{10}{16} \right] = \frac{5}{16}$$

Sol.142. (b)

$$\tan 75^\circ + \cot 75^\circ = 2 + \sqrt{3} + 2 - \sqrt{3} = 4$$

Sol.143. (b)

$$\cos 46^\circ \cos 47^\circ \cos 48^\circ \cos 49^\circ \dots \cos 135^\circ$$

We know, $\cos 90^\circ = 0$

$\therefore$ Given expression has $\cos 90^\circ$ and so its value is 0.

Sol.144. (b)

$$\sin 2\theta = \cos 3\theta$$

$$\Rightarrow \sin 2\theta = \sin(90^\circ - 3\theta)$$

$$2\theta = 90^\circ - 3\theta$$

$$\therefore \sin \theta = \sin 18^\circ = \frac{\sqrt{5} - 1}{4}$$

$$\Rightarrow 50 = 90^\circ$$

$$\Rightarrow \theta = 18^\circ$$

Sol.145. (d)

minimum value of

$$a^2 \sec^2 x + b^2 \csc^2 x \text{ is } (a + b)^2$$

so answer is $(5 + 6)^2 = 121$

Sol.146. (a)

minimum value of

$$a^2 \sec^2 x + b^2 \csc^2 x \text{ is } (a + b)^2$$

Sol.147. (b)

$\csc \theta = 29/21$, where $0 < \theta < 90^\circ$,

here perpendicular = 21, Hypotenuse = 29,

then by Pythagoras Base = 20

$$4 \sec \theta + 4 \tan \theta = 4 \left(\frac{29}{20} + \frac{21}{20} \right) = 10$$

Sol.148. (a)

for positive values of x

$$x + \frac{1}{x} \geq 2$$

and for negative values of x

$$x + \frac{1}{x} \leq -2$$

Sol.149. (c)

apply cosine formula in triangle

$$\cos \theta = \frac{1^2 + 1^2 - x^2}{2}$$

$$2 \cos \theta = 2 - x^2$$

$$x^2 = 2 - 2 \cos \theta$$

$$x^2 = 2.2 \sin^2 \left(\frac{\theta}{2} \right)$$

$$x = 2 \sin \left(\frac{\theta}{2} \right)$$

Sol.150. (a)

$$\tan A - \tan B = x, \frac{1}{\tan B} - \frac{1}{\tan A} = y$$

$$\tan A - \tan B = x, \frac{\tan A - \tan B}{\tan A \tan B} = y$$

$$\frac{x}{\tan A \tan B} = y \Rightarrow \tan A \tan B = \frac{x}{y}$$

$$\cot(A - B) = \frac{1}{\tan(A - B)} = \frac{1 + \tan A \tan B}{\tan A - \tan B}$$

$$= \frac{1 + \frac{x}{y}}{\frac{x}{y}} = \frac{x + y}{xy} = \frac{1}{x} + \frac{1}{y}$$

Sol.151. (a)

$$\sin(\alpha + \beta) + \sin(\alpha - \beta) - 2 \sin \alpha \cos \beta$$

$$= 2 \sin \alpha \cos \beta - 2 \sin \alpha \cos \beta = 0$$

Sol.152. (c)

$$\tan A = (1/2), \tan B = (1/3)$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B} = \frac{\frac{1}{2} - \frac{1}{3}}{1 + \frac{1}{2} \cdot \frac{1}{3}} = \frac{\frac{1}{6}}{\frac{5}{6}} = \frac{1}{5}$$

Sol.153. (c)

$$\cos 80^\circ + \cos 40^\circ - \cos 20^\circ$$

$$= 2 \cos 60^\circ \cos 20^\circ - \cos 20^\circ = 0$$

Sol.154. (d)

$$\frac{1}{\tan(A/2)} - \tan(A/2) = 2 \frac{1 - \tan^2(A/2)}{2 \tan(A/2)}$$

$$-3/2 \leq k \leq 1/2$$

$$k = -1, 0$$

Sol.185. (c)

$$\cot 6x = \cot(4x + 2x) = \frac{\cot 4x \cot 2x - 1}{\cot 4x + \cot 2x}$$

$$\cot 6x \cot 4x + \cot 6x \cot 2x = \cot 4x \cot 2x - 1$$

$$\cot 4x \cot 2x - \cot 6x \cot 4x - \cot 6x \cot 2x = 1$$

Sol.186. (d)

$$\tan x = -3/4$$

than $\sin x = 3/5$ in 2nd quad

and $\cos x = -4/5$ in 2nd quad

$$\sin x \cdot \cos x = -12/25$$

Sol.187. (c)

$$-\operatorname{cosec} 30 \cdot \cos 60 = -4$$

Sol.188. (c)

$$4 \sin^2 x = 3$$

$$\sin^2 x = 3/4$$

$$\sin x = \frac{\sqrt{3}}{2}$$

$$x = 60^\circ \text{ or } 120^\circ$$

$$\tan 3x = \tan 180^\circ$$

$$= 0$$

Sol.189. (c)

$$\sec x \cdot \operatorname{cosec} x = p$$

$$\Rightarrow \frac{2}{2 \sin x \cos x} = p$$

$$\frac{1}{\sin 2x} = \frac{p}{2}$$

$$\operatorname{cosec} 2x = p/2$$

We know that

$$\operatorname{cosec} \theta \geq 1$$

$$\text{so, } p/2 \geq 1$$

$$p \geq 2$$

Minimum value of p is 2

Sol.190. (c)

$\sin \theta$ and $\cos \theta$ are roots of equation

$$ax^2 + bx + c = 0$$

$$\sin \theta + \cos \theta = -\frac{b}{a}$$

$$\sin \theta \cdot \cos \theta = \frac{c}{a}$$

$$(\sin \theta + \cos \theta)^2 = \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta$$

$$\frac{b^2}{a^2} = 1 + 2 \frac{c}{a}$$

$$b^2 = a^2 + 2ac$$

$$a^2 - b^2 + 2ac = 0$$

Sol.191. (d)

$$\sin(A+B) = 1 \Rightarrow A+B = 90^\circ$$

$$\sin(A-B) = \frac{1}{2} \Rightarrow A-B = 30^\circ$$

$$A = 60^\circ, B = 30^\circ$$

$$\frac{\tan A}{\tan B} = \frac{\tan 60^\circ}{\tan 30^\circ} = \frac{\sqrt{3}}{1/\sqrt{3}} = 3$$

Sol.192. (a)

$$\sin x + \sin y = \cos x + \cos y$$

$$\tan \left(\frac{x+y}{2} \right) = 1$$

Sol.193. (c)

$$\tan \theta = -\frac{5}{12}$$

θ may lie in 2nd or 4th quadrant

Perpendicular = 5, Base = 12 (given)

Hypotheneous = 13 (by Pythagoras)

$$\sin \theta = \frac{5}{13} \text{ (if } \theta \text{ in 2nd quadrant)}$$

$$\sin \theta = -\frac{5}{13} \text{ (if } \theta \text{ in 4th quadrant)}$$

Sol.194. (b)

$$\cos^4 \frac{7\pi}{8} + \cos^4 \frac{5\pi}{8}$$

$$\cos^4 \left(\pi - \frac{\pi}{8} \right) + \cos^4 \left(\frac{\pi}{2} + \frac{\pi}{8} \right)$$

$$\cos^4 \frac{\pi}{8} + \sin^4 \frac{\pi}{8}$$

$$\left(\sin^2 \frac{\pi}{8} + \cos^2 \frac{\pi}{8} \right)^2 - 2 \sin^2 \frac{\pi}{8} \cdot \cos^2 \frac{\pi}{8}$$

$$1 - \frac{1}{2} \sin^2 \frac{\pi}{4} = 1 - \frac{1}{2} \left(\frac{1}{2} \right) = \frac{3}{4}$$

Sol.195. (a)

$$\sin^2 \left(\frac{\pi}{4} + \theta \right) - \sin^2 \left(\frac{\pi}{4} - \theta \right)$$

$$[\sin^2 A - \sin^2 B = \sin(A+B) \sin(A-B)]$$

$$\Rightarrow \sin \left(\frac{\pi}{4} + \theta + \frac{\pi}{4} - \theta \right) \cdot \sin \left(\frac{\pi}{4} + \theta - \frac{\pi}{4} - \theta \right)$$

$$= \sin \frac{\pi}{2} \cdot \sin 2\theta = \sin 2\theta$$

Sol.196. (b)

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

but given that $\tan \alpha + \tan \beta = 1 - \tan \alpha \tan \beta$

$$\text{so } \tan(\alpha + \beta) = 1$$

$$\Rightarrow \alpha + \beta = \frac{\pi}{4}$$

Sol.197. (b)

$$(1 + \tan \theta)(1 + \tan 9\theta) = 2$$

$$1 + \tan \theta + \tan 9\theta + \tan \theta \tan 9\theta = 2$$

$$\tan \theta + \tan 9\theta = 1 - \tan \theta \tan 9\theta$$

$$\therefore \tan 10\theta = \tan(9\theta + \theta)$$

$$= \frac{\tan 9\theta + \tan \theta}{1 - \tan 9\theta \tan \theta}$$

$$= 1$$

Sol.198. (b)

$$\sin 0^\circ + \sin 10^\circ + \sin 20^\circ + \dots + \sin 360^\circ$$

$$= \sin 0^\circ + \sin 10^\circ + \dots + \sin 170^\circ$$

$$+ \sin 180^\circ$$

$$+ \sin 190^\circ + \sin 200^\circ + \dots + \sin 350^\circ$$

$$+ \sin 360^\circ +$$

$$\therefore 0^\circ \text{ to } 180^\circ \Rightarrow \sin \theta = \text{positive}$$

$$180^\circ + 360^\circ \sin \theta = \text{Negative}$$

So sum will be zero

Sol.199. (b)

$$\cot^2 15 + \tan^2 15 = (2 + \sqrt{3})^2 + (2 - \sqrt{3})^2 = 14$$

Sol.200. (c)

$$\alpha + \beta = \frac{\pi}{4}$$

$$2\alpha + 2\beta = \frac{\pi}{2}$$

$$\tan(2\alpha + 2\beta) = \tan \frac{\pi}{2}$$

$$\frac{\tan 2\alpha + \tan 2\beta}{1 - \tan 2\alpha \tan 2\beta} = \infty$$

$$= \tan 2\alpha \cdot \tan 2\beta = 1$$

$$\frac{2 \tan \alpha}{1 - \tan^2 \alpha} \times \frac{2 \tan \beta}{1 - \tan^2 \beta} = 1$$

$$\frac{1 - \tan^2 \alpha}{1} \tan 2\beta = 1$$

$$1 - \left(\frac{1}{2} \right)^2$$

$$\tan 2\beta = \frac{3}{4}$$

$$\tan 2\beta = \frac{3}{4}$$

$$\tan 2\beta = \frac{3}{4}$$

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$$\tan 2\beta = \frac{3}{4}$$

$$\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = 1$$

$$\cos 2\theta = 1$$

Sol.202. (c)

$$\sin 2\theta = \cos 3\theta$$

$$\sin 2\theta = \sin(90^\circ - 3\theta)$$

$$2\theta + 3\theta = 90^\circ$$

$$\theta = 18^\circ$$

$$1 + 4 \sin 18 = 1 + 4 \left(\frac{\sqrt{5} - 1}{4} \right) = \sqrt{5}$$

Sol.203. (b)

$$\operatorname{cosec} \left(-\frac{73\pi}{3} \right) = -\operatorname{cosec} \left(\frac{73\pi}{3} \right) = -\operatorname{cosec} \left(24\pi + \frac{\pi}{3} \right)$$

$$= -\operatorname{cosec} \left(\frac{\pi}{3} \right) = -\frac{2}{\sqrt{3}}$$

Sol.204. (a)

$$\cos \left(\frac{5\pi}{17} \right) + \cos \left(\frac{7\pi}{17} \right) + 2 \cos \left(\frac{11\pi}{17} \right) \cos \left(\frac{\pi}{17} \right)$$

$$2 \cos \left(\frac{6\pi}{17} \right) \cos \left(\frac{\pi}{17} \right) + 2 \cos \left(\frac{11\pi}{17} \right) \cos \left(\frac{\pi}{17} \right)$$

$$2 \cos \left(\frac{\pi}{17} \right) \left[\cos \left(\frac{6\pi}{17} \right) + \cos \left(\frac{11\pi}{17} \right) \right]$$

$$2 \cos \left(\frac{\pi}{17} \right) \left[2 \cos \left(\frac{\pi}{2} \right) \cos \left(\frac{5\pi}{34} \right) \right] = 0$$

Sol.205. (b)

$$\tan \theta = \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$$

$$\tan \frac{3\pi}{8} = \frac{1 - \cos \frac{3\pi}{4}}{1 + \cos \frac{3\pi}{4}} = \frac{1 + \frac{1}{\sqrt{2}}}{1 - \frac{1}{\sqrt{2}}}$$

$$\Rightarrow \frac{\sqrt{2} + 1}{\sqrt{2} - 1} = \frac{\sqrt{2} + 1}{\sqrt{2} - 1} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1} = \sqrt{2} + 1$$

Sol.206. (c)

$$\cos 36^\circ - \sin 18^\circ = \frac{\sqrt{5} + 1}{4} - \frac{\sqrt{5} - 1}{4} = \frac{1}{2}$$

Sol.207. (b)

$$\sec x = \frac{25}{24}$$

$$\sec x = \frac{25}{24}$$

In forth quadrant $\tan x$ and $\sin x$ is negative

$$\tan x + \sin x = -\frac{7}{24} - \frac{7}{25} = -\frac{343}{600}$$

Sol.208. (a)

$$\tan^2 165^\circ + \cot^2 165^\circ = \tan^2 15^\circ + \cot^2 15^\circ$$

$$= (2 - \sqrt{3})^2 + (2 + \sqrt{3})^2 = 7 + 7 = 14$$

Sol.209. (a)

$$\sin \left(2n\pi + \frac{5\pi}{6} \right) \sin \left(2n\pi - \frac{5\pi}{6} \right)$$

$$\Rightarrow -\sin \left(\frac{5\pi}{6} \right) \sin \left(\frac{5\pi}{6} \right) = -\frac{1}{2} \cdot \frac{1}{2} = -\frac{1}{4}$$

Sol.210. (d)

$$1 + 2(\sin^2 x - \cos^2 x) = 1$$

$$\cos^2 x - \sin^2 x = \frac{1}{2}$$

$$\cos 2x = \frac{1}{2}$$

$$2x = 60^\circ \text{ or } 300^\circ \text{ or } 420^\circ \text{ or } 660^\circ$$

$$x = 30^\circ \text{ or } 150^\circ \text{ or } 210^\circ \text{ or } 330^\circ$$

Sol.211. (b)

$$3 \sin x - 3 \cos x + 4 \cos^3 x - 4 \sin^3 x$$

$$3 \sin x - 4 \sin^3 x - 3 \cos x + 4 \cos^3 x$$

$$3 \sin 3x - \cos 3x$$

Maximum value of $a \sin x + b \cos x$ is $\sqrt{a^2 + b^2}$

Sol.212. (c)

diameter of a circle inscribed in a regular polygon

$$d = a \cot\left(\frac{\pi}{n}\right) = \cot\left(\frac{\pi}{12}\right) = \cot 15^\circ = 2 + \sqrt{3}$$

Sol.213.

$$\sin x + \cos x + \tan x + \sec x + \csc x = 7$$

$$\sin x + \cos x + \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} + \frac{1}{\cos x} + \frac{1}{\sin x} = 7$$

$$\Rightarrow \sin x + \cos x + \frac{\sin^2 x + \cos^2 x}{\sin x \cos x} + \frac{\sin x + \cos x}{\sin x \cos x} = 7$$

$$\Rightarrow \sin x + \cos x + \frac{1}{\sin x \cos x} + \frac{\sin x + \cos x}{\sin x \cos x} = 7$$

$$\Rightarrow \sqrt{(\sin x + \cos x)^2} + \frac{1}{\sin x \cos x} + \frac{\sqrt{(\sin x + \cos x)^2}}{\sin x \cos x} = 7$$

$$\Rightarrow \sqrt{1 + \sin 2x} + \frac{2}{2 \sin x \cos x} + \frac{2\sqrt{1 + \sin 2x}}{2 \sin x \cos x} = 7$$

$$\Rightarrow \sqrt{1 + \sin 2x} + \frac{2}{\sin 2x} + \frac{2\sqrt{1 + \sin 2x}}{\sin 2x} = 7$$

$$\Rightarrow \sin 2x \sqrt{1 + \sin 2x} + 2 + 2\sqrt{1 + \sin 2x} = 7 \sin 2x$$

$$\Rightarrow \sqrt{1 + \sin 2x}(\sin 2x + 2) = 7 \sin 2x - 2$$

Squaring both side

$$\Rightarrow (1 + \sin 2x)(\sin^2 2x + 4 + 4 \sin 2x) = 49 \sin^2 2x - 28 \sin 2x + 4$$

$$\Rightarrow (\sin^2 2x + 4 + 4 \sin 2x) + (\sin 2x)(\sin^2 2x + 4 + 4 \sin 2x)$$

$$= 49 \sin^2 2x - 28 \sin 2x$$

Divide by $\sin 2x$ both sides

$$\Rightarrow (\sin 2x + 4) + (\sin^2 2x + 4 + 4 \sin 2x) = 49 \sin 2x - 28$$

$$\Rightarrow \sin^2 2x - 44 \sin 2x + 36 = 0$$

$$\sin^2 2x - 44 \sin 2x + 36 = 0$$

$$\Rightarrow \sin 2x = \frac{44 \pm \sqrt{44^2 - 4(36)}}{2} = \frac{44 \pm \sqrt{1936 - 144}}{2}$$

$$\Rightarrow \frac{44 \pm \sqrt{1792}}{2} = 22 \pm \sqrt{448} = 22 - 8\sqrt{7} = a - b\sqrt{c}$$

$$a - b + 2c = 22 - 8 + 14 = 28$$

Sol.215. (a)

$$\sin \beta = \sqrt{\sin \alpha \cos \alpha}$$

$$\cos 2\beta = 1 - 2 \sin^2 \beta = 1 - 2 \sin \alpha \cos \alpha$$

$$= (\sin^2 \alpha + \cos^2 \alpha) - 2 \sin \alpha \cos \alpha$$

$$= (\cos \alpha - \sin \alpha)^2$$

Sol.216. (b)

$$\tan \gamma = \frac{\sin \alpha + \cos \alpha}{2}$$

$$\sec 2\gamma = \frac{1}{\cos 2\gamma} = \frac{1 + \tan^2 \gamma}{1 - \tan^2 \gamma}$$

$$= \frac{1 + \left(\frac{\sin \alpha + \cos \alpha}{2}\right)^2}{1 - \left(\frac{\sin \alpha + \cos \alpha}{2}\right)^2}$$

$$= \frac{4 + \sin^2 \alpha + \cos^2 \alpha + 2 \sin \alpha \cos \alpha}{4 - \sin^2 \alpha - \cos^2 \alpha - 2 \sin \alpha \cos \alpha}$$

$$= \frac{5 + \sin 2\alpha}{3 - \sin 2\alpha}$$

Sol.217. (c)

$$\tan(\pi \cos \theta) = \cot(\pi \sin \theta)$$

$$\tan(\pi \cos \theta) = \tan\left(\frac{\pi}{2} - \pi \sin \theta\right)$$

$$\pi \cos \theta = \frac{\pi}{2} - \pi \sin \theta$$

$$\sin \theta + \cos \theta = \frac{1}{2}$$

$$\frac{1}{\sqrt{2}} \sin \theta + \frac{1}{\sqrt{2}} \cos \theta = \frac{1}{2\sqrt{2}}$$

$$\sin \frac{\pi}{4} \cos \theta + \cos \frac{\pi}{4} \sin \theta = \frac{1}{2\sqrt{2}}$$

$$\sin\left(\frac{\pi}{4} + \theta\right) = \frac{1}{2\sqrt{2}}$$

$$\sin^2\left(\frac{\pi}{4} + \theta\right) = \frac{1}{8}$$

$$8 \sin^2\left(\frac{\pi}{4} + \theta\right) = 1$$

Sol.218. (c)

$$\cos(\alpha + 2\beta) = \cos \alpha \cos 2\beta - \sin \alpha \sin 2\beta$$

$$= \cos \alpha (1 - 2 \sin^2 \beta) - 2 \sin \alpha \sin \beta \cos \beta$$

$$\sin \beta = \frac{1}{\sqrt{10}}, \cos \beta = \frac{3}{\sqrt{10}}, \sin \alpha = \frac{1}{\sqrt{50}}, \cos \alpha = \frac{7}{\sqrt{50}}$$

$$= \frac{7}{\sqrt{50}} \left(1 - 2 \frac{1}{10}\right) - 2 \frac{1}{\sqrt{50}} \cdot \frac{1}{\sqrt{10}} \cdot \frac{3}{\sqrt{10}}$$

$$= \frac{7}{\sqrt{50}} \left(\frac{8}{10}\right) - 2 \frac{1}{\sqrt{50}} \cdot \frac{3}{10}$$

$$= \frac{56 - 6}{10\sqrt{50}} = \frac{5}{5\sqrt{2}} = \frac{1}{\sqrt{2}}$$

Sol.219. (c)

$$p + q = \cos\left(\frac{\pi}{5}\right) \cos\left(\frac{2\pi}{5}\right) + \cos\left(\frac{4\pi}{5}\right) \cos\left(\frac{8\pi}{5}\right)$$

$$= \cos\left(\frac{\pi}{5}\right) \cos\left(\frac{2\pi}{5}\right) + \cos\left(\pi - \frac{\pi}{5}\right) \cos\left(2\pi - \frac{2\pi}{5}\right)$$

$$= \cos\left(\frac{\pi}{5}\right) \cos\left(\frac{2\pi}{5}\right) - \cos\left(\frac{\pi}{5}\right) \cos\left(\frac{2\pi}{5}\right) = 0$$

Sol.220. (a)

$$pq = \cos\left(\frac{\pi}{5}\right) \cos\left(\frac{2\pi}{5}\right) \cos\left(\frac{4\pi}{5}\right) \cos\left(\frac{8\pi}{5}\right)$$

$$pq = -\cos\left(\frac{\pi}{5}\right) \cos\left(\frac{2\pi}{5}\right) \cos\left(\frac{\pi}{5}\right) \cos\left(\frac{2\pi}{5}\right)$$

$$pq = -\left[\cos\left(\frac{\pi}{5}\right) \cos\left(\frac{2\pi}{5}\right)\right]^2 = -\cos^2 36^\circ \cos^2 72^\circ$$

$$pq = -\cos^2 36^\circ \sin^2 18^\circ$$

$$pq = -\left[\frac{\sqrt{5}+1}{4}\right]^2 \left[\frac{\sqrt{5}-1}{4}\right]^2$$

$$pq = -\left[\frac{5-1}{16}\right]^2 = -\frac{1}{16}$$

Sol.221. (d)

$$p = \frac{1}{3} - \frac{\tan 3x}{\tan x}$$

$$q = 1 - 3 \tan^2 x$$

$$p = \frac{1}{3} - \frac{(3 \tan x - \tan^3 x)}{(1 - 3 \tan^2 x)} = \frac{1}{3} - \frac{(3 - \tan^2 x)}{(1 - 3 \tan^2 x)}$$

$$= \frac{(1 - 3 \tan^2 x) - 3(3 - \tan^2 x)}{3(1 - 3 \tan^2 x)} = \frac{-8}{3q}$$

$$pq = -\frac{8}{3}$$

Sol.222. (c)

$$p = \frac{-8}{3(1 - 3 \tan^2 x)}$$

$$\frac{1}{p} = \frac{3(1 - 3 \tan^2 x)}{-8} = 0$$

$$1 - 3 \tan^2 x = 0$$

$$\tan x = \pm 1/\sqrt{3}$$

$$x = 30^\circ \text{ and } 150^\circ$$

Sol.223.

$$\sin x + \sin y = \sqrt{3}(\cos y - \cos x);$$

$$2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) = \sqrt{3} \cdot 2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right)$$

$$\cos\left(\frac{x-y}{2}\right) = \sqrt{3} \sin\left(\frac{x-y}{2}\right)$$

$$\tan\left(\frac{x-y}{2}\right) = \frac{1}{\sqrt{3}}$$

$$\frac{x-y}{2} = 30^\circ$$

$$x - y = 60^\circ \dots\dots\dots(i)$$

$$\text{and given that } x + y = 90^\circ \dots\dots\dots(ii)$$

$$x = 75^\circ \text{ and } y = 15^\circ$$

$$\sin 3x + \sin 3y = 2 \cos \frac{3x+3y}{2} \sin \frac{3x-3y}{2}$$

$$= 2 \cos \frac{3x+3y}{2} \sin \frac{3x-3y}{2} = 2 \cos 135^\circ \cos 90^\circ = 0$$

Sol.224. (b)

$$\cos^3 x + \cos^3 y$$

$$= (\cos x + \cos y)(\cos^2 x + \cos^2 y - \cos x \cos y)$$

$$\text{By using values of } x \text{ and } y \text{ by above question}$$

$$= (\cos 75^\circ + \cos 15^\circ)(\cos^2 75^\circ + \cos^2 15^\circ - \cos 75^\circ \cos 15^\circ)$$

$$= \left(\frac{\sqrt{3}-1}{2\sqrt{2}} + \frac{\sqrt{3}+1}{2\sqrt{2}}\right) \left(1 - \frac{\sqrt{3}-1}{2\sqrt{2}} \cdot \frac{\sqrt{3}+1}{2\sqrt{2}}\right)$$

$$= \left(\frac{2\sqrt{3}}{2\sqrt{2}}\right) \left(1 - \frac{3-1}{8}\right) = \left(\frac{\sqrt{3}}{\sqrt{2}}\right) \left(\frac{3}{4}\right) = \frac{3\sqrt{6}}{8}$$

Sol.225. (b)

$$\sin 10^\circ \cdot \sin 50^\circ + \sin 50^\circ \cdot \sin 250^\circ + \sin 250^\circ \cdot \sin 10^\circ$$

$$= \sin 10^\circ \cdot \sin 50^\circ - \sin 50^\circ \cdot \sin 70^\circ - \sin 70^\circ \cdot \sin 10^\circ$$

$$= \frac{1}{2} (2 \sin 10^\circ \sin 50^\circ - 2 \sin 50^\circ \sin 70^\circ - 2 \sin 70^\circ \sin 10^\circ)$$

$$= \frac{1}{2} (\cos 40^\circ - \cos 60^\circ + \cos 120^\circ - \cos 20^\circ + \cos 80^\circ - \cos 60^\circ)$$

$$= \frac{1}{2} (\cos 40^\circ - \cos 20^\circ + \cos 80^\circ - \frac{1}{2} - \frac{1}{2} - \frac{1}{2})$$

$$= \frac{1}{2} (2 \cos 60^\circ \cos 20^\circ - \cos 20^\circ - \frac{3}{2})$$

$$= \frac{1}{2} \left(-\frac{3}{2}\right) = -\frac{3}{4}$$

Sol.226. (a)

given

$$\alpha + \beta = \frac{5\pi}{4}$$

$$\tan(\alpha + \beta) = \tan \frac{5\pi}{4} = 1$$

$$\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta} = 1$$

$$\tan \alpha + \tan \beta = 1 - \tan \alpha \cdot \tan \beta$$

$$f(\theta) = \frac{1}{1 + \tan \theta}$$

$$f(\alpha)f(\beta) = \frac{1}{1+\tan\alpha} \cdot \frac{1}{1+\tan\beta}$$

$$f(\alpha)f(\beta) = \frac{1}{1+\tan\alpha+\tan\beta+\tan\alpha\tan\beta}$$

$$f(\alpha)f(\beta) = \frac{1}{1+1-\tan\alpha\tan\beta+\tan\alpha\tan\beta} = \frac{1}{2}$$

Sol.227. (b)

Given equation $x^2 - 6x + 8 = 0$

Roots of this equation are $\tan\alpha$ and $\tan\beta$

$$\tan\alpha + \tan\beta = 6$$

$$\tan\alpha \cdot \tan\beta = 8$$

$$\tan(\alpha + \beta) = \frac{6}{1-8} = -\frac{6}{7}$$

$$\cos 2(\alpha + \beta) = \frac{1 - \tan^2(\alpha + \beta)}{1 + \tan^2(\alpha + \beta)}$$

$$\cos 2(\alpha + \beta) = \frac{1 - \frac{36}{49}}{1 + \frac{36}{49}} = \frac{13}{85}$$

Sol.228. (c)

$$\tan 65^\circ + 2 \tan 45^\circ - 2 \tan 40^\circ - \tan 25^\circ$$

$$\tan 65^\circ - \cot 65^\circ + 2 - 2 \tan 40^\circ$$

$$\tan 65^\circ - \frac{1}{\tan 65^\circ} + 2 - 2 \tan 40^\circ$$

$$\Rightarrow -\left(\frac{1 - \tan^2 65^\circ}{\tan 65^\circ}\right) + 2 - 2 \tan 40^\circ$$

$$\Rightarrow -2 \frac{1}{\tan 130^\circ} + 2 - 2 \tan 40^\circ$$

$$\Rightarrow -2 \cot 130^\circ + 2 - 2 \tan 40^\circ$$

$$\Rightarrow -2 \cot(90^\circ + 40^\circ) + 2 - 2 \tan 40^\circ$$

$$\Rightarrow 2 \tan 40^\circ + 2 - 2 \tan 40^\circ = 2$$

Sol.229. (d)

$$\frac{\sqrt{3} \cos 10^\circ - \sin 10^\circ}{\sin 25^\circ \cos 25^\circ}$$

$$\Rightarrow 4 \times \frac{\frac{\sqrt{3}}{2} \cos 10^\circ - \frac{1}{2} \sin 10^\circ}{2 \sin 25^\circ \cos 25^\circ}$$

$$\Rightarrow 4 \times \frac{\sin 60^\circ \cos 10^\circ - \cos 60^\circ \sin 10^\circ}{2 \sin 25^\circ \cos 25^\circ}$$

$$\Rightarrow 4 \times \frac{\sin(60^\circ - 10^\circ)}{\sin 50^\circ} = 4 \frac{\sin 50^\circ}{\sin 50^\circ} = 4$$

Sol.230. (a)

$$\sin 9^\circ - \cos 9^\circ$$

$$\sqrt{(\sin 9^\circ - \cos 9^\circ)^2}$$

$$\Rightarrow \sqrt{\sin^2 9^\circ + \cos^2 9^\circ - 2 \sin 9^\circ \cos 9^\circ}$$

$$\Rightarrow \sqrt{1 - \sin 18^\circ}$$

$$\Rightarrow \sqrt{1 - \left(\frac{\sqrt{5}-1}{4}\right)} = \sqrt{\frac{5-\sqrt{5}}{4}} = \pm \frac{\sqrt{5-\sqrt{5}}}{2}$$

And we know that for 9° , $\sin\theta$ is less than $\cos\theta$

So $\sin 9^\circ - \cos 9^\circ$ is negative

$$\sin 9^\circ - \cos 9^\circ = -\frac{\sqrt{5-\sqrt{5}}}{2}$$

Sol.231. (b)

$$(\sin\theta - \cos\theta)^2 = 2$$

$$\sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta = 2$$

$$1 - 2\sin\theta\cos\theta = 2$$

$$\sin 2\theta = -1$$

$$\text{if } 2\theta = -90^\circ \text{ then } \theta = -45^\circ$$

$$\text{if } 2\theta = 270^\circ \text{ then } \theta = 135^\circ$$

there are two solutions between -180° to 180° .

Sol.232. (b)

maximum value of $a\cos x + b\sin x$ is $\sqrt{a^2 + b^2}$

so maximum value of $a\cos x + b\sin x + c$ will be

$$\sqrt{a^2 + b^2} + c$$

Sol.233. (c)

$$\tan \frac{\theta}{2} = \sqrt{\frac{1-\cos\theta}{1+\cos\theta}}$$

$$\text{put } \theta = 45^\circ$$

$$\tan 22.5^\circ = \sqrt{\frac{1-\frac{1}{\sqrt{2}}}{1+\frac{1}{\sqrt{2}}}} = \sqrt{\frac{\sqrt{2}-1}{\sqrt{2}+1}} = \sqrt{2}-1$$

that is irrational number

and

$$\cot 22.5^\circ = \sqrt{2} + 1$$

that is also an irrational number

$$\tan 22.5^\circ - \cot 22.5^\circ = -2$$

that is rational number.

Sol.234. (b)

$$\frac{x}{\cos\theta} = \frac{y}{\cos\left(\frac{2\pi}{3}-\theta\right)} = \frac{z}{\cos\left(\frac{2\pi}{3}+\theta\right)} = k$$

$$x = k \cos\theta,$$

$$y = k \cos\left(\frac{2\pi}{3}-\theta\right), z = k \cos\left(\frac{2\pi}{3}+\theta\right)$$

now $x + y + z$

$$= k \cos\theta + k \cos\left(\frac{2\pi}{3}-\theta\right) + k \cos\left(\frac{2\pi}{3}+\theta\right)$$

$$= k \left[\cos\theta + \cos\left(\frac{2\pi}{3}-\theta\right) + \cos\left(\frac{2\pi}{3}+\theta\right) \right]$$

$$= k \cos\theta + k \cos\left(\frac{2\pi}{3}\right) \cos\theta + k \sin\left(\frac{2\pi}{3}\right) \sin\theta +$$

$$k \cos\left(\frac{2\pi}{3}\right) \cos\theta - k \sin\left(\frac{2\pi}{3}\right) \sin\theta$$

$$= k \cos\theta + k \cos\left(\frac{2\pi}{3}\right) \cos\theta + k \cos\left(\frac{2\pi}{3}\right) \cos\theta$$

$$= k \cos\theta \left(1 + \cos\left(\frac{2\pi}{3}\right) + \cos\left(\frac{2\pi}{3}\right) \right)$$

$$= k \cos\theta \left(1 - \frac{1}{2} - \frac{1}{2} \right) = 0$$

Sol.235. (d)

$$p \tan(\theta - 30^\circ) = q \tan(\theta + 120^\circ)$$

$$\frac{p}{q} = \frac{\tan(\theta + 120^\circ)}{\tan(\theta - 30^\circ)}$$

$$\frac{p}{q} = \frac{\sin(\theta + 120^\circ) \cos(\theta - 30^\circ)}{\sin(\theta - 30^\circ) \cos(\theta + 120^\circ)}$$

$$\frac{p}{q} = \frac{\sin(\theta + 120^\circ) \cos(\theta - 30^\circ)}{\sin(\theta - 30^\circ) \cos(\theta + 120^\circ)}$$

apply componendo and dividendo

$$\frac{p+q}{p-q} = \frac{\sin(\theta + 120^\circ) \cos(\theta - 30^\circ) + \sin(\theta - 30^\circ) \cos(\theta + 120^\circ)}{\sin(\theta + 120^\circ) \cos(\theta - 30^\circ) - \sin(\theta - 30^\circ) \cos(\theta + 120^\circ)}$$

$$\text{and apply } \sin A \cos B + \cos A \sin B = \sin(A+B)$$

$$\frac{p+q}{p-q} = \frac{\sin(2\theta + 90^\circ)}{\sin(150^\circ)}$$

$$\frac{p+q}{p-q} = \frac{\sin(2\theta + 90^\circ)}{\sin(150^\circ)}$$

$$\frac{p+q}{p-q} = 2 \cos 2\theta$$

$$\frac{p+q}{p-q}$$

$$\frac{p+q}{p-q} = 2 \cos 2\theta$$

$$\frac{p+q}{p-q}$$

Sol.236. (c)

$$\cot 2x \cdot \cot 3x = 1$$

$$\tan 2x = \cot 3x$$

$$\tan A = \cot B \text{ for all } A + B = n\pi + \frac{\pi}{2}$$

$$2x + 3x = n\pi + \frac{\pi}{2}$$

$$5x = n\pi + \frac{\pi}{2}$$

$$x = \frac{n\pi}{5} + \frac{\pi}{10}$$

for $n = 0, 1, 2, 3, 4$ all values of x lie between $-\pi$ and π .

so there will be five values of x

$$\frac{\pi}{10}, \frac{3\pi}{10}, \frac{5\pi}{10}, \frac{7\pi}{10}, \frac{9\pi}{10}$$

Sol.237. (a)

$$\cos^{100}x - \sin^{100}x = 1$$

$$\cos^{100}x = 1 + \sin^{100}x$$

but maximum value of $\cos\theta$ is 1 so $\sin^{100}x$ must be zero.

If $\sin^{100}x = 0$ then $x = n\pi$

Sol.238. (c)

$$\sin 12^\circ \sin 48^\circ$$

$$\frac{1}{2} (2 \sin 12^\circ \sin 48^\circ)$$

$$\frac{1}{2} (\cos 36^\circ - \cos 60^\circ)$$

$$\frac{1}{2} \left(\frac{\sqrt{5}+1}{4} - \frac{1}{2} \right) = \frac{1}{2} \left(\frac{\sqrt{5}-1}{4} \right) = \frac{\sqrt{5}-1}{8}$$

Sol.239. (d)

$$\frac{\cos 17^\circ - \sin 17^\circ}{\cos 17^\circ + \sin 17^\circ}$$

divide numerator and denominator by $\cos 17^\circ$

$$\Rightarrow \frac{1 - \tan 17^\circ}{1 + \tan 17^\circ} = \tan(45^\circ - 17^\circ)$$

$$\Rightarrow \tan(28^\circ) = \cot(62^\circ)$$

Sol.240. (b)

$$p + q + r = \sin 35^\circ + \sin 25^\circ + \sin(-95^\circ)$$

$$= 2 \sin \frac{35^\circ + 25^\circ}{2} \cos \frac{35^\circ - 25^\circ}{2} - \sin(95^\circ)$$

$$= 2 \sin 30^\circ \cos 5^\circ - \cos 5^\circ = \cos 5^\circ - \cos 5^\circ = 0$$

Sol.241. (a)

$$pq + qr + rp$$

$$= \sin 35^\circ \sin 25^\circ + \sin 25^\circ \sin(-95^\circ) + \sin(-95^\circ) \sin 35^\circ$$

$$= \sin 35^\circ \sin 25^\circ - \sin 25^\circ \sin 95^\circ - \sin 95^\circ \sin 35^\circ$$

$$= \frac{1}{2} [2 \sin 35^\circ \sin 25^\circ - 2 \sin 25^\circ \sin 95^\circ - 2 \sin 95^\circ \sin 35^\circ]$$

$$= \frac{1}{2} \left[(\cos 10^\circ - \cos 60^\circ) \right]$$

$$= \frac{1}{2} \left[(\cos 120^\circ - \cos 70^\circ) + (\cos 130^\circ - \cos 60^\circ) \right]$$

$$= \frac{1}{2} \left[\left(\cos 10^\circ - \frac{1}{2} \right) + \left(-\frac{1}{2} - \cos 70^\circ \right) + \left(\cos 130^\circ - \frac{1}{2} \right) \right]$$

$$= \frac{1}{2} \left[\left(\cos 130^\circ + \cos 10^\circ - \cos 70^\circ - \frac{3}{2} \right) \right]$$

$$= \frac{1}{2} \left[\left(2 \cos 70^\circ \cos 60^\circ - \cos 70^\circ - \frac{3}{2} \right) \right]$$

$$= \frac{1}{2} \left[\left(\cos 70^\circ - \cos 70^\circ - \frac{3}{2} \right) \right] = \frac{-3}{4}$$

$$\frac{-3}{4}$$

Sol.242. (c)

$$(p + q + r)^2 = p^2 + q^2 + r^2 + 2(pq + qr + pr)$$

By above two solutions

$$0 = p^2 + q^2 + r^2 + 2 \left(-\frac{3}{4} \right)$$

$$p^2 + q^2 + r^2 = \frac{3}{2}$$

Sol.243. (a)

$$p = |\sin\alpha - \sin(\alpha - 90^\circ)|$$

$$p = |\sin\alpha + \cos\alpha|$$

we know that

$$-\sqrt{2} \leq \sin\alpha + \cos\alpha \leq \sqrt{2}$$

Then $0 \leq |\sin \alpha + \cos \alpha| \leq \sqrt{2}$
Minimum value is 0 at 135° or 315° .

Sol.244. (b)

$$p = |\sin \alpha + \cos \alpha|$$

we know that

$$-\sqrt{2} \leq \sin \alpha + \cos \alpha \leq \sqrt{2}$$

Then $0 \leq |\sin \alpha + \cos \alpha| \leq \sqrt{2}$

Maximum value is $\sqrt{2}$ at 45° or 225° .

Sol.245. (c)

$$\frac{p}{q} = \frac{\tan 2\alpha - \tan \alpha}{\cot \alpha - \cot 2\alpha} = \frac{\tan 2\alpha - \tan \alpha}{\frac{1}{\tan \alpha} - \frac{1}{\tan 2\alpha}}$$

$$= \frac{\tan 2\alpha - \tan \alpha}{\frac{1}{\tan 2\alpha - \tan \alpha}} = \tan 2\alpha \tan \alpha$$

Sol.246. (d)

$$p + q = (\tan 2\alpha - \tan \alpha) + (\cot \alpha - \cot 2\alpha)$$

$$= (\tan 2\alpha - \cot 2\alpha) + (\cot \alpha - \tan \alpha)$$

$$= \left(\tan 2\alpha - \frac{1}{\tan 2\alpha} \right) + \left(\frac{1}{\tan \alpha} - \tan \alpha \right)$$

$$= \left(\frac{\tan^2 2\alpha - 1}{\tan 2\alpha} \right) + \left(\frac{1 - \tan^2 \alpha}{\tan \alpha} \right)$$

$$= \left(\frac{\tan^2 2\alpha - 1}{\tan 2\alpha} \right) + 2 \left(\frac{1 - \tan^2 \alpha}{2 \tan \alpha} \right)$$

$$= \left(\frac{\tan^2 2\alpha - 1}{\tan 2\alpha} \right) + 2 \left(\frac{1}{\tan 2\alpha} \right)$$

$$= \left(\frac{\tan^2 2\alpha - 1 + 2}{\tan 2\alpha} \right) = \left(\frac{\tan^2 2\alpha + 1}{\tan 2\alpha} \right)$$

$$= 2 \left(\frac{\tan^2 2\alpha + 1}{2 \tan 2\alpha} \right) = 2 \left(\frac{1}{\sin 4\alpha} \right) = 2 \operatorname{cosec} 4\alpha$$

Sol.247. (c)

$$\frac{p}{q} = \tan 2\alpha \tan \alpha$$

$$\frac{p}{q} = \left(\frac{2 \tan \alpha}{1 - \tan^2 \alpha} \right) \tan \alpha$$

$$\frac{p}{2q} = \left(\frac{\tan^2 \alpha}{1 - \tan^2 \alpha} \right)$$

Apply Comp. and Divi.

$$\frac{p+2q}{p-2q} = \left(\frac{1}{2 \tan^2 \alpha - 1} \right)$$

$$\frac{p-2q}{p+2q} = (2 \tan^2 \alpha - 1)$$

$$\frac{p-2q}{p+2q} + 1 = 2 \tan^2 \alpha$$

$$\frac{2p}{p+2q} = 2 \tan^2 \alpha$$

$$= \tan^2 \alpha = \frac{p}{p+2q}$$

Sol.248. (c)

$$2 \sin \alpha + \cos \alpha = 2$$

$$4 \sin^2 \alpha + 4 \sin \alpha \cos \alpha + \cos^2 \alpha = 4$$

Divide by $\cos^2 \alpha$

$$4 \tan^2 \alpha + 4 \tan \alpha + 1 = 4 \sec^2 \alpha$$

$$4 \tan^2 \alpha + 4 \tan \alpha + 1 = 4(1 + \tan^2 \alpha)$$

$$4 \tan^2 \alpha + 4 \tan \alpha + 1 = 4 + 4 \tan^2 \alpha$$

$$2 \tan \alpha = 3$$

$$\tan \alpha = 3/4$$

Sol.249. (b)

$$\tan \alpha = 3/4$$

$$\text{so } \sin \alpha = 3/5 \text{ and } \cos \alpha = 4/5$$

$$= 2 \sin 2\alpha + \cos 2\alpha$$

$$= 4 \sin \alpha \cos \alpha + 2 \cos^2 \alpha - 1$$

$$= 4 \left(\frac{3}{5} \right) \left(\frac{4}{5} \right) + 2 \left(\frac{4}{5} \right)^2 - 1 = \frac{11}{5}$$

Sol.250. (c)

$$\frac{p}{q} = \frac{\sin A + \sin B}{\cos A + \cos B}$$

$$= \frac{2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)}{2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)} = \tan \left(\frac{A+B}{2} \right)$$

Sol.251. (d)

by above solution

$$\frac{p}{q} = \tan \left(\frac{A+B}{2} \right)$$

$$\frac{p^2}{q^2} = \tan^2 \left(\frac{A+B}{2} \right)$$

Now apply comp. and divi.

$$\frac{p^2 - q^2}{p^2 + q^2} = \frac{\tan^2 \left(\frac{A+B}{2} \right) - 1}{\tan^2 \left(\frac{A+B}{2} \right) + 1}$$

$$= \frac{1 - \tan^2 \left(\frac{A+B}{2} \right)}{1 + \tan^2 \left(\frac{A+B}{2} \right)} = -\cos 2 \left(\frac{A+B}{2} \right)$$

$$= -\cos (A+B) = \cos [\pi - (A+B)]$$

Sol.252. (c)

$$\left(\frac{\sqrt{3} \operatorname{cosec} 20^\circ}{4} - \frac{\cos 70^\circ}{4} \right)$$

$$\left(\frac{\sqrt{3}}{4 \sin 20^\circ} - \frac{1}{4 \cos 20^\circ} \right)$$

$$\frac{\sqrt{3} \cos 20^\circ - \sin 20^\circ}{4 \sin 20^\circ \cos 20^\circ}$$

$$= \frac{\frac{\sqrt{3}}{2} \cos 20^\circ - \frac{1}{2} \sin 20^\circ}{2 \sin 20^\circ \cos 20^\circ}$$

$$= \frac{\sin 60^\circ \cos 20^\circ - \cos 60^\circ \sin 20^\circ}{2 \sin 20^\circ \cos 20^\circ}$$

$$= \frac{\sin (60^\circ - 20^\circ)}{\sin 40^\circ} = \frac{\sin 40^\circ}{\sin 40^\circ} = 1$$

Sol.253. (b)

$$\frac{p^2 + q^2}{p^2 q^2} = \frac{1}{p^2} + \frac{1}{q^2}$$

$$\Rightarrow \frac{1}{\operatorname{cosec}^2 20^\circ} + \frac{1}{\sec^2 20^\circ} = \sin^2 20^\circ + \cos^2 20^\circ = 1$$

Sol.254. (c)

$$\cos (2x + 3y) = \frac{1}{2}, -\pi < (2x + 3y) < \pi$$

$$2x + 3y = 60^\circ \dots\dots\dots(i)$$

$$\text{Or } 2x + 3y = -60^\circ \dots\dots\dots(ii)$$

$$\cos (3x + 2y) = \frac{\sqrt{3}}{2}, -\pi < (3x + 2y) < \pi$$

$$3x + 2y = 30^\circ \dots\dots\dots(iii)$$

$$\text{Or } 3x + 2y = -30^\circ \dots\dots\dots(iv)$$

Add (i) and (ii)

$$5x + 5y = 90^\circ \Rightarrow x + y = 18^\circ$$

Add (i) and (iv)

$$5x + 5y = 30^\circ \Rightarrow x + y = 6^\circ$$

Add (ii) and (ii)

$$5x + 5y = -30^\circ \Rightarrow x + y = -6^\circ$$

Add (ii) and (iv)

$$5x + 5y = -90^\circ \Rightarrow x + y = -18^\circ$$

Sol.255. (c)

By above solution

Subtract (ii) from (iii)

$$x - y = 90^\circ$$

Subtract (ii) from (iv)

$$x - y = 30^\circ$$

Subtract (i) from (iii)

$$x - y = -30^\circ$$

Subtract (i) from (iv)

$$x - y = -90^\circ$$

Sol.256. (c)

both $\cot \theta$ and $\operatorname{cosec} \theta$ are negative in 4th quadrant

$\cot \theta = -\sqrt{3}$ and $\operatorname{cosec} \theta = -2$, exists only for $\theta = -30^\circ$

Sol.257. (c)

$$x + \frac{1}{x} = 2 \cos \theta$$

$$\left(x + \frac{1}{x} \right)^3 = (2 \cos \theta)^3$$

$$x^3 + \frac{1}{x^3} + 3 \left(x + \frac{1}{x} \right) = 8 \cos^3 \theta$$

$$x^3 + \frac{1}{x^3} + 3(2 \cos \theta) = 8 \cos^3 \theta$$

$$x^3 + \frac{1}{x^3} = 8 \cos^3 \theta - 6 \cos \theta$$

$$= 2(4 \cos^3 \theta - 3 \cos \theta) = 2 \cos 3\theta$$

Sol.258. (b)

$$\frac{\sin x}{\cos x} + \frac{1}{\cos x} = 2 \cos x$$

$$\sin x + 1 = 2 \cos^2 x$$

$$\sin x + 1 = 2 - 2 \sin^2 x$$

$$2 \sin^2 x + \sin x - 1 = 0$$

$$(2 \sin x - 1)(\sin x + 1) = 0$$

$$\sin x = \frac{1}{2} \text{ and } -1$$

$\sin x = \frac{1}{2}$ only one value $x = 30^\circ$ in first quadrant

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